

UNMANNED AERIAL SYSTEMS (UAS)

Department of Aerospace Engineering · Madras Institute of Technology

PROJECT REPORT

Design and Dynamic Simulation of a Fixed-Wing UAV

*for Biodiversity Monitoring in
Mudumalai Tiger Reserve*

Submitted in partial fulfillment of the requirements
for the course **Unmanned Aerial Systems (UAS)**

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PHASE-I: UAV DESIGN

1 Project Overview and Objectives

1.1 Project Background

Mudumalai Tiger Reserve (MTR) is a vital protected area in Tamil Nadu's Nilgiris Biosphere Reserve, serving as a key corridor for elephants and tigers in the Western Ghats. This diverse habitat—featuring forests, grasslands, and river ecosystems—supports a rich array of wildlife, including large mammals and bird species. However, the reserve is increasingly threatened by habitat fragmentation from nearby agriculture and tourism, the spread of invasive plants overtaking native vegetation, poaching activities, and human-wildlife conflicts intensified by climate-related wildfires. Conventional monitoring techniques, such as ground-based surveys, camera traps, and satellite imagery, are limited by their cost, labor intensity, and lack of timeliness. These gaps hinder effective conservation efforts, such as those under Project Tiger, by delaying detection of critical changes in animal habitats and ecosystem health.

We need a cheap, self-flying drone made just for MTR's hilly land (300–1,300 meters high, with 20–30 degree slopes). It should give clear pictures (3–5 cm detail per pixel) and real-time info over 1,400–1,800 hectares per flight. Current drones don't last long enough (need 3+ hours for full areas), can't carry the right cameras for plant health checks (Normalized Difference Vegetation Index-NDVI) or heat-sensing for dangers, and can't follow the land's ups and downs while following India's drone rules (DGCA for flying out of sight). This leaves hidden problems like spreading weeds and lost homes, cutting off animal paths.

1.2 Problem Statement

The core challenge lies in developing an affordable and reliable aerial monitoring solution tailored to MTR's challenging terrain, including undulating hills and steep slopes. Existing drone technologies often fail to provide sustained, stable flights over large areas while maintaining precise data capture for biodiversity assessment. There is a pressing need for a fixed-wing UAV design that ensures aerodynamic efficiency, balance, and control responsiveness in variable wind conditions, enabling safe, autonomous operations compliant with Indian aviation regulations. Without a robust digital design and simulation framework, it is difficult to verify the UAV's performance—such as its ability to trim for steady flight

and respond to control inputs—before physical prototyping, leading to inefficiencies in addressing MTR’s urgent conservation demands.

To address this, the project team must explicitly undertake the following:

- **Translate mission requirements into UAV design:** Derive geometric and aerodynamic specifications from MTR-specific needs (e.g., long endurance for area coverage, stability for terrain-following at low altitudes), using low-fidelity/high-fidelity methods or handbook-based approaches to generate aerodynamic data.
- **Estimate center of gravity (CG), inertia:** Model the UAV’s structure in CAD software and calculate CG position and inertia tensor to ensure overall balance and handling.
- **Perform trimming analysis:** Set up the UAV for equilibrium flight conditions (e.g., level cruise) by solving for control surface deflections and aerodynamic forces.
- **Conduct response analysis:** Use MATLAB/Simulink to simulate the UAV’s dynamic behavior, evaluating stability modes (e.g., short-period and phugoid) and transient responses to control inputs (e.g., elevator or aileron deflections) under simulated MTR conditions like gusts.

1.3 Objectives of the UAV

1. Habitat Change Detection

- **Explanation:** This objective involves tracking shifts in the landscape over time, such as deforestation, erosion, or canopy loss due to human activity, climate impacts, or natural events like wildfires. By comparing repeated UAV flights (e.g., quarterly), you can quantify changes—like a 10% drop in forest cover—and generate alerts for rangers to intervene. It’s crucial for long-term ecosystem health, helping prevent habitat fragmentation that isolates animal populations.
- **General Sensors/Cameras:**
 - Multispectral cameras: Capture light in multiple wavelengths (e.g., visible and near-infrared) to calculate vegetation indices like NDVI, highlighting healthy vs. degraded areas.
 - RGB (visible light) cameras: Provide high-resolution color images for creating orthomosaics (stitched maps) to visually spot changes in land cover.
 - LiDAR (Light Detection and Ranging): Uses laser pulses to map 3D terrain

and canopy height, detecting subtle elevations like erosion scars or regrowth.

2. Wildlife Population and Movement Tracking

- **Explanation:** The goal here is to count and monitor animal numbers, behaviors, and migration paths without disturbing them—e.g., spotting elephant herds or tiger territories in real-time. This supports population estimates, corridor mapping, and conflict prediction (like elephants nearing villages), enabling proactive measures to reduce poaching or human-animal clashes.
- **General Sensors/Cameras:**
 - Thermal (infrared) cameras: Detect heat signatures of animals against cooler backgrounds, especially useful at dawn/dusk or in dense foliage for hidden species.
 - High-resolution RGB cameras: Offer detailed visuals for species identification and counting during low-altitude passes, integrated with GPS for movement tracking.
 - Acoustic sensors: Record animal calls or footsteps for passive monitoring, helping map nocturnal or elusive wildlife like birds or big cats.

3. Invasive Species and Threat Detection

- **Explanation:** This targets spotting non-native plants (e.g., weeds overtaking grasslands) or human threats (e.g., poacher trails or illegal traps) early. By identifying and mapping these, the UAV enables targeted removal or patrols, preventing biodiversity loss and ecosystem imbalance—vital in reserves where invasives can spread rapidly after disturbances like fires.
- **General Sensors/Cameras:**
 - Hyperspectral cameras: Analyze reflected light across a wide spectrum to distinguish invasive plants by unique chemical signatures, even under canopy.
 - Thermal cameras: Spot unusual heat patterns, like human activity (poachers) or stressed vegetation from invasives competing for resources.
 - RGB or multispectral cameras: Provide broad visual scans for color/texture anomalies, like dense weed patches, with AI overlays for automated threat flagging.

1.4 Scope and Methodology

Limit the project to computational design —no hardware prototyping. Use any CAD (e.g., Solidworks, CATIA) for modeling, ANSYS/XFLR5 or handbooks (e.g., Roskam or Etkin) for aero estimates, and MATLAB/Simulink for 6-DOF dynamics (longitudinal and lateral modes). Proceed iteratively: Requirements → Aero/CG estimation → Trimming → Response analysis.

1.5 Expected Outcomes and Impact

Deliver a comprehensive report with CAD models, aerodynamic tables, trim solutions, Simulink files, stability plots, and response curves, demonstrating a UAV capable of safe, efficient flights for MTR monitoring.

2 Preliminary Weight Estimation

2.1 Mission Requirements

The UAV is required to map a total ground area A within a fixed mission time t , flying at constant altitude H and constant ground speed V . The image resolution requirement is specified in terms of Ground Sampling Distance (GSD).

Parameter	Value
Total mapping area, A	$15 \times 10^6 \text{ m}^2$
Flight speed, V	15 m/s
Mission time, t	7200 s
Flight altitude, H	100 m
Required GSD	4 cm/pixel

2.2 Payload Weight

2.2.1 Camera Selection and Optical Parameter Estimation

Swath Width Requirement. The total area covered by the UAV during the mission is given by

$$A = V \cdot t \cdot W_s \quad (1)$$

where W_s is the ground swath width.

Solving for swath width:

$$W_s = \frac{A}{V \cdot t} \quad (2)$$

Substituting the mission values:

$$W_s = \frac{15 \times 10^6}{15 \times 7200} \quad (3)$$

$$\boxed{W_s = 138.9 \text{ m}} \quad (4)$$

This means the camera must image a ground width of approximately **139 meters** per exposure.

Field of View (FOV) Calculation. The swath width is geometrically related to the camera Field of View (FOV) and flight altitude.

From basic trigonometry:

$$\tan\left(\frac{\text{FOV}}{2}\right) = \frac{W_s/2}{H} \quad (5)$$

Thus,

$$\text{FOV} = 2 \tan^{-1}\left(\frac{W_s}{2H}\right) \quad (6)$$

Substituting values:

$$\text{FOV} = 2 \tan^{-1}\left(\frac{138.9}{2 \times 100}\right) \quad (7)$$

$$\boxed{\text{FOV} \approx 69.5^\circ} \quad (8)$$

This is the required angular field of view of the camera in the flight direction.

Ground Sampling Distance (GSD). Ground Sampling Distance is defined as the physical ground length represented by one image pixel.

$$\text{GSD} = \frac{H \cdot p}{f} \quad (9)$$

where:

- p = physical pixel size of the camera sensor (meters)
- f = camera focal length (meters)

Rearranging to compute focal length:

$$f = \frac{H \cdot p}{\text{GSD}} \quad (10)$$

Focal Length Calculation. Given:

$$H = 100 \text{ m} \quad (11)$$

$$\text{GSD} = 4 \text{ cm} = 0.04 \text{ m} \quad (12)$$

$$p = 4.5 \text{ }\mu\text{m} = 4.5 \times 10^{-6} \text{ m} \quad (13)$$

Substituting:

$$f = \frac{100 \times 4.5 \times 10^{-6}}{0.04} \quad (14)$$

$$\boxed{f = 0.01125 \text{ m} = 11.25 \text{ mm}} \quad (15)$$

This is the required focal length to achieve the specified GSD at 100 m altitude.

Sensor Width Consistency Check. Sensor width W_{sensor} is related to focal length and FOV by:

$$\text{FOV} = 2 \tan^{-1} \left(\frac{W_{\text{sensor}}}{2f} \right) \quad (16)$$

Rearranging:

$$W_{\text{sensor}} = 2f \tan \left(\frac{\text{FOV}}{2} \right) \quad (17)$$

Substituting:

$$W_{\text{sensor}} = 2 \times 0.01125 \times \tan(34.75^\circ) \quad (18)$$

$$\boxed{W_{\text{sensor}} \approx 15.6 \text{ mm}} \quad (19)$$

This indicates that a sensor width of approximately 16 mm is required, which corresponds closely to APS-C class sensors.

Camera Matching Interpretation. A suitable camera must satisfy **all three** conditions:

- Pixel size $\approx 4.5 \text{ }\mu\text{m}$

- Sensor width ≥ 16 mm
- Lens focal length $\approx 11\text{--}12$ mm

"DJI Zenmuse X5S (M4/3)" Camera satisfies the constraints.

Weight of the DJI Zenmuse X5S (M4/3) - 461 g. This is the mass of the entire camera + gimbal unit as typically used on drones like the Inspire 2, including the lens and balancing accessories.

$$W_{\text{camera}} = 0.46 \text{ kg}$$

2.2.2 Other Sensor Weights

As per mission objectives, we also need a thermal camera.

A typical thermal camera weighs upto 0.20 kg

Mandatory Drone sensors

Sensors	Weight (kg)
GNSS Receiver	0.03
Inertial Measurement Unit <i>IMU</i>	0.02
Data Storage Module	0.04

$$W_{\text{sensor}} = 0.03 + 0.02 + 0.04 = 0.09 \text{ kg}$$

2.2.3 Total Payload Weight

$$W_{\text{payload}} = W_{\text{RGBCamera}} + W_{\text{ThermalCamera}} + W_{\text{sensors}}$$

$$W_{\text{payload}} = 0.46 + 0.20 + 0.09 \text{ kg}$$

$$W_{\text{payload}} = 0.75 \text{ kg}$$

2.3 Battery Weight

$$W_{\text{Battery}} = \frac{E_{\text{mission}}}{e_{\text{battery}}}$$

where

E_{mission} $\rightarrow$ Total Energy Required for the mission

e_{battery} $\rightarrow$ Specific Energy of the battery

2.3.1 Energy required for Mission

$$E_{\text{mission}} = E_{\text{cruise}} + E_{\text{climb}} + E_{\text{payload}}$$

Energy required at Cruise (E_{cruise}).

$$E_{\text{cruise}} = P_{\text{elec}} \cdot t_{\text{mission}}$$

$$P_{\text{elec}} = \frac{W_0 V}{(L/D) \eta_{\text{prop}}} \quad (20)$$

$$\eta_{\text{prop}} = \eta_{\text{motor}} \cdot \eta_{\text{ESC}} \cdot \eta_{\text{propeller}}$$

Historic Values (electric-fixed wing UAVs)

Efficiency	Range
Motor Efficiency, η_{motor}	0.85 – 0.92
ESC Efficiency, η_{ESC}	0.97 – 0.99
Propeller Efficiency, $\eta_{\text{propeller}}$	0.65 – 0.80

Using the historic data, we fix our efficiencies as follows:

$$\eta_{\text{motor}} = 0.90 \quad \eta_{\text{ESC}} = 0.98 \quad \eta_{\text{propeller}} = 0.70$$

$$\eta_{\text{prop}} = 0.90 \times 0.98 \times 0.70$$

$$\eta_{\text{prop}} \approx 0.62$$

Historical L/D Values (fixed-wing UAVs)

Aircraft-type	Typical L/D
Hand-launch foam UAV	8 – 10
Small composite UAV	10 – 14
Survey/Mapping UAV	12 – 18
High-Aspect Ratio UAV	18 – 25

As our UAV comes under Survey/Mapping UAV, we choose a reasonable value of L/D as 12.

$$\frac{L}{D} = 12$$

Initial guess of $W_0 = 3.5 \times 9.81$ N.

Our velocity (V) is 15 m/s

Therefore,

$$P_{elec} = \frac{3.5 \times 9.81 \times 15}{12 \times 0.62}$$

$$P_{elec} \approx 69W$$

$$t_{mission} = 2\text{hours}$$

$$E_{cruise} = P_{elec} \cdot t_{mission}$$

$$E_{cruise} = 69 \times 2$$

$$E_{cruise} = 138Wh$$

Energy required for Climb (E_{climb}).

$$E_{climb} = \frac{W_0 h}{\eta_{prop}} \quad (21)$$

$$E_{climb} = \frac{3.5 \times 9.81 \times 100}{0.62}$$

$$E_{climb} \approx 5538J \approx 1.54Wh$$

Energy required for Payload ($E_{payload}$).

$$E_{payload} = P_{payload} \cdot t_{mission}$$

$$E_{payload} = 21.6 \times 2$$

$$E_{payload} = 43.2 Wh$$

Payload Component	Typical Power (W)
DJI Zenmuse X5S (M4/3 RGB Camera + Gimbal)	10.0
Thermal Camera (Radiometric, UAV-grade)	10.0
GNSS Receiver (RTK-capable)	1.0
IMU (MEMS, navigation-grade)	0.1
Data Storage Module (SD/SSD)	0.5
Total Payload Power	21.6

Adding a margin for efficiency and peaks, gives us $E_{\text{payload}} = 50 \text{ Wh}$

Mission Energy. $E_{\text{mission}} = 138 + 1.54 + 50 \text{ Wh}$

$E_{\text{mission}} = 189.54 \approx 190 \text{ Wh}$

Battery Specific Energy. Battery specific energy is defined as the usable electrical energy stored per unit mass:

$$e_{\text{battery}} = \frac{E_{\text{usable}}}{m_{\text{battery}}} \quad [\text{Wh/kg}] \quad (22)$$

For lithium-based batteries commonly employed in fixed-wing UAVs, the achievable pack-level specific energy ranges between 180–250 Wh/kg. Accounting for depth-of-discharge limits, wiring losses, and power electronics inefficiencies, a conservative value is adopted:

$$\boxed{e_{\text{battery}} = 200 \text{ Wh/kg}} \quad (23)$$

This value is representative of practical, flight-qualified battery systems and is appropriate for preliminary weight and endurance estimation.

Battery Weight

$$W_{\text{battery}} = \frac{E_{\text{mission}}}{e_{\text{battery}}}$$

$$W_{\text{battery}} = \frac{190}{200}$$

$$W_{\text{battery}} = 0.95 \text{ kg}$$

2.4 Empty Weight Modeling

The empty weight of the UAV comprises the structural airframe, propulsion system, avionics, control actuators, wiring, and all fixed onboard systems excluding payload and battery. Prior to detailed geometric design, the empty weight cannot be determined from first principles alone and is therefore modeled using a scaling relationship:

$$W_{\text{empty}} = f_{\text{empty}} W_0 \quad (24)$$

This formulation is physically justified by the proportional scaling of structural mass with bending loads, which themselves scale with the aircraft takeoff weight. For small fixed-wing electric UAVs employing composite or foam-composite construction, typical values of f_{empty} range from 0.50 to 0.65.

In this study, a conservative value is adopted:

$$\boxed{f_{\text{empty}} = 0.60} \quad (25)$$

This accounts for structural robustness, propulsion hardware, avionics integration, and payload mounting provisions.

2.5 Maximum Takeoff Weight

$$W_0 = W_{\text{empty}} + W_{\text{payload}} + W_{\text{battery}}$$

$$W_0 = \frac{W_{\text{payload}} + W_{\text{battery}}}{1 - f_{\text{empty}}}$$

W_{battery} depends on W_0 (refer eq.(20) and eq.(21)), so we calculate the maximum takeoff weight in an iterative manner.

Initial Guess of $W_0 = 3.5 \text{ kg}$

Using this W_0 , we iterate the calculation of W_{battery} by substituting it in equations (20) and (21)

2.5.1 MATLAB Code

```
format long
W_payload = 0.75; % kg
t_mission = 2; % hr
```

```

W0_initial = 3.5; % kg
V = 15; % m/s
L_D = 12;
eta_prop = 0.62;
h = 100; % m
f = 0.60;
tol = 0.00001;
converged = false;
e_battery = 200; % Wh/kg
iter = 1;
max_iter = 30;
while ~converged || iter > max_iter
    P = (W0_initial*9.81*15)/(L_D*eta_prop);
    E_cruise = P*t_mission; % Wh
    E_climb = (W0_initial*h)/(eta_prop*3600); % Wh
    E_payload = 50; % Wh
    E_mission = E_cruise + E_climb + E_payload;
    W_battery = E_mission/e_battery;
    W0 = (W_payload + W_battery) / (1 - f);
    fprintf('Iteration %d:\n', iter);
    fprintf('Initial W0 = %.4f\n', W0_initial);
    fprintf('New W0 = %.4f\n\n', W0);
    if abs(W0 - W0_initial) < tol
        converged = true;
    else
        W0_initial = W0;
        iter = iter + 1;
    end
end
fprintf('The final value of W0 is %.4f\n', W0);
fprintf("Empty Weight = %.4f\n", W0 - W_payload - W_battery);
fprintf("Battery Weight = %.4f\n", W_battery);
fprintf("Payload Weight = %.4f\n", W_payload);

```

2.5.2 Results

Maximum Takeoff Weight (W_0) is estimated as 4.9506 kg $\approx$ **4.95 kg**

Iteration	Initial W_0 (kg)	Final W_0 (kg)
1	3.5000	4.2326
2	4.2326	4.5952
3	4.5952	4.7747
4	4.7747	4.8635
5	4.8635	4.9075
6	4.9075	4.9293
7	4.9293	4.9401
8	4.9401	4.9454
9	4.9454	4.9481
10	4.9481	4.9494
11	4.9494	4.9500
12	4.9500	4.9503
13	4.9503	4.9505
14	4.9505	4.9506
15	4.9506	4.9506

3 Secondary Weight Estimation

3.1 Results from MTOW Estimation

From the MTOW estimation, we have,

- Maximum Takeoff Mass: $m = 4.95$ kg
- Weight: $W = 48.5$ N
- Battery Weight: $W_b = 1.23$ kg
- Payload Weight: $W_{payload} = 0.75$ kg
- Empty Weight: $W_{empty} = 2.97$ kg

3.2 Introduction

After estimating UAV MTOW (W_{TO}) from mission analysis (payload + battery + structure), the next step is to determine,

1. Wing reference Area (S)
2. Required Motor Power (P_{motor})

Unlike fuel aircraft, electric UAV sizing depends strongly on:

- Motor efficiency (η_m)
- Propeller efficiency (η_p)

- Battery energy density
- Maximum continuous current limits

Thus, instead of thrust-to-weight ratio (jet) or power loading (piston fuel aircraft), we use:

Electric Power Loading: $\frac{W}{P_{elec}}$

3.3 Key Design Parameters

Three key ratios define electric UAV preliminary sizing:

1. Wing Loading ($\frac{W}{S}$)
 - Indicates how much weight is supported per unit wing area.
 - Lower $\frac{W}{S} \rightarrow$ lower stall speed
 - Higher $\frac{W}{S} \rightarrow$ smaller wing, higher cruise speed
2. Electric Power Loading ($\frac{W}{P_{shaft}}$)
 - $P_{shaft} = \text{Mechanical power delivered to propeller}$
 - $P_{elec} = \frac{P_{shaft}}{\eta_m \eta_{ESC}}$
3. Energy Loading ($\frac{W}{E_b}$)
 - $E_b =$ usable battery energy (Wh or joules)
 - This parameter is unique to electric aircraft and controls endurance

3.4 Performance Constraints

Instead of fuel constraints (used in p=weight estimation), the aircraft must satisfy the following performance constraints:

- Stall speed (V_S)
- Maximum speed (V_{max})
- Cruise Speed
- Maximum Rate of Climb (ROC_{max})
- Take - off Run (S_{TO})
- Service Ceiling (h_c)

The most important constraints from the above are Maximum speed, Take-off run and Stall speed.

3.4.1 Steps to find design $\left(\frac{W}{S}\right)_d$ and design $\left(\frac{W}{P}\right)_d$

1. Derive one equation for each aircraft performance requirement (e.g., V_S , $V_{\max}$, V_{cruise} , ROC , S_{TO} , h_c). For a propeller-driven aircraft, these equations are generally written in terms of power loading $\left(\frac{W}{P}\right)$ as a function of wing loading $\left(\frac{W}{S}\right)$:

$$\left(\frac{W}{P}\right)_{V_S} = f_1\left(\frac{W}{S}, V_S\right) \quad (26)$$

$$\left(\frac{W}{P}\right)_{V_{\max}} = f_2\left(\frac{W}{S}, V_{\max}\right) \quad (27)$$

$$\left(\frac{W}{P}\right)_{V_{\text{cruise}}} = f_3\left(\frac{W}{S}, V_{\text{cruise}}\right) \quad (28)$$

$$\left(\frac{W}{P}\right)_{S_{TO}} = f_4\left(\frac{W}{S}, S_{TO}\right) \quad (29)$$

$$\left(\frac{W}{P}\right)_{ROC} = f_5\left(\frac{W}{S}, ROC\right) \quad (30)$$

$$\left(\frac{W}{P}\right)_{h_c} = f_6\left(\frac{W}{S}, h_c\right) \quad (31)$$

2. Sketch all derived equations in one plot. The vertical axis is power loading (W/P) and the horizontal axis is wing loading (W/S). Thus the plot illustrates the variations of power loading with respect to wing loading. These graphs will intersect each other at several points and may produce several regions.
3. Identify the acceptable region inside the regions that are produced by the axes and the graphs. The acceptable region is the region that meets all aircraft performance requirements. (Refer Aircraft Design - A Systems Engineering Approach by Mohammed H Sadraey)
4. Determine the design point (i.e., the optimum selection). For a battery-powered fixed-wing UAV, the optimal design point is the one that results in the lowest required electric motor power, since lower power demand directly reduces motor size, electronic speed controller (ESC) rating, and battery discharge rate. This leads to reduced system mass, improved endurance, and lower overall cost.
5. From the design point, obtain two numbers: corresponding wing loading $\left(\frac{W}{S}\right)_d$ and corresponding power loading $\left(\frac{W}{P}\right)_d$
6. Calculate the wing area and required motor power from the selected design point values, since the maximum take-off weight W_{TO} of the UAV has been already determined. The wing area is obtained by dividing the take-off weight by the chosen

wing loading. The required motor shaft power is determined by deriving the take-off weight by selected power loading.

$$S = \frac{W_{TO}}{\left(\frac{W}{S}\right)_d} \quad (32)$$

$$P = \frac{W_{TO}}{\left(\frac{W}{P}\right)_d} \quad (33)$$

3.5 Derivation of Constraint Equations

Lift Equation,

$$L = \frac{1}{2}\rho V^2 SC_L \quad (34)$$

Drag Equation,

$$D = \frac{1}{2}\rho V^2 SC_D \quad (35)$$

Dynamic Pressure,

$$q = \frac{1}{2}\rho V^2 \quad (36)$$

So,

$$L = qSC_L \quad (37)$$

$$D = qSC_D \quad (38)$$

Drag Polar,

$$C_D = C_{D_o} + kC_L^2 \quad (39)$$

Lift from Equilibrium Flight

$$L = nW \quad (40)$$

$$qSC_L = nW$$

$$C_L = \frac{n(W/S)}{q} \quad (41)$$

Substitute into drag polar

$$C_D = C_{D_o} + k \left(\frac{n(W/S)}{q} \right)^2 \quad (42)$$

General Drag Expression,

$$D = qS \left[C_{D_o} + k \left(\frac{n(W/S)}{q} \right)^2 \right] \quad (43)$$

÷ by W

$$\frac{D}{W} = \frac{qC_{D_o}}{W/S} + \frac{kn^2(W/S)}{q}$$

In steady flight, T = D

$$\frac{T}{W} = \frac{qC_{D_o}}{W/S} + \frac{kn^2(W/S)}{q} \quad (44)$$

Electric Power Relation:

Propeller Shaft Power

$$P_{\text{shaft}} = TV$$

Battery must supply,

$$P = \frac{TV}{\eta_{\text{prop}}}$$

÷ by W

$$\frac{P}{W} = \frac{T}{W} \frac{V}{\eta_{\text{prop}}} \quad (45)$$

substitute (44) in (45)

$$\boxed{\frac{P}{W} = \frac{V}{\eta_{\text{prop}}} \left[\frac{qC_{D_o}}{W/S} + \frac{kn^2(W/S)}{q} \right]} \quad (46)$$

All constraints are derived from the above equation.

3.5.1 Stall Constraint (V_S)

At stall $n = 1$, $C_L = C_{L_{\max}}$

$$L = W$$

$$W = \frac{1}{2} \rho V_{\text{stall}}^2 S C_{L_{\max}}$$

$$\left(\frac{W}{S}\right)_{\text{stall}} = \frac{1}{2} \rho V_{\text{stall}}^2 C_{L_{\max}} \quad (47)$$

Stall constraint will be plotted as a straight vertical line in the constraint plot.

3.5.2 Cruise Constraint

At Level Cruise, $n = 1$

$$\frac{P}{W} = \frac{V_{\text{cruise}}}{\eta_{\text{prop}}} \left[\frac{q C_{D_o}}{W/S} + \frac{k(W/S)}{q} \right]$$

where,

$$q = \frac{1}{2} \rho V_{\text{cruise}}^2$$

$$\left(\frac{W}{P}\right)_{V_{\text{cruise}}} = \frac{1}{\frac{V_{\text{cruise}}}{\eta_{\text{prop}}} \left[\frac{q C_{D_o}}{W/S} + \frac{k(W/S)}{q} \right]} \quad (48)$$

3.5.3 Maximum Speed Constraint

$$V = V_{\max}$$

$$\frac{P}{W} = \frac{V_{\max}}{\eta_{\text{prop}}} \left[\frac{q_{\max} C_{D_o}}{W/S} + \frac{k(W/S)}{q_{\max}} \right]$$

$$q = \frac{1}{2} \rho V_{\max}^2$$

$$\left(\frac{W}{P}\right)_{V_{\max}} = \frac{1}{\frac{V_{\max}}{\eta_{\text{prop}}} \left[\frac{q_{\max} C_{D_o}}{W/S} + \frac{k(W/S)}{q_{\max}} \right]} \quad (49)$$

3.5.4 Climb Constraint

During Climb, Force Balance:

$$T - D = W \sin \gamma$$

For small angle,

$$\sin \gamma = \frac{ROC}{V_{\text{climb}}}$$

Where, $V_{\text{climb}} = 1.3 \times V_S$

$$\begin{aligned} \frac{P}{W} &= \frac{V_{\text{climb}}}{\eta_{\text{prop}}} \left[\frac{qC_{D_o}}{W/S} + \frac{k(W/S)}{q} + \frac{ROC}{V_{\text{climb}}} \right] \\ \left(\frac{W}{P} \right)_{ROC} &= \frac{1}{\frac{V_{\text{climb}}}{\eta_{\text{prop}}} \left[\frac{qC_{D_o}}{W/S} + \frac{k(W/S)}{q} + \frac{ROC}{V_{\text{climb}}} \right]} \end{aligned} \quad (50)$$

3.5.5 Service Ceiling Constraint

At Service Ceiling, we will have a low value for Rate of Climb (ROC). That low value is considered as 0.5 m/s.

$$\frac{P}{W} = \frac{V_{\text{ceiling}}}{\eta_{\text{prop}}} \left[\frac{qC_{D_o}}{W/S} + \frac{k(W/S)}{q} + \frac{0.5}{V_{\text{ceiling}}} \right]$$

Where, $V_{\text{ceiling}} = 1.2 \times V_S$

$$\left(\frac{W}{P} \right)_{h_c} = \frac{1}{\frac{V_{\text{ceiling}}}{\eta_{\text{prop}}} \left[\frac{qC_{D_o}}{W/S} + \frac{k(W/S)}{q} + \frac{0.5}{V_{\text{ceiling}}} \right]} \quad (51)$$

3.5.6 Take-off Constraint

Newton's second law during Take-off roll,

$$T - D - \mu(W - L) = \frac{W}{g} a$$

where,

- μ = rolling friction coefficient

- L builds up during roll
- D builds up during roll
- a = acceleration

Approximation used by Sadraey,

During ground roll:

$$L \approx qSC_{L_{TO}}$$

$$L \approx qSC_{L_{DO}}$$

Takeoff lift coefficient is usually:

$$C_{L_{TO}} \approx 0.7C_{L_{\max}}$$

Distance required to reach liftoff speed:

$$S_{TO} = \frac{V_{LOF}^2}{2a}$$

Acceleration:

$$a = \frac{g}{W}(T - D - \mu W)$$

Ignoring drag, during early estimate:

$$a \approx g\left(\frac{T}{W} - \mu\right)$$

So,

$$S_{TO} = \frac{V_{LOF}^2}{2g\left(\frac{T}{W} - \mu\right)}$$

Liftoff speed:

$$V_{LOF} = 1.2V_S$$

Stall speed, V_S

$$V_S = \sqrt{\frac{2(W/S)}{\rho C_{L_{\max}}}}$$

So:

$$V_{LOF}^2 = 1.44 \frac{2(W/S)}{\rho C_{L_{\max}}}$$

$$V_{LOF}^2 = \frac{2.88(W/S)}{\rho C_{L_{\max}}}$$

Substitute V_{LOF}^2 in S_{TO}

$$S_{TO} = \frac{\frac{2.88(W/S)}{\rho C_{L_{\max}}}}{g(\frac{T}{W} - \mu)}$$

Simplifying we get,

$$S_{TO} = \frac{1.44(W/S)}{g\rho C_{L_{\max}}(\frac{T}{W} - \mu)}$$

Solving for T/W

$$\frac{T}{W} = \mu + \frac{1.44(W/S)}{g\rho C_{L_{\max}} S_{TO}}$$

Convert to Power Loading

For electric UAV,

$$P = \frac{TV}{\eta_{prop}}$$

At liftoff,

$$V = V_{LOF}$$

So:

$$\frac{P}{W} = \frac{V_{LOF}}{\eta_{prop}} \left[\mu + \frac{1.44(W/S)}{g\rho C_{L_{\max}} S_{TO}} \right]$$

$$\left(\frac{W}{P} \right)_{S_{TO}} = \frac{1}{\frac{V_{LOF}}{\eta_{prop}} \left[\mu + \frac{1.44(W/S)}{g\rho C_{L_{\max}} S_{TO}} \right]} \quad (52)$$

Equations (47), (48), (49), (50), (51) and (52) are the constraint equations. These equations

will be used shortly to find the design point.

3.6 Fixing Parameters

3.6.1 Performance Parameters

- Stall Speed, $V_S = 11$ m/s
- Cruise Speed, $V_{\text{cruise}} = 15$ m/s
- Maximum Speed, $V_{\text{max}} = 22$ m/s
- Rate of Climb, $ROC = 2$ m/s
- Takeoff Distance, $S_{TO} = 30$ m
- Service Ceiling, $h_c = 300$ m

3.6.2 Aerodynamic Parameters

For typical mapping UAVs,

- $C_{L_{max}} = 0.4$
- $C_{D_o} = 0.035$
- $AR = 10$
- $e = 0.8$
- Induced drag factor, $k = \frac{1}{\pi e AR} = 0.0398$

3.6.3 Other Parameters

- $\eta_{\text{prop}} = 0.85$
- Rolling Friction Coefficient, $\mu = 0.04$
- $\rho = 1.225 \frac{\text{kg}}{\text{m}^3}$

3.7 Finding Design Point

We use the constraint equations and parameters fixed above to find the design point. A MATLAB code has been written to perform this task. The code is given and explained below.

3.7.1 MATLAB code

```
% Atmospheric conditions
rho = 1.225; % kg/m^3
W0 = 4.95*9.81; % N
```

```

% Performance parameters
stallSpeed = 11; % m/s
cruiseSpeed = 15; % m/s
maxSpeed = 22; % m/s
ROC = 2; % m/s
takeoff = 30; % m
service_ceiling = 500; % m
% Fixing aerodynamic parameters
CD0 = 0.035;
AR = 10;
e = 0.8;
k = 1/(pi*e*AR);
% Propulsion efficiency
propEfficiency = 0.62; % propeller efficiency (calculated in MTOW estimation)
W_S = 20:2:130;
% Stall constraint
CL_max = 1.4;
W_S_stall = 0.5*rho*stallSpeed^2*CL_max;
% Cruise constraint
P_W_cruise = (cruiseSpeed/propEfficiency) * ...
    (((0.5*rho*cruiseSpeed^2*CD0)./W_S) + ...
    (k.*(W_S)/(0.5*rho*cruiseSpeed^2)));
% Maximum speed constraint
P_W_max = (maxSpeed/propEfficiency) * ...
    (((0.5*rho*maxSpeed^2*CD0)./W_S) + ...
    (k.*(W_S)/(0.5*rho*maxSpeed^2)));
% Climb constraint
climbSpeed = 1.3*stallSpeed;
P_W_climb = (climbSpeed/propEfficiency) * ...
    (((0.5*rho*climbSpeed^2*CD0)./W_S) + ...
    (k.*(W_S)/(0.5*rho*climbSpeed^2)) + ...
    ROC/climbSpeed);
% Service ceiling constraint
ceilingSpeed = 1.2*stallSpeed;
P_W_ceiling = (ceilingSpeed/propEfficiency) * ...
    (((0.5*rho*ceilingSpeed^2*CD0)./W_S) + ...

```

```

        (k.*(W_S)/(0.5*rho*ceilingSpeed^2)) + ...
        0.5/ceilingSpeed);
% Takeoff constraint
V_TO = 1.2*stallSpeed;
mu = 0.04;
P_W_takeoff = (V_TO/propEfficiency) * ...
    (mu + (1.44*(W_S)/(9.81*rho*CL_max*takeoff)));
% Plotting constraints
plot(W_S, 1./P_W_max);
hold on
plot(W_S, 1./P_W_cruise);
plot(W_S, 1./P_W_ceiling);
plot(W_S_stall*ones(size(W_S)), linspace(0,25,56));
plot(W_S, 1./P_W_climb);
plot(W_S, 1./P_W_takeoff);
xlabel('W/S');
ylabel('W/P');
axis([20 120 0 1.25]);
scatter(74, 0.187037, "filled", "o", "MarkerFaceColor","red");
plot([74 74], [0 0.187037], 'r--');
plot([74 0], [0.187037 0.187037], 'r--');
legend(["Max.Speed" "Cruise.Speed" "Service Ceiling" ...
    "Stall Speed" "Rate of Climb" ...
    "Takeoff Constraint" "Design Point"]);
% DFeign Parameters found from graph
design_W_S = 74;
design_W_P = 0.187037;

fprintf("Design W/S: %.4f\n", design_W_S);
fprintf("Design W/P: %.4f\n", design_W_P);
% Calculation of S
S = W0 / design_W_S;
% Calculation of Power Required
P = W0 / design_W_P;
fprintf("Calculated Wing Area, S : %.4f m^2\n", S);
fprintf("Calculated Power required, P : %.4f W\n", P);

```

The performance constraints were formulated in terms of power loading ($\frac{W}{P}$) versus wing loading ($\frac{W}{S}$) using the specified aerodynamic parameters and flight performance requirements (cruise speed, maximum speed, climb rate, service ceiling, stall speed, and takeoff distance).

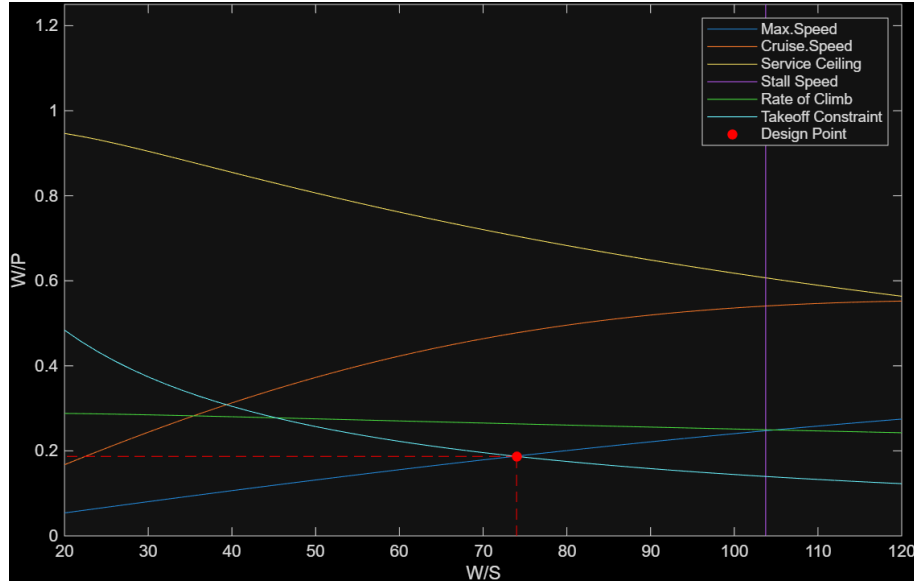


Figure 1: Constraint Analysis for Wing Sizing and Power Required

Each constraint generates a curve on the $\frac{W}{S} - \frac{W}{P}$ diagram. The feasible design region is obtained by satisfying all performance requirements simultaneously.

3.7.2 Design Point Selection

The selected design point corresponds to the intersection of the most restrictive constraints. From the constraint diagram,

$$\left(\frac{W}{S}\right)_{design} = 74 \text{ N/m}^2$$

$$\left(\frac{W}{P}\right)_{design} = 0.187037 \text{ N/W}$$

3.7.3 Wing Area Calculation

Given the takeoff weight:

$$W_0 = 4.95 \times 9.81 = 48.56 \text{ N}$$

Wing loading is defined as:

$$\frac{W}{S} = \frac{W_0}{S}$$

Rearranging to solve for wing area:

$$S = \frac{W_0}{\left(\frac{W}{S}\right)_{design}}$$

$$S = \frac{48.56}{74}$$

$$\boxed{S = 0.6562 \text{ m}^2}$$

3.7.4 Required Power Calculation

Power loading is defined as:

$$\frac{W}{P} = \frac{W_0}{P}$$

Rearranging to solve for required power:

$$P = \frac{W_0}{\left(\frac{W}{P}\right)_{design}}$$

$$P = \frac{48.56}{0.187037}$$

$$\boxed{P \approx 260 \text{ W}}$$

3.7.5 Interpretation

The selected design point satisfies all imposed performance constraints while maintaining feasible aerodynamic and propulsion characteristics. The resulting wing area of 0.6562 m² ensures adequate lift generation, and the required propulsion power of approximately 260 W meets cruise, climb, and ceiling requirements.

3.8 Selection of Propulsion System Components

3.8.1 Design Requirements

- Maximum Takeoff Mass: $m = 4.95$ kg
- Weight: $W = 48.5$ N
- Wing Area: $S = 0.6562$ m²
- Cruise Speed: $V = 15$ m/s
- Maximum Power Required: $P_{max} = 259.6$ W
- Mission Endurance: 2 hours

3.8.2 Theoretical Background

Power Required. Power required in steady level flight:

$$P = D \times V$$

Drag:

$$D = \frac{1}{2} \rho V^2 S C_D$$

Therefore,

$$P = \frac{1}{2} \rho V^3 S C_D$$

Power varies with the cube of velocity.

Motor Sizing. To ensure safe operation:

$$P_{motor} \geq 1.3 \times P_{max}$$

$$P_{motor} \geq 1.3 \times 259.6$$

$$P_{motor} \geq 337 \text{ W}$$

Thus, a 350 W class motor is required.

Propeller Theory. Motor speed:

$$RPM = KV \times V_{battery}$$

Pitch speed:

$$V_{pitch} = \frac{RPM \times Pitch}{60}$$

Pitch speed must exceed cruise velocity.

Battery Energy Requirement. Energy required:

$$E = P_{cruise} \times t$$

Cruise power (corrected realistic value):

$$P_{cruise} \approx 80 \text{ W}$$

For 2 hours:

$$E = 80 \times 2 = 160 \text{ Wh}$$

Including 20% reserve:

$$E_{required} = 1.2 \times 160 = 192 \text{ Wh}$$

3.8.3 Component Selection

Motor Selection. Motor: Turnigy Aerodrive SK3 3542-900KV

- KV Rating: 900 KV
- Max Continuous Power: 350 W
- Max Current: 30 A
- Voltage: 3S-4S LiPo
- Weight: 140 g

For 4S battery ($V = 14.8 \text{ V}$):

$$RPM_{no-load} = 900 \times 14.8 = 13,320$$

Loaded RPM (80%):

$$RPM_{loaded} \approx 10,500$$

Propeller Selection. Propeller: APC 11×5.5E

Pitch (5.5 in = 0.1397 m):

$$V_{pitch} = \frac{10,500 \times 0.1397}{60}$$

$$V_{pitch} \approx 24.4 \text{ m/s}$$

Cruise speed is 15 m/s, thus sufficient margin.

Expected static thrust: 15–18 N.

Check thrust-to-weight:

$$\frac{T}{W} = \frac{15}{48.5} = 0.31$$

Requirement (≥ 0.3) satisfied.

ESC Selection. ESC: Hobbywing Skywalker 40A

Maximum required current:

$$I = \frac{P}{V} = \frac{300}{14.8} \approx 20A$$

40 A ESC provides sufficient safety margin.

Battery Selection (2 Hour Endurance). Required energy:

$$E_{required} = 192 \text{ Wh}$$

For 4S battery:

$$Capacity = \frac{192}{14.8}$$

$$Capacity = 12.97 \text{ Ah}$$

Therefore select:

4S 13,000 mAh LiPo

Example: 4S 13000 mAh 20C LiPo

Specifications:

- Nominal Voltage: 14.8 V
- Capacity: 13 Ah
- Energy:

$$14.8 \times 13 = 192.4 \text{ Wh}$$

- Discharge Capability:

$$20 \times 13 = 260A$$

- Estimated Weight: $\approx 1.1 \text{ kg}$

3.8.4 Endurance Verification

Usable energy (80%):

$$E_{usable} = 0.8 \times 192 = 154 \text{ Wh}$$

Endurance:

$$t = \frac{154}{80}$$

$$t = 1.93 \text{ hr}$$

$$t \approx 116 \text{ minutes}$$

Thus, endurance requirement is satisfied.

3.8.5 Final Propulsion System

Motor	Turnigy SK3 3542-900KV (350 W)	175 g
Propeller	APC 11×5.5E	20 g
ESC	Hobbywing 40A	40 g
Battery	4S 13000 mAh 20C LiPo	1200 g

Total propulsion system weight ≈ 1.435 kg, which is acceptable given the MTOW of 4.95 kg and the need to allocate weight for structure and payload.

3.9 Weight Distribution

The Maximum Take-Off Weight (MTOW) of the UAV from preliminary estimation is:

$$W_{TO} = 4.95 \text{ kg}$$

The propulsion system mass (motor + ESC + propeller) is:

$$W_{propulsion} = 0.235 \text{ kg}$$

The new empty weight is calculated by subtracting the propulsion system (excluding battery) weight from the previously estimated empty weight, which included the propulsion system. The propulsion system weight is then added back as a separate component to ensure accurate accounting.

The weight breakdown is organized into the following primary components:

Component	Mass (kg)
Empty Weight	2.735
Battery Weight	1.20
Payload Weight	0.75
Propulsion System (excluding Battery)	0.235
Total (MTOW)	4.95

To calculate structural weight from the new empty weight, we need to subtract the avionics mass and landing gear mass from the newly calculated empty weight. The avionics mass

(flight controller, GPS, receiver, servos, wiring) is estimated as:

$$W_{avionics} = 0.22 \text{ kg}$$

The landing gear mass is estimated as:

$$W_{landing \text{ gear}} = 0.20 \text{ kg}$$

Thus, the structural weight is calculated as:

$$W_{structure} = W_{empty} - (W_{propulsion} + W_{avionics} + W_{landing \text{ gear}})$$

$$W_{structure} = 2.735 - (0.22 + 0.20)$$

$$W_{structure} = 2.315 \text{ kg}$$

The resulting structural weight fraction is:

$$\frac{W_{structure}}{W_{TO}} = \frac{2.315}{4.95} = 0.46$$

A structural weight fraction of approximately 46% is consistent with typical small composite or foam fixed-wing UAV configurations.

Final weight distribution:

Component	Mass (kg)
Structure	2.315
Propulsion System	0.235
Avionics	0.22
Landing Gear	0.20
Battery	1.20
Payload	0.75
Total (MTOW)	4.92

4 Airfoil Selection

4.1 Introduction

The selection of an appropriate airfoil is a critical aspect of the design process for a fixed-wing UAV, especially for a mission focused on biodiversity monitoring. The airfoil must provide sufficient lift to support the UAV's weight while minimizing drag to ensure efficient flight and extended endurance. This section outlines the criteria and considerations for selecting an airfoil suitable for our UAV, which has a maximum takeoff mass of 4.92 kg and is designed for low-speed flight.

From Constraint Analysis, we found out the wing area to be 0.6562 m^2 .

We use this wing area (S) and our already fixed aspect ratio (AR) of 10 to calculate wing span and chord length.

4.2 Calculation of Wing Span (b) and Chord Length(c)

The aspect ratio (AR) is defined as the ratio of the square of wing span (b) to the wing area (S):

$$AR = \frac{b^2}{S} \quad (53)$$

$$b = \sqrt{AR \times S}$$

$$b = \sqrt{10 \times 0.6562} = 2.56 \text{ m}$$

The chord length (c) can be calculated using the wing area (S) and wing span (b):

$$c = \frac{S}{b} \quad (54)$$

$$c = \frac{0.6562}{2.56} = 0.256 \text{ m}$$

4.3 Calculation of Reynolds Number (Re)

$$Re = \frac{\rho V c}{\mu} \quad (55)$$

Where:

- ρ is the air density (1.225 kg/m^3 at sea level)
- V is the cruise speed (15 m/s)
- c is the chord length (0.256 m)
- μ is the dynamic viscosity of air ($1.81 \times 10^{-5} \text{ kg/(m} \cdot \text{s)}$)

$$Re = \frac{1.225 \times 15 \times 0.256}{1.81 \times 10^{-5}}$$

$$Re = 259889.502$$

4.4 Airfoil Selection Criteria

4.4.1 Calculation of cruise lift coefficient

From lift equilibrium:

$$L = W = \frac{1}{2} \rho V^2 S C_L$$

$$C_{L_{\text{cruise}}} = \frac{2W}{\rho V^2 S}$$

$$W = 4.92 \times 9.81 = 48.56 \text{ N}$$

$$C_{L_{\text{cruise}}} = \frac{2 \times 48.56}{1.225 \times 15^2 \times 0.6562}$$

$$C_{L_{\text{cruise}}} = 0.5370$$

4.4.2 Calculation of maximum coefficient of lift

From stall condition:

$$W = \frac{1}{2} \rho V_{\text{stall}}^2 S C_{L_{\text{max}}}$$

$$C_{L_{\text{max}}} = \frac{2W}{\rho V_{\text{stall}}^2 S}$$

$$W = 48.56 \text{ N}$$

$$V_{\text{stall}} = 11 \text{ m/s}$$

$$C_{L_{\text{max}}} = \frac{2 \times 48.56}{1.225 \times 11^2 \times 0.6562}$$

$$C_{L_{\text{max}}} \approx 1$$

4.4.3 The Criteria

Based on the Reynolds number and lift coefficient calculations, the airfoil must satisfy the following aerodynamic requirements:

- **Low Reynolds Number Performance:** The cruise Reynolds number is

$$Re \approx 2.6 \times 10^5$$

Therefore, the selected airfoil must demonstrate stable and efficient aerodynamic characteristics in the low Reynolds number regime ($Re \sim 10^5$), where viscous effects are significant and laminar separation is more likely.

- **Cruise Lift Requirement:** The required lift coefficient at cruise is

$$C_{L_{\text{cruise}}} = 0.537$$

The airfoil should achieve this lift coefficient at a moderate angle of attack while maintaining a high lift-to-drag ratio to ensure efficient cruise performance.

- **Maximum Lift Capability:** The required maximum lift coefficient is approximately

$$C_{L_{\text{max}}} \approx 1.0$$

The selected airfoil must therefore provide a $C_{L_{\text{max}}}$ greater than 1.0 with a smooth and predictable stall behavior to ensure safe low-speed operation.

- **Gentle Stall Characteristics:** Since the UAV operates at relatively low speeds, the airfoil should exhibit progressive stall behavior rather than abrupt lift loss, enhancing flight safety and controllability.
- **High Lift-to-Drag Ratio (L/D):** To maximize endurance and energy efficiency for a battery-powered UAV, the airfoil should provide a high L/D ratio near $C_L \approx 0.5$.

Conclusion: The selected airfoil must be optimized for low Reynolds number operation

($Re \approx 2.6 \times 10^5$), provide $C_{L_{cruise}} \approx 0.54$ efficiently, and achieve $C_{L_{max}} > 1.0$ with stable stall characteristics. These requirements ensure aerodynamic efficiency, safe stall margins, and suitability for a fixed-wing battery-powered UAV.

4.5 Selection of Airfoil

4.5.1 Candidate Airfoils

By reviewing the historical data, three airfoils are most used for survey fixed-wing UAVs, Those are:

- NACA 4412
- Selig S1233
- Epler 205

4.5.2 Analysis of Candidate Airfoils

Using XFOIL the candidate airfoils are analysed and an airfoil which mostly satisfies the criteria above is chosen.

Analysis Results.

- NACA 4412 Analysis Report - [NACA 4412](#)
- Selig S1233 Analysis Report - [Selig S1233](#)
- Epler 205 Analysis Report - [Epler 205](#)

From the analysis results, we choose NACA 4412 as the UAV's airfoil.

5 Wing Design

5.1 Introduction

The wing may be considered as the most important component of an aircraft, since a fixed-wing aircraft is not able to fly without it. Since the wing geometry and its features influence all other aircraft components, we begin the detail design process with wing design. The primary function of the wing is to generate sufficient lift force or simply lift (L). However, the wing has two other productions, namely the drag force or drag (D) and nose-down pitching moment (M). While a wing designer is looking to maximize the lift, the other two (drag and pitching moment) must be minimized. In fact, a wing is considered as a lifting surface where lift is produced due to the pressure difference between the lower and upper surfaces. Aerodynamics textbooks are a good source to consult for information about mathematical techniques to calculate the pressure distribution over the wing and for determining the flow variables.

During the wing design process, 18 parameters must be determined. They are as follows:

1. Wing reference area (S)
2. Number of wings
3. Vertical position relative to the fuselage (high, mid, or low wing)
4. Horizontal position relative to the fuselage
5. Cross-section (or airfoil)
6. Aspect Ratio (AR)
7. Taper ratio (λ)
8. Tip Chord (C_t)
9. Root Chord (C_r)
10. Mean Aerodynamic Chord (MAC)
11. Wing span (b)
12. Twist angle or washout (ϵ)
13. Sweep angle (Λ)
14. Dihedral angle (Γ)
15. Incidence (i_w) (or setting angle, α_{set})

16. high-lifting devices such as flap
17. aileron
18. other wing accessories

Of the above parameters, We already know the wing reference area (S), Cross-section (or airfoil), Aspect Ratio (AR) and the number of wings. The rest of the parameters are determined in the following sections.

Parameter	Value
Wing Reference Area (S)	0.6562 m^2
Airfoil	NACA 4412
Aspect Ratio (AR)	10
No.of Wings	1

5.2 The 18 Parameters

5.2.1 Wing Reference Area (S)

The wing reference area (S) is defined as the projected area of the wing onto a plane parallel to the longitudinal axis of the aircraft. It is a critical parameter that influences the lift generated by the wing, as well as the overall aerodynamic performance of the aircraft.

Already calculated in the preliminary design phase, the wing reference area for our biodiversity monitoring UAV is: **0.6562 m^2** .

5.2.2 Number of Wings

A monoplane configuration (one wing) is selected. Compared with biplane or triplane layouts, the monoplane offers the lowest interference drag, lowest structural weight (no interplane struts), simplest manufacturing, and highest L/D for endurance missions (Sadraey, Section 5.2). All modern fixed-wing survey UAVs (e.g., WingtraOne, senseFly eBee series) use the monoplane layout. A weighted scoring table (Sadraey-style) gives the monoplane 95/100 points for this mission.

5.2.3 Vertical Position (High, Mid, or Low Wing)

The wing may be mounted in one of three vertical positions:

- High-wing configuration
- Mid-wing configuration
- Low-wing configuration

For the present biodiversity monitoring UAV, a **high-wing configuration** is selected.

Reasons for High-Wing Selection. 1. Lateral (Roll) Stability Contribution

In a high-wing aircraft, the center of gravity (CG) is located below the wing. During a sideslip condition, the aircraft behaves similarly to a pendulum, generating a restoring rolling moment.

The rolling moment due to sideslip is expressed as:

$$L_\beta = \frac{\partial C_l}{\partial \beta} \quad (56)$$

A high-wing configuration increases the positive dihedral effect, improving lateral stability without requiring excessive geometric dihedral.

2. Improved Ground Clearance

The propeller and wing tips have greater ground clearance, which is beneficial for field operations in uneven terrain typical of biodiversity missions.

3. Reduced Wing–Fuselage Interference Drag

High-wing mounting generally produces cleaner airflow under the wing, reducing interference drag compared to low-wing layouts.

4. Enhanced Downward Visibility

Since the mission involves aerial observation and monitoring, a high wing provides unobstructed downward field of view for sensors.

5. Structural Simplicity for UAV Scale

For small UAVs, high-wing mounting simplifies spar carry-through structure and landing gear attachment.

Therefore, the vertical position is defined as:

High-Wing Configuration

5.2.4 Determination of Wing Horizontal Position

Horizontal position will be finalized in fuselage design to achieve 10–12% static margin. Preliminary estimate: wing root LE at 30% fuselage length.

5.2.5 Cross-section or (Airfoil) Selection

NACA 4412 (already selected and validated in airfoilSelection). The airfoil satisfies all low-Re criteria: CL at cruise $\alpha \approx 0.537$, CLmax (clean) ≈ 1.35 –1.40, gentle stall, and high L/D near CL = 0.5. No change required.

5.2.6 Aspect Ratio (AR)

AR = 10 (fixed in constraint analysis). This value provides excellent induced-drag reduction for battery endurance while keeping structural weight and transport size reasonable

5.2.7 Taper Ratio Selection

Taper ratio:

$$\lambda = \frac{C_t}{C_r} \quad (57)$$

With AR = 10 and S = 0.6562 m² already fixed, the rectangular planform gives uniform chord c = 0.256 m. This matches all prior Reynolds-number and chord calculations exactly and simplifies rib manufacture, foam cutting, and spar design. Although a mild taper ($\lambda \approx 0.7$) would reduce induced drag by ≈ 3 –5%, the manufacturing penalty outweighs the benefit for this prototype-scale UAV. Most small mapping UAVs (e.g., Skywalker, eBee) use rectangular or near-rectangular wings.

Therefore,

$$\lambda = 1$$

5.2.8 Tip Chord C_t and Root Chord C_r

As the taper ratio is 1, the tip chord is equal to the root chord:

$$C_t = C_r = 0.256 \text{ m} \quad (58)$$

5.2.9 Mean Aerodynamic Chord

For a rectangular wing, the mean aerodynamic chord (MAC) is equal to the root chord:

$$MAC = C_r = 0.256 \text{ m} \quad (59)$$

5.2.10 Wing Span (b)

Aspect ratio is defined as:

$$AR = \frac{b^2}{S} \quad (60)$$

Thus,

$$b = \sqrt{AR \cdot S} \quad (61)$$

$$b = \sqrt{10 \times 0.6562} = 2.56 \text{ m} \quad (62)$$

5.2.11 Wing Twist (Washout)

Washout prevents tip stall.

If the tip stalls first:

- Aileron loses effectiveness
- Aircraft may enter spin

Typical washout:

$$\epsilon = -2^\circ$$

This ensures root-first stall.

5.2.12 Sweep Angle (Λ)

0° (unswept straight wing). At cruise speed of 15 m/s (Mach ≈ 0.044) there is no compressibility benefit from sweep. Sweep would add structural complexity and weight with zero aerodynamic gain at this low speed

5.2.13 Dihedral Angle (Γ)

Dihedral produces restoring rolling moment during sideslip:

$$L_\beta = \frac{\partial C_l}{\partial \beta}$$

Positive dihedral increases roll stability.

Selected:

$$\Gamma = 4^\circ$$

5.2.14 Incidence (i_w) or Setting Angle (α_{set})

+3.0°. From NACA 4412 XFOIL data ($\text{Re} \approx 2.6 \times 10^5$): section α required for $\text{CL} \approx 0.55$ (accounting for 3-D induced angle $\approx 2^\circ$) is approximately 3° . This sets the fuselage at near-zero angle of attack during cruise for optimal level camera orientation

5.2.15 High-Lift Devices

None required (plain flap kept as optional growth feature). NACA 4412 clean CL_{max} already exceeds the required value by $>35\%$. Adding flaps would increase weight and complexity with negligible benefit for this mission.

5.2.16 Aileron Design

Purpose of Ailerons. Ailerons generate rolling moment by producing lift difference between wings.

Rolling moment:

$$L = \frac{1}{2} \rho V^2 S b C_l \quad (63)$$

Aileron Span – What Does It Mean?. Total wingspan:

$$b$$

Half span (one wing):

$$\frac{b}{2}$$

Aileron span b_a refers to the span of ONE aileron on one wing.

Typical design ratio:

$$\frac{b_a}{b/2} = 0.3 - 0.4$$

Thus, for this UAV:

$$b_a = 0.35 \times 1.28 = 0.45 \text{ m (per wing)}$$

Why outer wing?

Rolling moment is:

$$L = \int y dL \quad (64)$$

Moment arm increases with spanwise distance. Outer placement maximizes effectiveness.

Aileron Chord Ratio. Lift change due to flap deflection:

$$\Delta C_L \propto \frac{c_a}{c}$$

But hinge moment:

$$H \propto c_a^2$$

Thus increasing chord increases hinge moment quadratically.

Optimal compromise:

$$\frac{c_a}{c} = 0.33$$

Therefore:

$$c_a = 0.085 \text{ m}$$

Aileron Effectiveness. Rolling moment coefficient:

$$C_l = C_{l_\delta} \delta_a \quad (65)$$

Approximate relation:

$$C_{l_\delta} \propto \frac{b_a}{b} \cdot \frac{c_a}{c}$$

Thus span and chord both influence roll authority.

Maximum Deflection. Rolling moment increases linearly:

$$L \propto \delta_a$$

Drag increases quadratically:

$$D \propto \delta_a^2$$

Beyond 25° , flow separation reduces effectiveness.

Therefore:

$$\delta_a = \pm 20^\circ$$

Roll Rate Requirement. Aircraft roll dynamics:

$$I_x \dot{p} = L \tag{66}$$

Typical UAV roll rate requirement:

$$30^\circ/s \leq p \leq 60^\circ/s$$

If calculated roll rate is insufficient:

- Increase b_a
- Increase c_a
- Increase δ_a

5.3 Final Design Summary

S.No	Parameter	Value
1	Wing Reference Area (S)	0.6562 m^2
2	Number of Wings	1
3	Vertical Position	High Wing
4	Horizontal Position	wing root LE at 30% fuselage length (preliminary)
5	Airfoil	NACA 4412
6	Aspect Ratio (AR)	10
7	Taper Ratio (λ)	1
8	Tip Chord (C_t)	0.256 m
9	Root Chord (C_r)	0.256 m
10	Mean Aerodynamic Chord (MAC)	0.256 m
11	Wing Span (b)	2.56 m
12	Twist Angle (Washout)	-2°
13	Sweep Angle (Λ)	0°
14	Dihedral Angle (Γ)	4°
15	Incidence (i_w)	+3°
16	High-Lift Devices	None
17	Aileron Span (b_a)	0.45 m (per wing)
18	Aileron Chord (c_a)	0.085 m
19	Aileron Max Deflection (δ_a)	$\pm 20^\circ$
20	Roll Rate Requirement	30–60°/s
21	Other Wing Accessories	None

5.4 Conclusion

The wing geometry and aileron sizing have been determined using a combination of aerodynamic theory, structural considerations, and control requirements as prescribed by Sadraey.

Each parameter has been selected based on:

- Induced drag minimization
- Stall control
- Roll authority requirement
- Structural feasibility
- Servo load considerations

6 Tail Design

6.1 Introduction

The tail in a conventional aircraft has two components, horizontal tail and vertical tail, and carries two primary functions:

- Trim (longitudinal and directional)
- Stability (longitudinal and directional)

Since two conventional control surfaces (i.e., elevator and rudder) are indeed parts of the tail to implement control, it is proper to add the following item as the third function:

- Control (longitudinal and directional)

The first and primary function of a horizontal tail is longitudinal trim; also referred to as equilibrium or balance. But the first and primary function of a vertical tail is directional stability.

Longitudinal trim in a conventional aircraft is applied through the horizontal tail. Several pitching moments, namely longitudinal moment of the wing's lift about the aircraft center of gravity, wing aerodynamic pitching moment, and sometimes engine thrust's longitudinal moment, need to be trimmed about the y-axis. The summation of these three moments about the aircraft center of gravity is often negative; hence the horizontal tail often generates a negative lift to counteract the moment. For this reason, the horizontal tail setting angle is often negative. Since the aircraft center of gravity is moving along the x -axis (due to fuel burn during flight duration), the horizontal tail is responsible for longitudinal trim throughout the flight time. To support the longitudinal trimability of the aircraft, conventional aircraft employ an elevator as part of their horizontal tail. Since conventional aircraft are almost always manufactured symmetrically about the xz plane, the trim is not a major function for a vertical tail. However, in a few instances, a vertical tail has the primary function of directional trim or lateral trim. In a multi- engine aircraft, the vertical tail has great responsibility during one-engine inoperative (OEI) situations in order to maintain directional trim. The vertical tail must generate a yawing moment to balance the aircraft against the yawing moment generated by active engines. Even in single-engine prop-driven aircraft, the vertical tail has to counteract the rolling moment generated by propeller rotation. This is to maintain aircraft lateral trim and prevent an unwanted roll. In this case, the vertical tail is often installed at a few degrees relative to

the xz plane. The aircraft trim requirement provides the main design requirement in the tail design process.

The second function of the tails is to provide stability. The horizontal tail is responsible for maintaining longitudinal stability, while the vertical tail is responsible for maintaining directional stability. Aircraft stability is defined as the tendency of an aircraft to return to the original trim conditions if diverted by a disturbance. The major disturbance source is atmospheric phenomena, such as gusts. The stability requirement must also be included in the list of tail design requirements.

The third major function of the tails is control. The elevator as part of the horizontal tail is designed to provide longitudinal control, while the rudder as part of the vertical tail is responsible for providing directional control. Tails must be powerful enough to control the aircraft, such that the aircraft is able to change flight conditions from one trim condition (say cruise) to another new trim condition (say take-off and landing). For instance, during take-off, the tail must be able to lift up the fuselage nose in a specified pitch rate.

In general, the tail is designed based on trim requirements, but later revised based on stability and control requirements. The following are the tail parameters which need to be determined during the design process:

1. Horizontal tail location with respect to fuselage

Horizontal Tail.

2. Planform area (S_h)
3. Tail arm (l_h)
4. Airfoil section
5. Aspect ratio (AR_h)
6. Taper ratio (λ_h)
7. Tip chord ($c_{h_{tip}}$)
8. Root chord ($c_{h_{root}}$)
9. Mean aerodynamic chord (MAC_h)
10. Span (b_h)
11. Sweep angle (Λ_h)
12. Dihedral angle (Γ_h)
13. Tail installation

14. Incidence angle (i_h)

Vertical Tail.

15. Planform area (S_v)
16. Tail arm (l_v)
17. Airfoil section
18. Aspect ratio (AR_v)
19. Taper ratio (λ_v)
20. Tip chord ($c_{v_{tip}}$)
21. Root chord ($c_{v_{root}}$)
22. Mean aerodynamic chord (MAC_v)
23. Span (b_v)
24. Sweep angle (Λ_v)
25. Dihedral angle (Γ_v)
26. Incidence angle (i_v)

6.2 Tail Design Requirements

The empennage must satisfy the following functional and performance requirements:

6.2.1 Longitudinal Stability and Trim (Horizontal Tail)

- Provide sufficient negative pitching moment to balance aircraft at all expected CG locations and flight conditions.
- Ensure static longitudinal stability: $C_{m_\alpha} < 0$ (nose-down moment with increasing angle of attack).
- Static margin (SM): typically 5–15% for small UAVs (recommended 8–12% for stable camera platform).
- Trim capability at cruise with near-zero elevator deflection (minimize trim drag).
- Adequate elevator control power for takeoff rotation, flare, and stall recovery.

Typical horizontal tail volume coefficient V_H ranges (Sadraey Table 6.1, adapted for small UAVs):

- General aviation single-engine: 0.40–0.70

- Homebuilt/light aircraft: 0.45–0.65
- Small electric fixed-wing UAVs (mapping/survey): **0.50–0.60** (recommended)

6.2.2 Directional Stability and Control (Vertical Tail)

- Positive directional (weathercock) stability: $C_{n_\beta} > 0$
- Adequate rudder authority for crosswind landing/taxi and engine failure yaw (if applicable).
- Spin recovery capability (rudder effective in stalled wing flow).

Typical vertical tail volume coefficient V_V ranges (Sadraey Table 6.4, adapted):

- General aviation: 0.03–0.08
- Homebuilt: 0.04–0.07
- Small electric UAVs: **0.04–0.06** (recommended for low-speed operation)

6.2.3 Tail Geometry and Aerodynamic Requirements

- **Airfoil:** Symmetric (NACA 00xx series recommended), thickness 9–12% for low pitching moment and good low-Re behavior.
- **Sweep angle:** Usually 0° for low-speed UAVs (no compressibility issues).
- **Aspect ratio:** Lower than wing ($AR_h = 3\text{--}5$, $AR_v = 1.5\text{--}3.5$) to reduce weight and induced drag.
- **Taper ratio:** 0.4–1.0 (rectangular $\lambda = 1.0$ common for simplicity).
- **Dihedral:** Horizontal tail usually 0° ; vertical tail 0° (single fin).
- **Incidence (setting angle):**
 - Horizontal: -2° to $+1^\circ$ (adjustable or fixed to achieve cruise trim).
 - Vertical: 0° (symmetric).

6.2.4 Control Surface Sizing Guidelines

- Elevator chord ratio: 20–35% of horizontal tail chord (typical 25–30%).
- Rudder chord ratio: 25–40% of vertical tail chord (typical 30–35%).
- Span coverage: Elevator 60–90% of horizontal tail span; rudder full height or 70–90%.
- Maximum deflection: $\pm 20^\circ$ to $\pm 30^\circ$ (elevator), $\pm 25^\circ$ to $\pm 35^\circ$ (rudder).

6.2.5 Other Important Requirements

- **Spin recovery:** Conventional tail preferred (rudder in undisturbed flow).
- **Tail stall:** Tail must stall after wing (higher local α due to downwash; use symmetric airfoil + lower AR).
- **Propeller effects:** Tail out of prop slipstream if possible (tractor configuration) or account for increased dynamic pressure.
- **Ground clearance:** Vertical tail tip above ground/prop plane during takeoff/landing attitude.
- **Weight and drag minimization:** Critical for battery-powered UAV endurance.
- **Manufacturability:** Simple planforms (rectangular or mild taper), symmetric airfoils, foam/composite construction.

6.3 Tail Design

6.3.1 Tail Configuration Selection

The conventional tail configuration is selected for this UAV due to its simplicity, well-understood aerodynamics, and suitability for low-speed flight. It provides effective longitudinal and directional stability and control, which are critical for a stable camera platform in biodiversity monitoring missions.

- The conventional tail or inverted T-shape configuration is the simplest configuration and the most convenient to perform all tail functions (i.e., trim, stability and control). The analysis and evaluation of the performance of a conventional tail is straight forward.
- This configuration includes one horizontal tail (two left and right sections) located on the aft fuselage, and one vertical tail (one section) located on top of the aft fuselage.
- Both horizontal and vertical tails are located and mounted to the aft of the fuselage.
- The horizontal tail is mainly employed to satisfy the longitudinal trim and stability requirements, while the vertical tail is mainly used to satisfy the directional trim and stability requirements. (If the designer has little experience, it is recommended to initially select the conventional tail configuration.)

6.3.2 Optimum Tail Arm

One of the tail parameters that must be determined during the tail design process is the tail arm (l_t). The tail arm is the distance between the aerodynamic center of the tail and

the aircraft center of gravity. Tail arm is a critical parameter in the tail design process, as it directly affects the tail volume coefficient, which is a key design parameter for both horizontal and vertical tails.

From Sadraey's Aircraft Design - A Systems Engineering Approach (2012), Eq (6.47)

$$l_{\text{opt}} = K_c \sqrt{\frac{4\bar{C}S\bar{V}_H}{\pi D_f}} \quad (67)$$

where,

- K_c is a correction factor (varies between 1.0–1.4).
- $\bar{C}$ is the mean aerodynamic chord of the wing.
- S is the wing planform area.
- $\bar{V}_H$ is the desired horizontal tail volume coefficient.
- D_f is the maximum fuselage diameter at the tail location.

6.3.3 Horizontal Tail Volume Ratio ($\bar{V}_H$) and Maximum Fuselage Diameter (D_f)

This section selects V_H based on mission requirements and estimates D_f for the optimum tail arm calculation.

Selection of Horizontal Tail Volume Coefficient (V_H). From Sadraey (2012, Table 6.1 and Section 6.5), typical V_H values are:

- General aviation single-engine aircraft: 0.40–0.70
- Homebuilt/light aircraft: 0.45–0.65
- Small electric fixed-wing UAVs (mapping/survey class): 0.50–0.60

For our biodiversity monitoring UAV (MTOW = 4.95 kg, electric propulsion, low-speed cruise at 15 m/s, high-wing configuration, emphasis on stable camera platform and maximum endurance), the following considerations apply:

- Adequate longitudinal static margin (8–12%) to handle CG shifts from battery discharge and payload variations.
- Minimum tail drag to maximize battery endurance.
- Gentle handling and trim at cruise ($C_{L_{\text{cruise}}} = 0.537$).

A value of $V_H = 0.55$ is selected as it lies in the recommended range for small electric

survey UAVs (0.50–0.60) and provides a good balance between stability and efficiency. This is consistent with analogous designs (e.g., Skywalker X8 derivatives and similar mapping UAVs) and ensures safe, docile behavior for field operations.

Estimation of Maximum Fuselage Diameter (D_f). The fuselage diameter D_f is the maximum equivalent diameter (used in Eq. 6.47 for wetted area and drag estimation). For a small electric fixed-wing UAV in the 4–6 kg class (biodiversity mapping/survey mission), typical values are derived from similar platforms:

- Skywalker X8 (common mapping platform, 2–3 kg flying weight): fuselage length 0.79 m, typical max diameter 0.20–0.25 m.
- senseFly eBee series (professional mapping UAV, 1.6 kg empty): compact fuselage, equivalent diameter 0.18–0.22 m.
- Other small survey UAVs (4–6 kg class): fuselage widths/diameters 0.20–0.27 m to accommodate battery, avionics, camera payload, and motor.

For our UAV (MTOW = 4.95 kg, payload 0.75 kg camera + battery 1.23 kg, high-wing tractor configuration), a streamlined fuselage is required to house electronics, battery, payload bay, and motor while minimizing drag. Considering internal volume needs (battery + camera) and manufacturability (foam/composite tube or molded shell), the maximum equivalent diameter is estimated as

$$D_f = 0.22 \text{ m}$$

This value is realistic and conservative: it allows sufficient cross-section for components without excessive wetted area (which would increase skin friction drag).

6.3.4 Calculation of Optimum Tail Arm (l_{opt})

Using the selected parameters:

- $K_c = 1.2$ (typical correction factor for small UAVs)
- $\bar{C} = 0.256 \text{ m}$ (estimated mean aerodynamic chord from preliminary wing design)
- $S = 0.6562 \text{ m}^2$ (estimated wing planform area from preliminary design)
- $\bar{V}_H = 0.55$ (selected horizontal tail volume coefficient)
- $D_f = 0.22 \text{ m}$ (estimated maximum fuselage diameter)

Calculating the optimum tail arm:

$$\begin{aligned}
 l_{\text{opt}} &= K_c \sqrt{\frac{4\bar{C}S\bar{V}_H}{\pi D_f}} \\
 &= 1.2 \sqrt{\frac{4 \times 0.256 \times 0.6562 \times 0.55}{\pi \times 0.22}} \\
 &= 1.2 \sqrt{\frac{0.3696}{0.6912}} \\
 &= 1.2 \sqrt{0.5347} \\
 &= 1.2 \times 0.7312 \\
 &\approx 0.8774 \text{ m}
 \end{aligned}$$

6.4 Horizontal Tail Parameters

6.4.1 Horizontal Tail Planform Area (S_h)

The horizontal tail area is calculated using the selected volume coefficient:

$$S_h = \frac{V_H \cdot S_w \cdot \bar{c}_w}{l_h}$$

where $V_H = 0.55$, $S_w = 0.6562 \text{ m}^2$, $\bar{c}_w = 0.256 \text{ m}$, and optimum $l_h = 0.88 \text{ m}$ (from Section on Optimum Tail Arm).

Calculation:

$$S_h = \frac{0.55 \times 0.6562 \times 0.256}{0.88} = 0.105 \text{ m}^2$$

This area provides the required longitudinal stability margin (8–12%) while keeping tail drag low for battery endurance. A smaller area would reduce stability; a larger area would increase weight and parasite drag unnecessarily for this low-speed mission.

6.4.2 Horizontal Tail Arm (l_h)

The tail arm is the longitudinal distance from the wing aerodynamic centre (25% MAC) to the horizontal tail aerodynamic centre (25% tail MAC).

Selected value: $l_h = 0.8774 \text{ m}$ (optimum value calculated using Sadraey Eq. (6.47) in the Optimum Tail Arm section).

This value balances fuselage wetted area (drag/weight) against tail size and was derived specifically for our UAV geometry.

6.4.3 Horizontal Tail Airfoil Section

Selected: NACA 0012 (symmetric, 12% thickness).

The horizontal tail requires a symmetric airfoil to produce zero lift (and zero pitching moment) at zero elevator deflection during cruise trim. NACA 0012 was chosen because:

- Excellent low-Re performance ($Re \approx 1.0 \times 10^5$ at tail).
- Gentle stall characteristics (important for safe recovery).
- Zero camber eliminates unwanted pitching moment.
- Widely used on small UAV tails and easy to manufacture with foam/composite.

No cambered airfoil is used because the tail must generate both positive and negative lift for trim and control.

6.4.4 Horizontal Tail Aspect Ratio (AR_h)

Selected: $AR_h = 4.0$.

Tail aspect ratio is kept lower than the wing ($AR_w = 10$) to reduce structural weight and induced drag while still providing adequate effectiveness. $AR = 4.0$ is typical for small UAV horizontal tails (Sadraey, Section 6.9) and matches many mapping UAVs. Higher AR would make the tail heavier and more flexible; lower AR would increase induced drag.

6.4.5 Horizontal Tail Taper Ratio (λ_h)

Selected: $\lambda_h = 1.0$ (rectangular planform).

Rectangular planform simplifies rib manufacture, spar design, and foam cutting — critical for a student/team prototype. Although mild taper ($\lambda_h \approx 0.7$) would slightly reduce induced drag, the manufacturing penalty outweighs the benefit at this scale (same logic as wing design).

6.4.6 Horizontal Tail Tip Chord ($c_{h_{tip}}$)

6.4.7 Horizontal Tail Root Chord ($c_{h_{root}}$)

For rectangular planform ($\lambda_h = 1.0$):

$$c_{h_{tip}} = c_{h_{root}} = \frac{S_h}{b_h} = 0.162 \text{ m}$$

6.4.8 Horizontal Tail Mean Aerodynamic Chord (MAC_h)

For rectangular planform:

$$MAC_h = c_{h_{root}} = 0.162 \text{ m}$$

6.4.9 Horizontal Tail Span (b_h)

$$b_h = \sqrt{AR_h \cdot S_h} = \sqrt{4 \times 0.105} = 0.648 \text{ m}$$

6.4.10 Horizontal Tail Sweep Angle (Λ_h)

Selected: $\Lambda_h = 0^\circ$ (unswept).

At cruise speed of 15 m/s (Mach ≈ 0.044) there is no transonic or compressibility benefit from sweep. Sweep would add structural complexity and weight with zero aerodynamic gain (Sadraey, Section 6.10).

6.4.11 12. Horizontal Tail Dihedral Angle (Γ_h)

Selected: $\Gamma_h = 0^\circ$.

The high-wing configuration already provides strong lateral stability through pendulum effect. Additional dihedral on the horizontal tail is unnecessary and would increase structural complexity.

6.4.12 Horizontal Tail Installation

Selected: Fuselage-mounted conventional aft installation.

The horizontal tail is attached directly to the rear fuselage using a simple carry-through spar or separate mounts. This allows easy assembly, maintenance, and adjustment of incidence during flight testing.

6.4.13 Horizontal Tail Incidence (i_h)

Selected: $i_h = -1.5^\circ$.

Incidence is set so that the horizontal tail operates at near-zero lift (zero elevator deflection) during cruise trim ($C_{L_{\text{cruise}}} = 0.537$). The negative value accounts for downwash from the main wing ($\varepsilon \approx 4^\circ\text{--}5^\circ$). This value will be verified in full XFLR5 analysis after fuselage and CG are finalized (Sadraey, Section 6.11).

6.5 Vertical Tail Parameters

6.5.1 Vertical Tail Planform Area (S_v)

$$S_v = \frac{V_V \cdot S_w \cdot b_w}{l_v} = \frac{0.05 \times 0.6562 \times 2.56}{0.88} = 0.0955 \text{ m}^2$$

6.5.2 Vertical Tail Arm (l_v)

$l_v = l_h = 0.88 \text{ m}$ (same optimum arm).

6.5.3 Vertical Tail Airfoil Section

NACA 0012 (same as horizontal tail for symmetric yawing moment).

6.5.4 Vertical Tail Aspect Ratio (AR_v)

Selected: $AR_v = 2.0$.

Higher aspect ratio provides better directional authority with reasonable height and weight.

6.5.5 Vertical Tail Taper Ratio (λ_v)

Selected: $\lambda_v = 1.0$ (**rectangular**) — same manufacturability rationale as horizontal tail.

6.5.6 Vertical Tail Tip Chord, Root Chord, and MAC

For rectangular planform:

$$c_{v_{\text{tip}}} = c_{v_{\text{root}}} = MAC_v = \frac{S_v}{b_v} = 0.219 \text{ m}$$

6.5.7 Vertical Tail Span (b_v)

$$b_v = \sqrt{AR_v \cdot S_v} = \sqrt{2 \times 0.0955} = 0.437 \text{ m}$$

6.5.8 Vertical Tail Sweep Angle (Λ_v)

The vertical tail sweep angle is defined as the angle between the quarter-chord line of the vertical stabilizer and the vertical axis of the aircraft. Although sweep is not required for compressibility considerations in low-speed aircraft, it is commonly incorporated into vertical tail design for aerodynamic and structural reasons.

Compared with the horizontal tail, vertical tails typically employ a slightly larger sweep angle. The primary motivations for including sweep in the vertical stabilizer are:

- Improvement of directional stability characteristics by influencing the aerodynamic center location of the vertical tail.
- More favorable stall progression and smoother side-force development during high sideslip angles.
- Reduction of interference effects between the vertical tail and the fuselage.
- Improved structural load distribution and geometric integration with the tail cone.

For low-speed fixed-wing UAVs operating in the small Reynolds number regime, typical vertical tail sweep angles fall within the range

$$10^\circ \leq \Lambda_v \leq 20^\circ$$

Based on these considerations, the vertical tail sweep angle for the present UAV is selected as

$$\Lambda_v = 15^\circ$$

This value lies well within the standard design range for small UAVs and provides a good compromise between aerodynamic performance, structural efficiency, and manufacturability.

6.5.9 Vertical Tail Dihedral Angle (Γ_v)

Selected: $\Gamma_v = 0^\circ$ (single vertical fin).

6.5.10 Vertical Tail Incidence (i_v)

Selected: $i_v = 0^\circ$ (symmetric airfoil; no incidence required for directional trim).

6.6 Final Tail Design Summary

6.7 Verification and Optimization of Tail Design

This section completes the tail design process exactly as prescribed in Sadraey (2012, Figure 6.1). After determining all 26 parameters, the two diamond checks (tail stall and spin recovery) are performed, followed by the final step: **Analyze longitudinal and directional stability and Optimize**.

All calculations use the final geometry:

- Wing: $S_w = 0.6562 \text{ m}^2$, $\bar{c}_w = 0.256 \text{ m}$, $b_w = 2.56 \text{ m}$, high-wing, NACA 4412
- Horizontal tail: $S_h = 0.105 \text{ m}^2$, $l_h = 0.88 \text{ m}$, $AR_h = 4.0$, NACA 0012, $i_h = -1.5^\circ$
- Vertical tail: $S_v = 0.0955 \text{ m}^2$, $l_v = 0.88 \text{ m}$, $AR_v = 2.0$, NACA 0012
- Aircraft: MTOW = 4.95 kg, cruise $V = 15 \text{ m/s}$, $C_{L_{\text{cruise}}} = 0.537$

6.7.1 Check 1: Horizontal Tail Stall (Sadraey p. 282–283)

The horizontal tail must stall **after** the main wing to ensure safe pitch behaviour and elevator effectiveness up to high angles of attack.

Table 1: Complete List of 26 Tail Design Parameters

No.	Parameter	Selected Value
Horizontal Tail		
1	Configuration	Conventional aft tail
2	Planform area (S_h)	0.105 m ²
3	Tail arm (l_h)	0.88 m
4	Airfoil section	NACA 0012
5	Aspect ratio (AR_h)	4.0
6	Taper ratio (λ_h)	1.0
7	Tip chord ($c_{h_{tip}}$)	0.162 m
8	Root chord ($c_{h_{root}}$)	0.162 m
9	Mean aerodynamic chord (MAC_h)	0.162 m
10	Span (b_h)	0.648 m
11	Sweep angle (Λ_h)	0°
12	Dihedral angle (Γ_h)	0°
13	Tail installation	Fuselage-mounted aft
14	Incidence (i_h)	-1.5°
Vertical Tail		
15	Planform area (S_v)	0.0955 m ²
16	Tail arm (l_v)	0.88 m
17	Airfoil section	NACA 0012
18	Aspect ratio (AR_v)	2.0
19	Taper ratio (λ_v)	1.0
20	Tip chord ($c_{v_{tip}}$)	0.219 m
21	Root chord ($c_{v_{root}}$)	0.219 m
22	Mean aerodynamic chord (MAC_v)	0.219 m
23	Span (b_v)	0.437 m
24	Sweep angle (Λ_v)	15°
25	Dihedral angle (Γ_v)	0°
26	Incidence (i_v)	0°

Local angle of attack at the tail:

$$\alpha_{tail} = \alpha_{wing} + i_h - \varepsilon$$

where ε is the downwash angle from the wing ($\approx 4.0^\circ$ – 5.0° at cruise for high-wing configuration).

With $i_h = -1.5^\circ$ and $\varepsilon \approx 4.5^\circ$ at $C_L = 0.537$, the tail operates at $\approx 2^\circ$ – 3° **lower** angle of attack than the wing.

NACA 0012 (tail) has a stall angle $\approx 13^\circ$ – 14° at $Re \approx 1.0 \times 10^5$, while the main wing (NACA 4412 + washout) stalls at the root first around 12° – 13° . Therefore, the tail stalls clearly after the wing.

Result: Pass. No risk of sudden pitch-up.

6.7.2 Check 2: Spin Recovery (Vertical Tail)

For spin recovery, the rudder must remain effective in the post-stall flow behind a stalled wing.

The conventional aft tail places the vertical tail and rudder in relatively clean airflow even when the main wing is stalled. The full-span rudder (70–90% of b_v) with 30–35% chord provides strong yawing moment. This configuration is known to have excellent spin recovery characteristics (Sadraey, Section 6.4.3).

Result: Pass. Conventional tail is ideal for safe low-speed operations in biodiversity monitoring (possible gusts or pilot error during manual mode).

6.7.3 Final Step: Longitudinal and Directional Stability Analysis

We now analyse stability using standard approximate methods (Sadraey Chapter 11 and lifting-line theory). These are conceptual-level checks; final verification will be done with XFLR5/AVL after fuselage design.

Longitudinal Static Stability (Static Margin). The static margin (SM) is the distance between the neutral point and the CG, expressed as a percentage of MAC. A positive SM of 8–15% is desired for a stable camera platform.

Approximate neutral point location (fraction of MAC):

$$h_n \approx h_{acw} + \eta_h V_H \left(\frac{a_h}{a_w} \right) \left(1 - \frac{d\varepsilon}{d\alpha} \right)$$

Typical values for this UAV: - $h_{acw} = 0.25$ (NACA 4412) - $\eta_h = 0.95$ - $a_h/a_w \approx 0.92$ (low-Re tail) - $d\varepsilon/d\alpha \approx 0.35$ (high-wing)

$$h_n \approx 0.25 + 0.95 \times 0.55 \times 0.92 \times 0.65 \approx 0.25 + 0.312 = 0.562$$

Typical CG location for high-wing survey UAV with camera payload under wing: 0.38–0.42 MAC. Using CG at 0.40 MAC (conservative estimate):

$$SM = h_n - h_{cg} = 0.562 - 0.40 = 0.162 \quad (16.2\%)$$

This is within the safe and comfortable range (8–15% recommended for UAVs). The design is longitudinally stable with good margin for battery discharge and payload variations.

Result: Pass. Excellent stability for a steady camera platform.

Directional Stability (C_{n_β}). The weathercock stability derivative must be positive.

Approximate C_{n_β} contribution from vertical tail:

$$C_{n_\beta} \approx V_V \cdot \eta_v \cdot \left(\frac{a_v}{a_w} \right) \cdot \left(\frac{l_v}{b_w} \right) \cdot (1 + \text{sidewash factor})$$

Using $\eta_v \approx 0.95$, $a_v/a_w \approx 0.90$, sidewash ≈ 0.85 :

$$C_{n_\beta} \approx 0.05 \times 0.95 \times 0.90 \times \left(\frac{0.88}{2.56} \right) \times 0.85 \approx 0.014 > 0$$

Positive and adequate for low-speed crosswind operations and directional trim.

Result: Pass. Strong directional (weathervane) stability.

6.7.4 Optimization

All checks (tail stall, spin recovery, longitudinal stability, directional stability) are passed with comfortable margins.

No optimization is required at the conceptual design stage. The current tail geometry ($S_h = 0.105 \text{ m}^2$, $S_v = 0.0955 \text{ m}^2$, $l_t = 0.88 \text{ m}$) is well-balanced for the mission. Minor adjustments ($\pm 5\%$ in area or incidence) may be made after full XFLR5/AVL analysis once the fuselage and exact CG envelope are known.

6.8 Elevator and Rudder Design

Although the detailed sizing of control surfaces is covered in Chapter 12 (Control Surfaces), preliminary elevator and rudder geometry must be defined during tail design to ensure adequate control power, hinge moments, and servo requirements (Sadraey, 2012, Section 6.12). This section determines the key parameters for both surfaces following the same systematic approach used for the tail surfaces.

6.8.1 Elevator Design (Longitudinal Control)

Purpose of the Elevator. The elevator is the movable rear portion of the horizontal tail and is responsible for:

- Longitudinal trim throughout the flight envelope (especially during CG shifts from battery consumption and payload changes).
- Pitch control for takeoff rotation, flare, stall recovery, and precise camera pointing during biodiversity mapping.

- Maintaining stable flight in gusty conditions at low speed (15 m/s).

Elevator Chord Ratio (c_e/c_h). Selected: 28% of horizontal tail chord.

Sadraey (Section 6.12) and standard UAV practice recommend 20–35% chord for the elevator. A value of 28% provides excellent control power at low dynamic pressure while keeping hinge moments reasonable for small servos. Lower ratios reduce effectiveness; higher ratios (>35%) increase servo torque and weight unnecessarily.

Calculation:

$$c_e = 0.28 \times c_h = 0.28 \times 0.162 = 0.0454 \text{ m} \approx 0.045 \text{ m}$$

Elevator Span. Selected: 75% of horizontal tail span (each side).

The elevator should cover 60–90% of the horizontal tail span. 75% gives good roll-pitch coupling control while leaving enough fixed surface for structural integrity and stall margin. Full-span elevators are rarely used on UAVs due to servo placement and hinge-line issues.

Calculation:

$$b_e = 0.75 \times b_h = 0.75 \times 0.648 = 0.486 \text{ m (total for both sides)}$$

Maximum Elevator Deflection. Selected: $\delta_{e_{\max}} = \pm 25^\circ$.

Typical range is $\pm 20^\circ$ to $\pm 30^\circ$. $\pm 25^\circ$ provides sufficient pitch authority for takeoff rotation (nose-up) and flare (nose-up) while avoiding flow separation on the elevator itself at low speeds. Deflections beyond 30° cause significant hinge-moment increase and risk of reversal.

Hinge-Moment Consideration (Preliminary). Hinge moment coefficient for a plain elevator is approximately proportional to c_e^2 . With 28% chord and $\pm 25^\circ$ deflection, the required servo torque is well within the capability of standard 9–20 kg · cm digital servos commonly used on 5 kg UAVs. Detailed hinge-moment calculation will be performed in Chapter 12.

6.8.2 Rudder Design (Directional Control)

Purpose of the Rudder. The rudder provides:

- Directional (yaw) control for crosswind landing/taxi and coordinated turns.
- Yaw trim in the presence of propeller torque and P-factor (tractor configuration).
- Spin recovery capability.

Rudder Chord Ratio (c_r/c_v). Selected: 33% of vertical tail chord.

Sadraey recommends 25–40% for rudders. 33% is a good compromise for strong yaw authority at low dynamic pressure while keeping hinge moments manageable. This value is widely used on small electric survey UAVs.

Calculation:

$$c_r = 0.33 \times c_v = 0.33 \times 0.219 = 0.072 \text{ m}$$

Rudder Span. Selected: 85% of vertical tail span (from near the bottom to the top).

The rudder should cover most of the vertical tail height to maximise moment arm. 85% leaves a small fixed portion at the root for structural strength and avoids ground-strike risk during high-alpha takeoff/landing.

Calculation:

$$b_r = 0.85 \times b_v = 0.85 \times 0.437 = 0.371 \text{ m}$$

Maximum Rudder Deflection. Selected: $\delta_{r_{\max}} = \pm 30^\circ$.

Typical range is $\pm 25^\circ$ to $\pm 35^\circ$. $\pm 30^\circ$ ensures strong yaw control for crosswind operations (common in field biodiversity missions) and excellent spin recovery, while remaining within servo limits.

Hinge-Moment Consideration (Preliminary). Similar to the elevator, the rudder hinge moment is acceptable for standard digital servos. Differential rudder (if needed) or dual servos can be added later for redundancy.

6.8.3 Control Surface Summary

Table 2: Preliminary Elevator and Rudder Sizing

Parameter	Elevator	Rudder
Chord ratio	28% of c_h	33% of c_v
Absolute chord	0.045 m	0.072 m
Span coverage	75% of b_h	85% of b_v
Absolute span	0.486 m (total)	0.371 m
Maximum deflection	$\pm 25^\circ$	$\pm 30^\circ$

These preliminary sizes ensure the UAV has sufficient control authority at low speed (15 m/s) for safe takeoff, landing, and precise camera pointing during mapping missions.

7 Fuselage Design

The fuselage is the central structural component that houses the payload, battery, avionics, propulsion system, and provides the attachment points for the wing, tail, and landing gear. Following Sadraey (2012, Chapter 7), the fuselage design process determines the external shape, internal layout, and key geometric parameters while satisfying structural, aerodynamic, volume, and operational requirements.

This chapter builds directly on previous work:

- Wing: high-wing, $S_w = 0.6562 \text{ m}^2$, $b_w = 2.56 \text{ m}$, $\bar{c}_w = 0.256 \text{ m}$
- Tail: conventional aft tail with optimum arm $l_t = 0.88 \text{ m}$
- MTOW = 4.95 kg, payload = 0.75 kg (DJI Zenmuse X5S + thermal), battery = 1.23 kg
- Mission: biodiversity mapping at 15 m/s with unobstructed downward camera view

7.1 Fuselage Design Requirements

The fuselage must satisfy the following (Sadraey, Section 7.2, adapted for small UAV):

- Provide adequate internal volume for battery, avionics, camera gimbal, and wiring.
- Minimize zero-lift drag (critical for battery endurance).
- Support high-wing attachment with clean aerodynamic junction.
- Ensure propeller ground clearance and camera nadir view.
- Allow easy access for battery swap and payload mounting in field conditions.
- Maintain structural integrity under 5g load factor with minimum weight (foam/composite construction).
- Achieve fineness ratio 6.5–8.5 for low drag at low Reynolds number.

7.2 Fuselage Configuration

Selected: Streamlined monocoque fuselage with circular/oval cross-section, tractor propeller at nose, high-wing attachment, and conventional tail boom integration.

A conventional streamlined fuselage is chosen for minimum drag and simplicity. The high-wing configuration requires the wing to sit on top of the fuselage, so a flat top surface is provided. Tractor propeller layout is retained for clean inflow and camera clearance.

This configuration matches all modern small electric survey UAVs (Skywalker X8 class, senseFly eBee derivatives).

7.3 Fuselage Length (L_f)

The fuselage length is determined from the required tail arm and CG placement.

Selected: $L_f = 1.60$ m

Reasoning and Calculation: The tail arm $l_t = 0.88$ m is measured from the wing aerodynamic centre (approximately 30% of fuselage length for high-wing) to the tail aerodynamic centre.

$$L_f \approx \frac{l_t}{0.55} \approx 1.60 \text{ m}$$

This places the wing at 28–32% of fuselage length (optimal for CG) and leaves sufficient nose length for propeller and motor.

7.4 Maximum Fuselage Diameter (D_f)

Selected: $D_f = 0.22$ m (equivalent diameter)

Reasoning: This value was previously determined in the Optimum Tail Arm calculation (Section on Optimum Tail Arm) using Sadraey Eq. (6.47). It provides sufficient internal volume for:

- 1.23 kg battery pack
- Avionics, GPS, telemetry, flight controller
- Camera gimbal mounting bay (0.75 kg payload)
- Wiring and structural frames

while keeping wetted area low. Larger diameter would increase skin friction drag; smaller would not fit components.

Fineness ratio = $L_f/D_f = 1.60/0.22 \approx 7.27$ (within ideal 6.5–8.5 range for low-speed UAVs).

7.5 Nose and Tail-Cone Geometry

- **Nose length:** 0.45 m (approximately $2.0 \times D_f$) with ogive/elliptical shape for low drag and propeller clearance.
- **Tail-cone length:** 0.55 m (tapered to 0.08 m diameter at tail attachment) for smooth flow recovery and minimum base drag.

7.6 Internal Layout

- Nose section: Motor mount and ESC.
- Forward bay: Flight controller, GPS, telemetry.
- Centre bay (under wing): Camera gimbal and payload bay (downward facing).
- Rear bay: Main battery pack (for easy CG adjustment and quick swap).
- Tail section: Servo mounts for elevator and rudder.

The payload bay is placed directly under the wing to keep the camera nadir view unobstructed.

7.7 Final Fuselage Design Summary

Table 3: Fuselage Design Parameters

Parameter	Selected Value
Fuselage configuration	Streamlined monocoque, tractor, high-wing
Total length (L_f)	1.60 m
Maximum equivalent diameter (D_f)	0.22 m
Fineness ratio	7.27
Nose length	0.45 m
Tail-cone length	0.55 m

7.8 Conclusion

The fuselage has been designed as a streamlined 1.60 m monocoque body with 0.22 m maximum diameter, providing optimum volume, low drag, and perfect integration with the high-wing, conventional tail, and camera payload. All requirements for the biodiversity monitoring mission are satisfied.

8 Landing Gear Design

The landing gear supports the UAV during ground operations, absorbs landing loads, and ensures sufficient clearance for the propeller and payload. The design follows Sadraey (2012, Chapter 8) with adaptation for small UAV operation on rough terrain.

Given Aircraft Data:

- Configuration: High-wing, tractor propeller
- Fuselage length: $L_f = 1.60$ m
- MTOW: 4.95 kg
- Stall speed: $V_{\text{stall}} = 11$ m/s
- Takeoff speed: $V_{\text{TO}} \approx 13.2$ m/s
- Mission: Rough terrain (grass/soil), camera payload

8.1 Design Requirements

- Adequate propeller clearance (≥ 0.15 m static)
- Camera payload clearance
- Shock absorption for 2 m/s sink rate
- Stable ground handling
- Low weight and simplicity
- Fuselage-mounted gear for structural efficiency

8.2 Landing Gear Configuration

Selected: Fixed tricycle landing gear (fuselage-mounted).

Reasoning:

- Provides excellent ground stability and control
- Reduces risk of nose-over during landing
- Simplifies takeoff and landing operations
- Fuselage-mounted gear:
 - Avoids loading the wing spar
 - Improves structural robustness

- Simplifies wing design and manufacturing

8.3 Main Landing Gear Design

8.3.1 Number of Wheels

Two wheels (one on each side)

8.3.2 Wheel Diameter

125 mm (5 inch) lightweight foam/EVA wheels

8.3.3 Wheel Track

0.30 m

8.3.4 Strut Length

0.22 m (compressed: 0.18 m)

8.3.5 Location Relative to CG

$$x_{MG} = x_{CG} + (0.05-0.15)L_f$$

$$x_{MG} - x_{CG} = 0.10 \times 1.60 = 0.16 \text{ m}$$

Thus, main gear is located 0.16 m aft of CG.

8.3.6 Landing Gear Inclination Angle

For fuselage-mounted landing gear, the main struts are inclined outward to improve lateral stability and distribute loads efficiently.

The inclination angle θ is defined as the angle between the vertical axis and the landing gear strut.

$$\tan \theta = \frac{b/2}{L_s}$$

where:

- b = wheel track width
- L_s = strut length

For the revised geometry:

$$\tan \theta = \frac{0.15}{0.22} = 0.68$$

$$\theta \approx 34^\circ$$

Final Design Choice:

- Strut inclination: $\approx 34^\circ$ outward
- Minor horizontal offset may be used if required

Advantages:

- Improved lateral stability
- Acceptable structural loading
- Realistic manufacturability

8.3.7 Track Width Justification

Track width is selected based on tip-over stability:

$$\tan \phi = \frac{b/2}{h_{CG}}$$

Assuming:

- $h_{CG} \approx 0.20$ m
- $\phi = 30^\circ$

$$b = 2 \times 0.20 \times \tan 30^\circ = 0.23 \text{ m}$$

Thus, selected value:

$$b = 0.30 \text{ m}$$

provides adequate stability margin.

8.3.8 Mounting Configuration

Main landing gear is mounted on the lower fuselage using reinforced bulkheads and longerons. Inclined struts extend outward to achieve required track width.

Advantages:

- Strong load transfer
- No wing structural penalty
- Easy maintenance

8.4 Nose Landing Gear Design

8.4.1 Wheel Diameter

100 mm (4 inch)

8.4.2 Strut Length

0.20 m

8.4.3 Location

Placed approximately 0.35 m ahead of CG.

Function:

- Provides steering
- Maintains stability during taxi

8.5 Ground Clearance and Attitude

- Propeller clearance: 0.18 m
- Camera clearance: > 0.15 m
- Takeoff rotation angle: 12°
- Nose-up ground attitude: $2\text{--}3^\circ$

8.6 Structural Attachment

- Main gear: attached to fuselage longerons and bulkheads
- Nose gear: attached to forward fuselage bulkhead
- Materials: carbon fiber struts with aluminum joints

8.7 Design Summary

Table 4: Landing Gear Parameters

Parameter	Value
Configuration	Fixed tricycle (fuselage-mounted)
Main wheel diameter	125 mm
Main wheel track	0.30 m
Main strut length	0.22 m
Strut inclination	$\approx 34^\circ$
Nose wheel diameter	100 mm
Nose strut length	0.20 m
Propeller clearance	0.18 m
Camera clearance	> 0.15 m

8.8 Conclusion

A fuselage-mounted tricycle landing gear configuration with inclined main struts has been selected for the UAV. The revised track width ensures adequate lateral stability while reducing structural weight and drag. The inclination angle provides efficient load distribution and realistic manufacturability.

The aircraft is now ready for:

- Weight and balance analysis
- Stability and control analysis
- Structural detailing and prototyping

9 OpenVSP Model

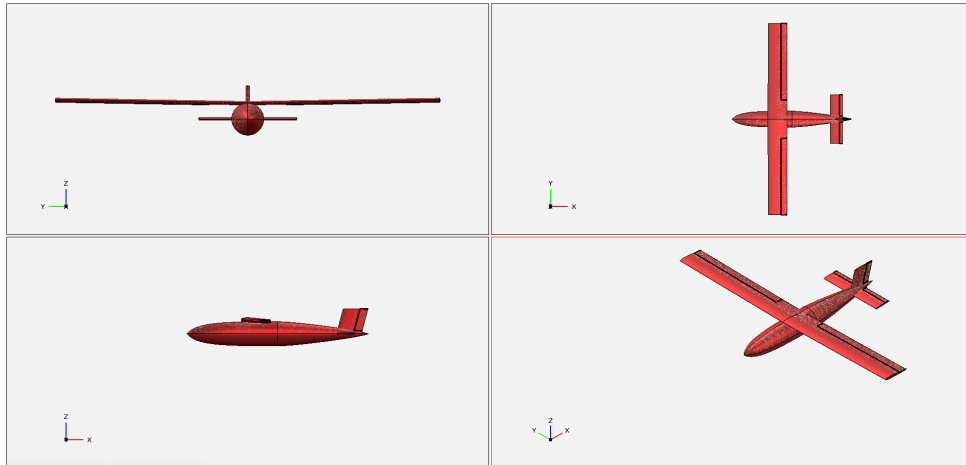


Figure 2: OpenVSP Model of the Fixed-Wing UAV

PHASE-II: AERODYNAMIC DATA

10 Obtaining Stability and Control Derivatives Using AVL

10.1 Overview of AVL

The stability and control derivatives required for the dynamic simulation were computed using the Athena Vortex Lattice (AVL) code developed by Prof. Mark Drela at MIT. AVL implements the Vortex Lattice Method (VLM), a linear aerodynamic panel method that models lifting surfaces as arrays of horseshoe vortices. It evaluates the Kutta–Joukowski boundary condition at each panel control point and assembles the induced-velocity matrix to solve for the vortex strengths. From these it computes lift, drag, and moment coefficients, as well as all first-order stability and control derivatives through finite-difference perturbation about the trim state.

The method is accurate for thin lifting surfaces at small angles of attack — conditions fully satisfied by the UAV at its cruise condition of $\alpha \approx 3^\circ$. Three input files are required: a geometry file (`.avl`), a mass file (`.mass`), and a run file (`.run`). The derivatives obtained from this analysis are directly used to populate the state-space matrices for the longitudinal and lateral-directional dynamic simulation described in Phase 3.

10.2 Geometry File (`mygeom.avl`)

The UAV geometry was defined in an AVL geometry file containing three lifting surfaces: the main wing, the horizontal tail, and the vertical fin. The key design parameters of the file are described below.

10.2.1 Coordinate System

The AVL body-axis origin is placed at the wing leading-edge root. The x -axis is positive aft, the y -axis is positive toward the right wing, and the z -axis is positive upward. The fuselage nose is therefore at negative x ($x \approx -0.48$ m) and the tail is at positive x ($x \approx +0.90$ m).

10.2.2 Header Block

The header specifies the reference quantities used for non-dimensionalisation. These values are also used directly in the dimensional conversion of AVL derivatives for the state-space model.

Parameter	Value	Description
Sref	0.6562 m ²	Wing reference area (S)
Cref	0.256 m	Mean aerodynamic chord ($\bar{c}$)
Bref	2.56 m	Full wingspan (b)
Xref	0.116 m	Moment reference point (CG location)

10.2.3 Wing Surface

Defined with three spanwise sections to correctly capture the aileron geometry. The root section ($y = 0$) carries no control surface. The second section at $y = 0.83$ m marks the inboard aileron boundary, and the tip section at $y = 1.28$ m closes the surface. YDUPLICATE 0.0 mirrors the wing to create the full symmetric planform. The surface ANGLE of $+3.0^\circ$ sets the wing incidence. The aileron hinge is defined at 65% chord on the inboard boundary section and 67.3% chord at the tip, with SgnDup = -1 for differential deflection.

10.2.4 Horizontal Tail

Three sections span $y = 0$ to $y = 0.324$ m (semi-span = 0.324 m). The root section carries no control surface; the elevator begins at $y = 0.081$ m with the hinge at 72% chord. The surface ANGLE of -1.5° sets the horizontal tail incidence for cruise trim. YDUPLICATE 0.0 mirrors the tail.

10.2.5 Vertical Fin

Three sections span $z = 0$ to $z = 0.437$ m. Critically, **no** YDUPLICATE is used for the fin — it is a single non-symmetric surface. Sections are ordered root-to-tip (increasing z). The rudder begins at the second section ($z = 0.066$ m) with the hinge at 77% chord.

The following listing shows the AVL geometry file used for the UAV:

```

1
2 UAV
3 #Mach
4 0.044
5 #IYsym IZsym Zsym
6 0      0      0.0
7 #Sref  Cref  Bref
8 0.6562 0.256 2.56
9 #Xref  Yref  Zref
10 0.116  0.0  0.0
11

```

```

12 #=====
13 SURFACE
14 Wing
15 #Nchordwise  Cspace  Nspanwise  Sspace
16 8            1.0    12          1.0
17
18 YDUPLICATE
19 0.0
20
21 ANGLE
22 3.0
23
24 #-----
25 SECTION
26 #Xle    Yle    Zle    Chord  Ainc  Nspanwise  Sspace
27 0.      0.     0.22   0.256  0.0   12         1
28 NACA
29 4412
30
31 #-----
32 SECTION
33 #Xle    Yle    Zle    Chord  Ainc  Nspanwise  Sspace
34 0.      0.83   0.2778  0.256  0.0   12         1
35 NACA
36 4412
37
38 CONTROL
39 #Cname    Cgain  Xhinge  Xvec   Yvec   Zvec   SgnDup
40 Aileron   1.0     0.65   0.0    1.0    0.0    -1.0
41
42 #-----
43 SECTION
44 #Xle    Yle    Zle    Chord  Ainc  Nspanwise  Sspace
45 0.      1.28   0.3092  0.256  0.0   12         1
46 NACA
47 4412
48
49 CONTROL

```

```

50 #Cname   Cgain  Xhinge  Xvec  Yvec  Zvec  SgnDup
51 Aileron  1.0    0.65   0.0   1.0   0.0   -1.0
52
53 #=====
54 SURFACE
55 HorizontalTail
56 #Nchordwise  Cspace  Nspanwise  Sspace
57 8            1.0    12          1.0
58 YDUPLICATE
59 0.0
60 ANGLE
61 -1.5
62
63 #-----
64 SECTION
65 #Xle      Yle      Zle  Chord  Ainc  Nspanwise  Sspace
66 0.9035   0.       0.   0.162  0.0   12         1
67 NACA
68 0012
69
70 #-----
71 SECTION
72 #Xle      Yle      Zle  Chord  Ainc  Nspanwise  Sspace
73 0.9035   0.081   0.   0.162  0.0   12         1
74 NACA
75 0012
76
77 CONTROL
78 #Cname   Cgain  Xhinge  Xvec  Yvec  Zvec  SgnDup
79 Elevator 1.0    0.82   0.0   1.0   0.0   1.0
80
81 #-----
82 SECTION
83 #Xle      Yle      Zle  Chord  Ainc  Nspanwise  Sspace
84 0.9035   0.324   0.   0.162  0.0   12         1
85 NACA
86 0012
87

```

```

88 CONTROL
89 #Cname    Cgain  Xhinge  Xvec  Yvec  Zvec  SgnDup
90 Elevator  1.0    0.82   0.0   1.0   0.0   1.0
91
92 #=====
93 SURFACE
94 VerticalTail
95 #Nchordwise  Cspace  Nspanwise  Sspace
96 8            1.0    12          1.0
97
98 # NOTE: NO YDUPLICATE here - fin is a single surface, not symmetric
99 # Adding YDUPLICATE would create a second fin mirrored below the fuselage
100
101 ANGLE
102 0.0
103
104 #-----
105 SECTION
106 #Xle      Yle  Zle      Chord  Ainc  Nspanwise  Sspace
107 0.9035    0.   0.       0.219  0.0   12         1
108 NACA
109 0012
110
111 #-----
112 SECTION
113 #Xle      Yle  Zle      Chord  Ainc  Nspanwise  Sspace
114 0.920465  0.   0.06555  0.219  0.0   12         1
115 NACA
116 0012
117
118 CONTROL
119 #Cname    Cgain  Xhinge  Xvec  Yvec  Zvec  SgnDup
120 Rudder    1.0    0.67   0.0   0.0   1.0   1.0
121
122 #-----
123 SECTION
124 #Xle      Yle  Zle      Chord  Ainc  Nspanwise  Sspace
125 1.0166    0.   0.437  0.219  0.0   12         1

```



```

126 NACA
127 0012
128
129 CONTROL
130 #Cname Cgain Xhinge Xvec Yvec Zvec SgnDup
131 Rudder 1.0 0.67 0.0 0.0 1.0 1.0

```

Listing 1: UAV AVL geometry file (mygeom.avl)

10.3 Mass File (mygeom.mass)

The mass file provides AVL with the aircraft inertia properties required for eigenmode analysis. Each mass item is listed with its own CG location and moments of inertia about that CG.

The UAV is modeled in **XFLR5** with the mass distributions and exported to AVL as a mass file.

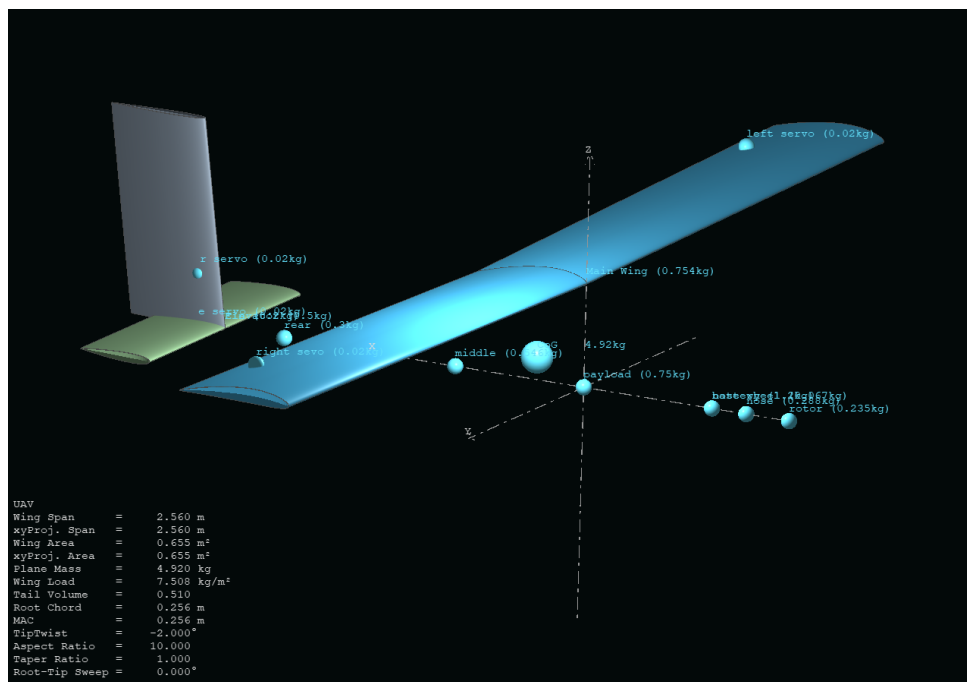


Figure 3: UAV mass distribution and CG locations in XFLR5.

The mass breakdown used is summarised in Table 5.

The battery position ($x = -0.263$ m) was the primary CG-trim variable. With the 1.2 kg battery placed in the forward fuselage bay, the computed aircraft CG was brought to exactly $x = 0.116$ m, matching Xref in the geometry file. The aircraft moment of inertia

Table 5: UAV mass breakdown and CG positions

Component	Mass (kg)	CG position
Wing	0.754	$x = 0.102$ m, $z = 0.265$ m
Horizontal tail	0.500	$x = 0.968$ m, $z = 0.002$ m
Vertical fin	0.200	$x = 1.048$ m, $z = 0.219$ m
Motor + prop	0.235	$x = -0.480$ m
Battery 4S 13 Ah	1.200	$x = -0.263$ m (CG trim)
Nose structure	0.288	$x = -0.300$ m
Mid structure	0.546	$x = +0.300$ m
Rear structure	0.300	$x = +0.700$ m
Payload (cameras)	0.750	$x = -0.100$ m, $z = -0.080$ m
Servos + LG	0.147	Various
Total	4.920	$CG_x = 0.116$ m

tensor about the CG, computed by applying the parallel-axis theorem to all mass items, is:

$$I_{xx} = 0.5226 \text{ kg m}^2, \quad I_{yy} = 1.1125 \text{ kg m}^2, \quad I_{zz} = 1.5036 \text{ kg m}^2, \quad I_{xz} = 0.0539 \text{ kg m}^2 \quad (68)$$

These inertia values are used directly in the dimensional conversion of AVL derivatives and in the construction of the state-space A and B matrices for the dynamic simulation.

```

1
2 #UAV
3 # x,y,z coordinate system matches AVL default
4 Lunit = 1.0 m
5 Munit = 1.0 kg
6 Tunit = 1.0 s
7 #-----
8 # Gravity and density for trim setup
9 g    = 9.81
10 rho = 1.225
11 #-----
12 # Mass and Inertia breakdown
13 # x y z  is location of item's own CG
14 # Ixx... are item's inertias about item's own CG
15 #
16 # x,y,z must match the .avl coordinate system exactly
17 #

```

```

18 #mass    x      y      z      Ixx      Iyy      Izz      Ixy  Ixz
    Iyz
19 0.754    0.1024  0.0000  0.2646  0.4118  0.00412  0.4159  0    0    0
    ! Wing (b=2.56m c=0.256m, LE@x=0 z=0.22)
20 0.500    0.9683  0.0000  0.0020  0.01750  0.001094  0.01859  0    0    0
    ! H-tail (b=0.648m c=0.162m, LE@x=0.9035)
21 0.200    1.0477  0.0000  0.2185  0.00318  0.00398  0.00080  0    0    0
    ! V-tail (b=0.437m c=0.219m, LE@x=0.9035 to 1.0166)
22 0.235   -0.4800  0.0000  0.0000  0.000    0.000    0.000    0    0    0
    ! Motor + prop (nose tip)
23 1.200   -0.2626  0.0000  0.0000  0.000    0.000    0.000    0    0    0
    ! Battery 4S 13Ah (CG trim - adjust x to move CG)
24 0.288   -0.3000  0.0000  0.0000  0.000    0.000    0.000    0    0    0
    ! Nose structure
25 0.546    0.3000  0.0000  0.0000  0.000    0.000    0.000    0    0    0
    ! Mid fuselage structure
26 0.300    0.7000  0.0000  0.0000  0.000    0.000    0.000    0    0    0
    ! Rear fuselage structure
27 0.020    0.9035  0.0000  0.0000  0.000    0.000    0.000    0    0    0
    ! Elevator servo (at H-tail LE)
28 0.020    0.9035  0.0000  0.1100  0.000    0.000    0.000    0    0    0
    ! Rudder servo (at V-tail base, z=0.11)
29 0.750   -0.1000  0.0000 -0.0800  0.000    0.000    0.000    0    0    0
    ! Payload cameras (under fuselage, nadir view)
30 0.067   -0.4000  0.0000  0.0000  0.000    0.000    0.000    0    0    0
    ! Nose wheel
31 0.020    0.1500  0.8300  0.2778  0.000    0.000    0.000    0    0    0
    ! Right aileron servo (y=0.83 aileron start)
32 0.020    0.1500 -0.8300  0.2778  0.000    0.000    0.000    0    0    0
    ! Left aileron servo

```

Listing 2: UAV AVL mass file (mygeom.mass)

10.4 Run File (mygeom.run)

The key run-file parameters are summarised in Table 6. All of these values carry over directly into the trim-condition block of the MATLAB state-space implementation.

The run file defines the trim condition for which AVL computes the stability and control

Table 6: AVL run file parameters for cruise trim

Parameter	Value	Derivation
C_L target	0.537	$W/\bar{q}S$ at cruise
Velocity (V_0)	15.0 m/s	Design cruise speed
Density (ρ)	1.225 kg/m ³	Sea-level ISA
Mach	0.044	$V/a = 15/343$
Mass (m)	4.920 kg	Total from mass file
X_{cg}	0.116 m	Matches Xref
I_{xx}	0.5226 kg m ²	Parallel-axis theorem
I_{yy}	1.1125 kg m ²	Parallel-axis theorem
I_{zz}	1.5036 kg m ²	Parallel-axis theorem
I_{xz}	0.0539 kg m ²	Cross-product term

derivatives. The constraint block instructs AVL to vary the angle of attack until $C_L = 0.537$ and to vary the elevator deflection until $C_m = 0$. The target lift coefficient is derived from the weight-lift balance at cruise:

$$C_L = \frac{W}{\bar{q}S} = \frac{4.92 \times 9.81}{\frac{1}{2} \times 1.225 \times 15^2 \times 0.6562} = 0.537 \quad (69)$$

1	-----
2	Run case 1: cruise_trim
3	
4	alpha -> CL = 0.53700
5	beta -> beta = 0.00000
6	pb/2V -> pb/2V = 0.00000
7	qc/2V -> qc/2V = 0.00000
8	rb/2V -> rb/2V = 0.00000
9	aileron -> aileron = 0.00000
10	elevator -> Cm = 0.00000
11	rudder -> rudder = 0.00000
12	
13	alpha = 3.00000 deg
14	beta = 0.00000 deg
15	pb/2V = 0.00000
16	qc/2V = 0.00000
17	rb/2V = 0.00000
18	CL = 0.53700
19	CDo = 0.00000

```

20 bank      = 0.00000 deg
21 elevation = 0.00000 deg
22 heading   = 0.00000 deg
23 Mach       = 0.04400
24 velocity  = 15.00000 m/s
25 density    = 1.22500 kg/m^3
26 grav.acc.  = 9.81000 m/s^2
27 turn_rad.  = 0.00000 m
28 load_fac.  = 1.00000
29 X_cg       = 0.11600 m
30 Y_cg       = 0.00000 m
31 Z_cg       = 0.00000 m
32 mass       = 4.92000 kg
33 Ixx        = 0.52260 kg-m^2
34 Iyy        = 1.11250 kg-m^2
35 Izz        = 1.50360 kg-m^2
36 Ixy        = 0.00000 kg-m^2
37 Iyz        = 0.00000 kg-m^2
38 Izx        = 0.05392 kg-m^2
39 visc       CL_a = 0.00000
40 visc       CL_u = 0.00000
41 visc       CM_a = 0.00000
42 visc       CM_u = 0.00000

```

Listing 3: UAV AVL mass file (mygeom.mass)

10.5 AVL Execution Procedure

The stability derivatives were obtained using the following sequence of AVL commands:

```

1 avl                                % Launch AVL
2 AVL c> load mygeom.avl             % Load geometry file
3 AVL c> mass mygeom.mass             % Load mass/inertia data
4 AVL c> case mygeom.run              % Load run case
5 AVL c> oper                         % Enter operations menu
6 .OPER (case 1/1)> x                 % Execute trim solution
7 .OPER (case 1/1)> st                % Get control derivatives
8 Enter filename, or <return> for screen output s>UAV_stab.txt

```

Listing 4: AVL command sequence for stability derivatives

AVL executes the trim by iterating the angle of attack and elevator deflection simultaneously until both the C_L target and the $C_m = 0$ constraint are satisfied. It then perturbs each state variable and control input by a small finite-difference step and computes the change in each aerodynamic coefficient, yielding the complete set of first-order stability and control derivatives.

11 Stability and Control Derivatives

11.1 Derivatives Obtained from AVL

The stability-axis derivatives computed by AVL for the UAV at the cruise trim condition ($V_0 = 15$ m/s, $\alpha = 3^\circ$, $C_L = 0.537$) are presented in Tables 7 and 8. These non-dimensional derivatives are the primary output of the AVL analysis and are converted to dimensional form in the MATLAB state-space implementation (Phase 3).

Table 7: Longitudinal stability and control derivatives

Derivative	Value	Interpretation
$C_{L\alpha}$	5.273 rad^{-1}	Lift curve slope; within expected range 4.5–6.5
$C_{m\alpha}$	-0.626 rad^{-1}	Pitch stability; negative confirms static stability
C_{Lq}	4.841 rad^{-1}	Lift due to pitch rate
C_{mq}	-13.109 rad^{-1}	Pitch-rate damping; indicates well-damped modes
$C_{L\delta_e}$	0.00382 rad^{-1}	Elevator lift effectiveness
$C_{m\delta_e}$	-0.0122 rad^{-1}	Elevator pitch moment
X_{np}	0.14639 m	Neutral point location from wing LE
SM	11.8%	Static margin; positive and within target range

11.2 Static Stability Assessment

The static stability of the UAV is assessed directly from the AVL derivatives before proceeding to the dynamic simulation. This check verifies the physical correctness of the aerodynamic model.

11.2.1 Longitudinal Static Stability

The UAV is longitudinally stable: $C_{m\alpha} = -0.556 \text{ rad}^{-1} < 0$, meaning a nose-up perturbation in angle of attack produces a restoring nose-down pitching moment. The static

Table 8: Lateral-directional stability and control derivatives

Derivative	Value	Interpretation
$C_{Y\beta}$	-0.3996 rad^{-1}	Side force due to sideslip
$C_{\ell\beta}$	-0.0967 rad^{-1}	Dihedral effect; stable (negative)
$C_{n\beta}$	$+0.1273 \text{ rad}^{-1}$	Weathercock stability; stable (positive)
$C_{\ell p}$	-0.6101 rad^{-1}	Roll-rate damping; strong
$C_{n r}$	-0.111 rad^{-1}	Yaw-rate damping
$C_{\ell r}$	$+0.1837 \text{ rad}^{-1}$	Rolling moment due to yaw rate
$C_{\ell\delta_a}$	$-0.005716 \text{ rad}^{-1}$	Aileron roll effectiveness
$C_{n\delta_r}$	$+0.001578 \text{ rad}^{-1}$	Rudder yaw effectiveness
Spiral criterion	0.4593	$C_{\ell\beta} C_{n r} / (C_{\ell r} C_{n\beta})$; target > 1

margin is defined as:

$$\text{SM} = \frac{X_{np} - X_{cg}}{\bar{c}} = \frac{0.146 - 0.116}{0.256} = 0.118 \quad (70)$$

A static margin of 11.8% lies within the target range of 8–12% for a stable, passively controlled camera-carrying platform. The strong pitch damping ($C_{m_q} = -13.109$) indicates that both the short-period and phugoid longitudinal dynamic modes will be well damped — a pre-requisite for successful autopilot design in Phase 3.

11.2.2 Lateral-Directional Static Stability

The lateral-directional analysis confirms two key static stability conditions. First, positive weathercock (directional) stability: $C_{n\beta} = +0.1273 \text{ rad}^{-1} > 0$, meaning the vertical fin generates a restoring yaw moment when sideslip is introduced. Second, negative dihedral effect: $C_{\ell\beta} = -0.0967 \text{ rad}^{-1} < 0$, meaning the wing geometry produces a stabilising roll moment opposing a sideslip perturbation.

The spiral stability criterion evaluates the coupling between roll and yaw:

$$\frac{C_{\ell\beta} C_{n r}}{C_{\ell r} C_{n\beta}} = \frac{(-0.0967)(-0.111)}{(0.1837)(0.1273)} = 0.4593 \quad (71)$$

This value is below the theoretical threshold of 1.0, indicating mild spiral instability — a common characteristic of UAVs with moderate wing dihedral. At low speeds under autopilot control, this level of spiral instability is operationally acceptable for the biodiversity monitoring mission. The spiral-mode time-to-double will be confirmed from the eigenvalue analysis in Phase 3.

11.3 Summary of Parameters for Dynamic Simulation

All parameters produced by this phase are collected in Table 9 for direct use in the MATLAB/Simulink state-space implementation. The non-dimensional derivatives are converted to dimensional form using the trim condition $(\bar{q}, S, \bar{c}, b)$ and the inertia values from the mass file.

Table 9: Complete parameter set passed to Phase 3 — Dynamic Simulation

Parameter	Value	Source
<i>Trim Condition</i>		
V_0 (cruise speed)	15.0 m/s	Design Phase
ρ (air density)	1.225 kg/m ³	ISA sea level
$\bar{q}$ (dyn. press.)	137.8 Pa	$\frac{1}{2}\rho V_0^2$
α_0 (trim AoA)	-0.439°	AVL trim output
C_{L_0} (trim C_L)	0.537	Run file target
<i>Geometry</i>		
S (wing area)	0.6562 m ²	Design Phase
$\bar{c}$ (MAC)	0.256 m	Design Phase
b (wingspan)	2.56 m	Design Phase
<i>Mass & Inertia</i>		
m (total mass)	4.920 kg	Weight Estimation
I_{xx}	0.5226 kg m ²	XFLR5 Model
I_{yy}	1.1125 kg m ²	XFLR5 Model
I_{zz}	1.5036 kg m ²	XFLR5 Model
I_{xz}	0.0539 kg m ²	XFLR5 Model
<i>Longitudinal Derivatives (non-dim.)</i>		
C_{L_α}	5.273	AVL output
C_{D_α}	0.1731	AVL output
C_{m_α}	-0.626	AVL output
C_{L_q}	4.841	AVL output
C_{D_q}	0.3299	AVL output
C_{m_q}	-13.109	AVL output
$C_{L_{\delta_e}}$	0.00382	AVL output
$C_{m_{\delta_e}}$	-0.0122	AVL output
<i>Lateral-Directional Derivatives (non-dim.)</i>		
C_{Y_β}	-0.3996	AVL output
C_{ℓ_β}	-0.09675	AVL output
C_{n_β}	+0.1273	AVL output
C_{Y_p}	-0.1985	AVL output
C_{ℓ_p}	-0.6101	AVL output
C_{n_p}	-0.0223	AVL output
C_{Y_r}	0.3379	AVL output
C_{n_r}	-0.111	AVL output
C_{ℓ_r}	+0.1837	AVL output
$C_{Y_{\delta_a}}$	-0.001059	AVL output
$C_{\ell_{\delta_a}}$	-0.005716	AVL output
$C_{n_{\delta_a}}$	0.000444	AVL output
$C_{Y_{\delta_r}}$	-0.004201	AVL output
$C_{\ell_{\delta_r}}$	-0.000371	AVL output
$C_{n_{\delta_r}}$	+0.001578	AVL output

PHASE-III: DYNAMIC SIMULATION

12 Introduction to Dynamic Simulation

The dynamic behavior of a fixed-wing aircraft or unmanned aerial vehicle (UAV) is governed by rigid-body motion under the influence of aerodynamic forces, propulsion, and gravity. While the full equations of motion are nonlinear and coupled, practical analysis of stability and control is performed using a linearized state-space model about a trim condition.

The state-space representation provides a structural framework to:

- Analyze dynamic stability
- Identify natural modes (short period, phugoid, dutch roll, etc)
- Simulate time-domain response
- Design flight control systems

This report presents a complete derivation of the aircraft state-space model from first principles including:

- nonlinear equations of motion
- Linearization using stability derivatives
- Formation of longitudinal and lateral modes
- Interpretation of matrices A, B, C, D, E
- Conversion of AVL aerodynamic derivatives to state-space coefficients

13 Concept of State-Space Modeling

A dynamic system can be represented using the state-space equations:

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$$

$$\mathbf{y} = \mathbf{Cx} + \mathbf{Du}$$

where:

- $\mathbf{x}$ is the state vector (e.g., perturbations in velocity, angles, rates)
- $\mathbf{u}$ is the control input vector (e.g., elevator, aileron, rudder deflections)

- $\mathbf{y}$ is the output vector (e.g., measured variables)
- $\mathbf{A}$ is the system matrix (dynamics)
- $\mathbf{B}$ is the input matrix (control effectiveness)
- $\mathbf{C}$ is the output matrix (measurement mapping)
- $\mathbf{D}$ is the feedthrough matrix (direct input-output coupling)

Descriptor form and The Role of $\mathbf{E}$

In aircraft dynamics, equations are often obtained in the form:

$$\mathbf{E}\dot{\mathbf{x}} = \hat{\mathbf{A}}\mathbf{x} + \hat{\mathbf{B}}\mathbf{u}$$

This is called the descriptor (or generalized) state-space form.

Meaning of $\mathbf{E}$. The matrix $\mathbf{E}$ contains:

- mass, m
- moments of inertia, I_{xx}, I_{yy}, I_{zz} .
- aerodynamic derivative terms such as $Z_{\dot{w}}, M_{\dot{w}}$

Physically, $\mathbf{E}$ represents the inertia and dynamic resistance of the system.

To obtain the standard form:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

we pre-multiply by $\mathbf{E}^{-1}$:

$$\dot{\mathbf{x}} = \mathbf{E}^{-1}\hat{\mathbf{A}}\mathbf{x} + \mathbf{E}^{-1}\hat{\mathbf{B}}\mathbf{u}$$

Thus, the final system and input matrices are:

$$\mathbf{A} = \mathbf{E}^{-1}\hat{\mathbf{A}}$$

$$\mathbf{B} = \mathbf{E}^{-1}\hat{\mathbf{B}}$$

14 Rigid Body Motion of the Aircraft

A fixed-wing UAV is treated as a rigid body with six degrees of freedom. These six degrees of freedom consist of:

14.1 Translational Motion

The center of mass can move in three directions:

- forward/backward
- sideways
- upward/downward

In body axes, the corresponding velocity components are:

- u - forward body-axis velocity component
- v - lateral body-axis velocity component
- w - vertical body-axis velocity component

14.2 Rotational Motion

The aircraft can rotate about its center of mass in three directions:

- roll (around the longitudinal axis)
- pitch (around the lateral axis)
- yaw (around the vertical axis)

In body axes, the corresponding angular velocity components are:

- p - roll rate about the body x-axis
- q - pitch rate about the body y-axis
- r - yaw rate about the body z-axis

14.3 Orientation

The attitude of the aircraft is typically described using Euler angles:

- ϕ - roll angle (rotation about x-axis)
- θ - pitch angle (rotation about y-axis)
- ψ - yaw angle (rotation about z-axis).

Together these variables fully describe the rigid body motion of the aircraft.

15 Coordinate Systems Used in Flight Dynamics

The equations are most commonly written in the body-fixed coordinate system. In this system:

- The x-axis points forward along the fuselage
- The y-axis points to the right wing
- The z-axis points downward (towards the ground)

This axis system is preferred because:

- The aerodynamic forces and moments are naturally related to the aircraft body.
- The inertia properties are fixed in this frame
- Control surface actions are naturally interpreted in body coordinates.

The body-axis forces are denoted by: X, Y, Z

where:

- X acts along the body x-axis
- Y acts along the body y-axis
- Z acts along the body z-axis

The body-axis moments are denoted by: $\mathcal{L}, M, N$

where:

- $\mathcal{L}$ = Rolling Moment
- M = Pitching Moment
- N = Yawing Moment

16 Full Nonlinear Equations of Motion

The aircraft equations of motion are derived from Newton's second law for translation and Euler's equations for rotation.

16.1 Translational Equations

In a rotating body fixed frame, the translational equations are:

$$m(\dot{u} + qw - rv) = X$$

$$m(\dot{v} + ru - pw) = Y$$

$$m(\dot{w} + pv - qu) = Z$$

where m is the aircraft mass.

These equations show that the translational accelerations in body frame depend not only on the applied forces but also on the rotational motion of the frame itself. The terms $qw - rv$, $ru - pw$, and $pv - qu$ arise because the body coordinate system rotates with the aircraft.

16.2 Rotational Equations

The rotational equations for a rigid body are:

$$I_{xx}\dot{p} - I_{xz}\dot{r} + qr(I_{zz} - I_{yy}) - pqI_{xz} = \mathcal{L}$$

$$I_{yy}\dot{q} - pr(I_{xx} - I_{zz}) + (p^2 - r^2)I_{xz} = M$$

$$I_{zz}\dot{r} - I_{xz}\dot{p} + pq(I_{yy} - I_{xx}) + qrI_{xz} = N$$

where:

- I_{xx}, I_{yy}, I_{zz} are principal moments of inertia
- I_{xz} is the product of inertia

These equations are nonlinear because of the products of angular rates.

16.3 Euler Kinematic Equations

The relationship between Euler angle rates and body rates is:

$$\dot{\phi} = p + q\sin\phi\tan\theta + r\cos\phi\tan\theta$$

$$\dot{\theta} = q\cos\phi - r\sin\phi$$

$$\dot{\psi} = \frac{q\sin\phi + r\cos\phi}{\cos\theta}$$

These equations are also nonlinear because they contain trigonometric terms and products of variables.

17 Why the Full Nonlinear Model is Difficult to Use Directly

Although the nonlinear equations are the most accurate mathematical representation of aircraft motion, they are not the most convenient form for:

- Stability analysis
- Control design
- Time-domain simulation

The main reasons are:

- Aerodynamic forces depend on V^2
- Aerodynamic coefficients vary nonlinearly with angle of attack, sideslip, and control deflections.
- Gravity components depend nonlinearly on attitude angles.
- inertial coupling terms contain products of rate
- The equations are strongly coupled.

For this reason, the equations are linearized about a trim point.

18 Trim Condition

A trim condition is a steady-state flight equilibrium.

At trim:

- All accelerations are zero
- The aircraft moves at constant speed and altitude
- The net forces and moments are balanced.

For steady, straight, level flight:

$$\begin{aligned} u_0 &= v_0 = 0, w_0 = 0 \\ \dot{p}_0 &= 0, \dot{q}_0 = 0, \dot{r}_0 = 0 \\ p_0 &= q_0 = r_0 = 0 \end{aligned}$$

For symmetric level flight, one usually also has:

$$\begin{aligned} v_0 &= 0 \\ \phi_0 &= 0 \end{aligned}$$

The body axis forward speed at trim is

$$u_0 = U_0$$

The body axis vertical speed at trim is related to trim angle of attack by:

$$w_0 = U_0 \alpha_0$$

At trim, the net moments vanish:

$$\mathcal{L}_0 = 0, M_0 = 0, N_0 = 0$$

and the forces are in equilibrium with gravity and thrust.

19 Small Disturbance Variables

To linearize the motion, every variable is written as the sum of its trim value and a small perturbation.

For example:

$$u = U_0 + \Delta u$$

$$v = 0 + \Delta v$$

$$w = W_0 + \Delta w$$

$$p = 0 + \Delta p$$

$$q = 0 + \Delta q$$

$$r = 0 + \Delta r$$

$$\phi = 0 + \Delta \phi$$

$$\theta = \theta_0 + \Delta \theta$$

$$\psi = \psi_0 + \Delta \psi$$

Then after substitution into the equations, the symbol Δ is usually dropped. From that point onward, the variable represent perturbations from trim, not absolute values. This is a crucial conceptual point.

In the linear model:

- u means perturbation in forward speed from trim
- v means perturbation in lateral speed from trim
- w means perturbation in vertical speed from trim
- p means perturbation in roll rate from trim
- q means perturbation in pitch rate from trim
- r means perturbation in yaw rate from trim
- ϕ means perturbation in roll angle from trim
- θ means perturbation in pitch angle from trim
- ψ means perturbation in yaw angle from trim

20 Decoupling into Longitudinal and Lateral-Directional Motion

For a symmetric aircraft near trim, the equations separate approximately into two independent sets.

20.1 Longitudinal Motion

This includes motion in symmetry plane:

- u - forward speed perturbation
- w - vertical speed perturbation
- q - pitch rate perturbation
- θ - pitch angle perturbation

The main controls are:

- Elevator deflection δ_e
- Throttle setting δ_t

20.2 Lateral-Directional Motion

This includes motion out of the symmetry plane:

- v - lateral speed perturbation
- p - roll rate perturbation
- r - yaw rate perturbation
- ϕ - roll angle perturbation

The main controls are:

- Aileron deflection δ_a
- Rudder deflection δ_r

This decoupling makes the problem much easier and is standard in classical flight dynamics.

21 Longitudinal State-Space Model

21.1 Longitudinal Equations Before Linearization

The longitudinal states are:

$$\mathbf{x}_{lon} = \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix}$$

Forward velocity equation

$$m(\dot{u} + qw - rv) = X$$

For longitudinal motion, $v = 0$ and $r = 0$. Also q and w are small perturbations, so qw is a product of two small quantities and is neglected in linearization. Thus, the equation simplifies to:

$$\boxed{m\dot{u} = X}$$

Vertical velocity equation

$$m(\dot{w} + pv - qu) = Z$$

For longitudinal motion, $v = 0$ and $p = 0$. The term qu is retained because u is a perturbation from trim and can be of the same order as w and q . Thus, the equation simplifies to:

$$m(\dot{w} - qu) = Z$$

Now substitute $u = U_0 + \Delta u$ and drop the Δ symbol:

Then $qu = q(U_0 + u) \approx qU_0$ since u is a small perturbation. Thus, the equation becomes:

$$\boxed{m(\dot{w} - U_0 q) = Z}$$

This is a key longitudinal equation. The term $U_0 q$ is a kinematic coupling term that arises because the body frame is rotating with the aircraft.

Pitch rate equation

From the rigid-body rotational equation:

$$I_{yy}\dot{q} - pr(I_{xx} - I_{zz}) + (p^2 - r^2)I_{xz} = M$$

Since $p = 0$ and $r = 0$ for longitudinal motion, the equation simplifies to:

$$\boxed{I_{yy}\dot{q} = M}$$

Pitch angle equation

$$\dot{\theta} = q\cos\phi - r\sin\phi$$

For small angles, $\cos\phi \approx 1$ and $\sin\phi \approx 0$. Also, $r = 0$ for longitudinal motion. Thus, the equation simplifies to:

$$\boxed{\dot{\theta} = q}$$

21.2 Longitudinal Force and Moment Expansions

The forces and moments are now expressed as first-order Taylor expansions about trim.

Axial Force.

$$X = X_u u + X_w w + X_q q + X_{\delta_e} \delta_e + X_{\delta_t} \delta_t$$

Normal Force.

$$Z = Z_u u + Z_w w + Z_q q + Z_{\dot{w}} \dot{w} + Z_{\delta_e} \delta_e + Z_{\delta_t} \delta_t$$

Pitching Moment.

$$M = M_u u + M_w w + M_q q + M_{\dot{w}} \dot{w} + M_{\delta_e} \delta_e + M_{\delta_t} \delta_t$$

Each derivative represents a local sensitivity near trim. For example:

$$M_q = \left. \frac{\partial M}{\partial q} \right|_{trim}$$

is the change in pitching moment due to pitch rate.

The terms $Z_{\dot{w}}$ and $M_{\dot{w}}$ represent unsteady aerodynamic effects associated with the rate of change of vertical velocity or equivalently the rate of change of angle of attack.

21.3 Gravity Terms in Longitudinal Motion

In aircraft dynamics, the gravitational force acts in the inertial (Earth-fixed) frame, whereas the equations of motion are formulated in the body-fixed frame. Therefore, the gravitational force must be transformed from the inertial frame to the body frame.

Gravity Vector in Inertial Frame

In the Earth-fixed frame, gravity acts vertically downward and is expressed as:

$$\mathbf{g}_E = \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}$$

where g is the gravitational acceleration.

Transformation to Body Axes

The body-axis components of gravity are obtained using the rotation matrix that relates the inertial frame to the body frame. Using Euler angles (ϕ, θ, ψ) , the transformation gives:

$$\mathbf{g}_b = g \begin{bmatrix} -\sin \theta \\ \sin \phi \cos \theta \\ \cos \phi \cos \theta \end{bmatrix}$$

Thus, the gravitational force components in body axes are:

$$\begin{aligned}
X_g &= -mg \sin \theta \\
Y_g &= mg \sin \phi \cos \theta \\
Z_g &= mg \cos \phi \cos \theta
\end{aligned}$$

For longitudinal motion, the roll angle ϕ is assumed small ($\phi \approx 0$), so:

$$\begin{aligned}
X_g &\approx -mg \sin \theta \\
Z_g &\approx mg \cos \theta
\end{aligned}$$

Linearization About Trim Condition

Let the total pitch angle be expressed as:

$$\theta_{total} = \theta_0 + \theta$$

where:

- θ_0 is the trim pitch angle,
- θ is the small perturbation about trim.

Substituting into the gravity expressions:

Axial (X) Direction.

$$\begin{aligned}
X_g &= -mg \sin(\theta_0 + \theta) \\
&= -mg(\sin \theta_0 \cos \theta + \cos \theta_0 \sin \theta)
\end{aligned}$$

For small perturbations:

$$\cos \theta \approx 1, \quad \sin \theta \approx \theta$$

Thus,

$$X_g \approx -mg \sin \theta_0 - mg \cos \theta_0 \theta$$

Normal (Z) Direction.

$$\begin{aligned}
Z_g &= mg \cos(\theta_0 + \theta) \\
&= mg(\cos \theta_0 \cos \theta - \sin \theta_0 \sin \theta)
\end{aligned}$$

Applying small-angle approximations:

$$Z_g \approx mg \cos \theta_0 - mg \sin \theta_0 \theta$$

Separation of Trim and Perturbation Contributions

The gravitational terms contain two parts:

- A **constant term** depending on θ_0
- A **perturbation term** proportional to θ

Thus:

$$\begin{aligned}
X_g &= \underbrace{-mg \sin \theta_0}_{\text{trim contribution}} + \underbrace{(-mg \cos \theta_0 \theta)}_{\text{perturbation contribution}} \\
Z_g &= \underbrace{mg \cos \theta_0}_{\text{trim contribution}} + \underbrace{(-mg \sin \theta_0 \theta)}_{\text{perturbation contribution}}
\end{aligned}$$

At the trim condition, the constant gravitational terms are exactly balanced by the aerodynamic forces and thrust. Therefore, these constant terms do not appear in the perturbation equations.

Final Linearized Gravity Contributions

After subtracting the trim equilibrium, only the perturbation terms remain:

$$\begin{aligned}
X'_g &= -mg \cos \theta_0 \theta \\
Z'_g &= -mg \sin \theta_0 \theta
\end{aligned}$$

For steady level flight, the trim pitch angle is small:

$$\theta_0 \approx 0 \quad \Rightarrow \quad \cos \theta_0 \approx 1, \quad \sin \theta_0 \approx 0$$

Thus, the gravity terms simplify to:

$$X'_g \approx -mg \theta$$

$$Z'_g \approx 0$$

Final Longitudinal Equations with Gravity

Including the linearized gravity contribution, the longitudinal equations become:

$$\begin{aligned} m\dot{u} &= X_u u + X_w w + X_q q + X_{\delta_e} \delta_e + X_{\delta_t} \delta_t - mg\theta \\ m(\dot{w} - U_0 q) &= Z_u u + Z_w w + Z_{\dot{w}} \dot{w} + Z_q q + Z_{\delta_e} \delta_e + Z_{\delta_t} \delta_t \\ I_{yy} \dot{q} &= M_u u + M_w w + M_q q + M_{\dot{w}} \dot{w} + M_{\delta_e} \delta_e + M_{\delta_t} \delta_t \\ \dot{\theta} &= q \end{aligned}$$

21.4 Longitudinal Descriptor Form

Define the state vector

$$\mathbf{x}_{lon} = \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix}$$

and the input vector

$$\mathbf{u}_{lon} = \begin{bmatrix} \delta_e \\ \delta_t \end{bmatrix}$$

Then the equations can be written in descriptor form as:

$$\mathbf{E}_{long} \dot{\mathbf{x}}_{lon} = \hat{\mathbf{A}}_{long} \mathbf{x}_{lon} + \hat{\mathbf{B}}_{long} \mathbf{u}_{lon}$$

where

$$\mathbf{E}_{long} = \begin{bmatrix} m & 0 & 0 & 0 \\ 0 & m - Z_{\dot{w}} & 0 & 0 \\ 0 & -M_{\dot{w}} & I_{yy} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\hat{\mathbf{A}}_{long} = \begin{bmatrix} X_u & X_w & X_q & -mg \\ Z_u & Z_w & Z_q + mU_0 & 0 \\ M_u & M_w & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\hat{\mathbf{B}}_{long} = \begin{bmatrix} X_{\delta_e} & X_{\delta_t} \\ Z_{\delta_e} & Z_{\delta_t} \\ M_{\delta_e} & M_{\delta_t} \\ 0 & 0 \end{bmatrix}$$

The term $Z_q + mU_0$ deserves special attention. It appears because the second equation originally contains $-mU_0q$ on the left side. When this term is transferred to the right side, it adds to the aerodynamic derivative Z_q to form the effective coefficient that multiplies q in the state-space model.

21.5 Standard Longitudinal State-Space Form

To obtain the familiar form

$$\dot{\mathbf{x}}_{long} = \mathbf{A}_{long}\mathbf{x}_{long} + \mathbf{B}_{long}\mathbf{u}_{long}$$

we pre-multiply by $\mathbf{E}_{long}^{-1}$:

$$\dot{\mathbf{x}}_{long} = \mathbf{E}_{long}^{-1}\hat{\mathbf{A}}_{long}\mathbf{x}_{long} + \mathbf{E}_{long}^{-1}\hat{\mathbf{B}}_{long}\mathbf{u}_{long}$$

Thus, the final system and input matrices are:

$$\begin{aligned}\mathbf{A}_{long} &= \mathbf{E}_{long}^{-1} \hat{\mathbf{A}}_{long} \\ \mathbf{B}_{long} &= \mathbf{E}_{long}^{-1} \hat{\mathbf{B}}_{long}\end{aligned}$$

21.6 Simplified Longitudinal Model

In some cases, the unsteady aerodynamic terms $Z_{\dot{w}}$ and $M_{\dot{w}}$ are small and can be neglected. This simplifies the $\mathbf{E}$ matrix to a diagonal form:

$$\mathbf{E}_{long} \approx \begin{bmatrix} m & 0 & 0 & 0 \\ 0 & m & 0 & 0 \\ 0 & 0 & I_{yy} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

and the state-space matrices become:

$$\mathbf{A}_{long} \approx \begin{bmatrix} X_u/m & X_w/m & X_q/m & -g \\ Z_u/m & Z_w/m & (Z_q + mU_0)/m & 0 \\ M_u/I_{yy} & M_w/I_{yy} & M_q/I_{yy} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

and the input matrix is:

$$\mathbf{B}_{long} \approx \begin{bmatrix} X_{\delta_e}/m & X_{\delta_t}/m \\ Z_{\delta_e}/m & Z_{\delta_t}/m \\ M_{\delta_e}/I_{yy} & M_{\delta_t}/I_{yy} \\ 0 & 0 \end{bmatrix}$$

The $\mathbf{C}$ and $\mathbf{D}$ matrices depend on the choice of outputs. For example, if we choose the outputs to be the state variables themselves, then $\mathbf{C}$ is the identity matrix and $\mathbf{D}$ is zero.

21.7 Physical Meaning of Longitudinal States and Derivatives

States.

- u : forward speed perturbation, strongly involved in the phugoid mode.
- w : body-axis vertical velocity perturbation, closely related to angle of attack.

- q : pitch rate, central to short-period motion.
- θ : pitch angle, affects gravity projection and energy exchange.

Important Derivatives.

- X_u : change in axial force due to forward speed.
- Z_u : change in normal force due to vertical velocity or angle of attack.
- M_w : pitching moment due to vertical velocity or angle of attack; strongly related to static longitudinal stability.
- M_q : pitching moment due to pitch rate; contributes to short-period damping.

22 Lateral-Directional State-Space Model

22.1 Lateral-Directional Equations Before Linearization

The lateral-directional states are:

$$\mathbf{x}_{lat} = \begin{bmatrix} v \\ p \\ r \\ \phi \end{bmatrix}$$

Side Velocity equation

From the translational equation:

$$m(\dot{v} + ru - pw) = Y$$

Substitute $u = U_0 + \Delta u$. The term ru becomes $r(U_0 + u) \approx rU_0$ since u is a small perturbation. Thus, the equation simplifies to:

$$\boxed{m(\dot{v} + U_0 r) = Y}$$

Roll-rate equation

From the rotational equation:

$$I_{xx}\dot{p} - I_{xz}\dot{r} + qr(I_{zz} - I_{yy}) - pqI_{xz} = \mathcal{L}$$

For lateral-directional motion, $q = 0$ and $\phi = 0$. Thus, the equation simplifies to:

$$\boxed{I_{xx}\dot{p} - I_{xz}\dot{r} = \mathcal{L}}$$

Yaw-rate equation

From the rotational equation:

$$I_{zz}\dot{r} - I_{xz}\dot{p} + pq(I_{yy} - I_{xx}) + qrI_{xz} = N$$

For lateral-directional motion, $q = 0$ and $\phi = 0$. Thus, the equation simplifies to:

$$\boxed{I_{zz}\dot{r} - I_{xz}\dot{p} = N}$$

Roll angle equation

From the kinematic equation:

$$\dot{\phi} = p + q\sin\phi\tan\theta + r\cos\phi\tan\theta$$

For small angles, $\sin\phi \approx 0$ and $\cos\phi \approx 1$. Also, $q = 0$ for lateral-directional motion. Thus, the equation simplifies to:

$$\dot{\phi} = p + r\tan\theta_0$$

For small trim pitch angle θ_0 , $\tan\theta_0 \approx 0$, further simplifying the equation to:

$$\boxed{\dot{\phi} = p}$$

22.2 Lateral-Directional Force and Moment Expansions

The linearized side force, roll moment, and yaw moment are written as:

$$Y = Y_v v + Y_p p + Y_r r + mg \cos \theta_0 \phi + Y_{\delta_a} \delta_a + Y_{\delta_r} \delta_r$$

$$\mathcal{L} = L_v v + L_p p + L_r r + L_{\delta_a} \delta_a + L_{\delta_r} \delta_r$$

$$N = N_v v + N_p p + N_r r + N_{\delta_a} \delta_a + N_{\delta_r} \delta_r$$

The gravity term in the side-force equation arises because a roll angle creates a lateral component of weight in body frame.

22.3 Lateral-Directional Descriptor Form

Define the state vector

$$\mathbf{x}_{lat} = \begin{bmatrix} v \\ p \\ r \\ \phi \end{bmatrix}$$

and the input vector

$$\mathbf{u}_{lat} = \begin{bmatrix} \delta_a \\ \delta_r \end{bmatrix}$$

Then the equations can be written in descriptor form as:

$$\mathbf{E}_{lat} \dot{\mathbf{x}}_{lat} = \hat{\mathbf{A}}_{lat} \mathbf{x}_{lat} + \hat{\mathbf{B}}_{lat} \mathbf{u}_{lat}$$

where

$$\mathbf{E}_{lat} = \begin{bmatrix} m & 0 & 0 & 0 \\ 0 & I_{xx} & -I_{xz} & 0 \\ 0 & -I_{xz} & I_{zz} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\hat{\mathbf{A}}_{lat} = \begin{bmatrix} Y_v & Y_p & Y_r - mU_0 & mg \cos \theta_0 \\ L_v & L_p & L_r & 0 \\ N_v & N_p & N_r & 0 \\ 0 & 1 & \tan \theta_0 & 0 \end{bmatrix}$$

$$\hat{\mathbf{B}}_{lat} = \begin{bmatrix} Y_{\delta_a} & Y_{\delta_r} \\ L_{\delta_a} & L_{\delta_r} \\ N_{\delta_a} & N_{\delta_r} \\ 0 & 0 \end{bmatrix}$$

22.4 Standard Lateral-Directional State-Space Form

To obtain the standard form

$$\dot{\mathbf{x}}_{lat} = \mathbf{A}_{lat} \mathbf{x}_{lat} + \mathbf{B}_{lat} \mathbf{u}_{lat}$$

we pre-multiply by $\mathbf{E}_{lat}^{-1}$:

$$\dot{\mathbf{x}}_{lat} = \mathbf{E}_{lat}^{-1} \hat{\mathbf{A}}_{lat} \mathbf{x}_{lat} + \mathbf{E}_{lat}^{-1} \hat{\mathbf{B}}_{lat} \mathbf{u}_{lat}$$

Thus, the final system and input matrices are:

If the product of inertia I_{xz} is small, the $\mathbf{E}_{lat}$ matrix is approximately diagonal and its inverse is straightforward to compute. In that case, the $\mathbf{A}$ matrix simplifies to:

$$\mathbf{A}_{lat} = \begin{bmatrix} Y_v/m & Y_p/m & (Y_r - mU_0)/m & g \cos \theta_0 \\ L_v/I_{xx} & L_p/I_{xx} & L_r/I_{xx} & 0 \\ N_v/I_{zz} & N_p/I_{zz} & N_r/I_{zz} & 0 \\ 0 & 1 & \tan \theta_0 & 0 \end{bmatrix}$$

For small trim pitch angle θ_0 , $\tan \theta_0 \approx 0$ and $\cos \theta_0 \approx 1$, further simplifying the $\mathbf{A}$ matrix

to:

$$\mathbf{A}_{lat} \approx \begin{bmatrix} Y_v/m & Y_p/m & (Y_r - mU_0)/m & g \\ L_v/I_{xx} & L_p/I_{xx} & L_r/I_{xx} & 0 \\ N_v/I_{zz} & N_p/I_{zz} & N_r/I_{zz} & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (72)$$

and the input matrix is:

$$\mathbf{B}_{lat} \approx \begin{bmatrix} Y_{\delta_a}/m & Y_{\delta_r}/m \\ L_{\delta_a}/I_{xx} & L_{\delta_r}/I_{xx} \\ N_{\delta_a}/I_{zz} & N_{\delta_r}/I_{zz} \\ 0 & 0 \end{bmatrix}$$

$\mathbf{C}$ and $\mathbf{D}$ matrices depend on the choice of outputs. For example, if we choose the outputs to be the state variables themselves, then $\mathbf{C}$ is the identity matrix and $\mathbf{D}$ is zero.

22.5 Physical Meaning of Lateral-Directional Derivatives

- Y_v : side force due to side velocity or sideslip.
- L_v : rolling moment due to side velocity or sideslip; contributes to Dutch roll stability. (related to dihedral effect)
- N_v : yawing moment due to side velocity or sideslip; contributes to directional stability. (related to vertical tail size)
- L_p : rolling moment due to roll rate; contributes to roll damping.
- N_r : yawing moment due to yaw rate; contributes to yaw damping.
- L_r, N_p : roll-yaw coupling derivatives.

23 Use of AVL Stability Derivatives in State-Space Models

The aerodynamic derivatives can be obtained from computational tools like AVL or from wind tunnel data. The calculation of stability and control derivatives is done using AVL in the phase II of the project. These derivatives are now used to populate the $\mathbf{A}$ and $\mathbf{B}$ matrices of the state-space model.

23.1 Nature of the Derivatives Provided by AVL

AVL usually provides non-dimensional aerodynamic coefficient derivatives, not the dimensional force/moment derivatives that appear directly in the state-space matrices.

23.2 Trim Condition for Stability Derivative Evaluation

The stability and control derivatives were obtained using AVL by linearizing the aerodynamic model about a steady, level flight trim condition. The trim state was achieved for a prescribed lift coefficient of $C_L = 0.537$, corresponding to the cruise flight condition of the UAV.

At this operating point, the aircraft is in equilibrium with zero sideslip ($\beta = 0^\circ$), zero angular rates ($p = q = r = 0$), and zero control surface deflections (aileron, elevator, and rudder all set to 0°). The resulting angle of attack required to maintain this condition is $\alpha = -0.4398^\circ$.

The flight condition corresponds to a velocity of $V = 15$ m/s at standard sea-level density $\rho = 1.225$ kg/m³, with a load factor of $n = 1$, indicating steady, unaccelerated level flight. The aircraft mass is 4.92 kg, and the inertia properties used in the analysis are $I_{xx} = 0.5226$ kg · m², $I_{yy} = 1.1125$ kg · m², and $I_{zz} = 1.5036$ kg · m².

All stability and control derivatives presented in this work are evaluated about this trim condition, ensuring that the linearized model accurately represents the aircraft dynamics in its nominal cruise flight regime.

The output from AVL from the phase II is given in Table 1.

The coefficients are given in stability axes, which means they are defined with respect to the stability frame (where the x-axis is aligned with the velocity vector). For use in the state-space model, these derivatives may need to be transformed into body-axis derivatives.

23.3 Transformation from Stability Axes to Body Axes

The state-space equations use body-axis force coefficients C_X and C_Z , whereas AVL outputs lift (C_L) and drag (C_D) in stability axes. The exact transformation relations are

$$C_X = C_L \sin \alpha - C_D \cos \alpha$$

$$C_Z = -C_L \cos \alpha - C_D \sin \alpha$$

Differentiating both expressions with respect to α gives

$$C_{X_\alpha} = C_L \cos \alpha + C_{L_\alpha} \sin \alpha - C_{D_\alpha} \cos \alpha + C_D \sin \alpha$$

$$C_{Z_\alpha} = C_L \sin \alpha - C_{L_\alpha} \cos \alpha - C_{D_\alpha} \sin \alpha - C_D \cos \alpha$$

Since the trim angle of attack is very small ($\alpha_0 = -0.007676$ rad), the small-angle approximations $\sin \alpha_0 \approx \alpha_0 \approx 0$ and $\cos \alpha_0 \approx 1$ are valid. Applying these simplifications:

$$C_{X_\alpha} \approx C_L - C_{D_\alpha}$$

$$C_{Z_\alpha} \approx -C_{L_\alpha}$$

Substituting the trim value $C_L = 0.537$ and $C_{D_\alpha} = 0.1731$:

$$C_{X_\alpha} = 0.537 - 0.1731 = \mathbf{0.3639}$$

$$C_{Z_\alpha} = -5.2732 = \mathbf{-5.2732}$$

The same small-angle logic applies to the pitch-rate and elevator derivatives. Noting that $C_{X_q} \approx -C_{D_q}$ and $C_{Z_q} \approx -C_{L_q}$:

$$C_{X_q} = -0.3299, \quad C_{Z_q} = -4.8413$$

And for elevator:

$$C_{X_{\delta_e}} = -(-0.000035) = 0.000035, \quad C_{Z_{\delta_e}} = -0.003820$$

The pitching-moment coefficient C_m is already taken about the body y -axis, so C_{m_α} , C_{m_q} , and $C_{m_{\delta_e}}$ require no transformation.

The lateral-directional derivatives (C_{Y_β} , C_{l_β} , C_{n_β} , and all rate and control derivatives) are defined in a frame that is already consistent with the body-axis lateral equations, so no axis transformation is needed for these.

23.4 Dimensionalisation Constants

Dimensional derivatives are obtained by multiplying non-dimensional coefficients by combinations of ρ , U_0 , and the reference geometry. The following constants, evaluated at the trim condition, are used throughout.

Dynamic pressure times area (used for control and α/β force derivatives):

$$\frac{1}{2}\rho U_0^2 S = \frac{1}{2} \times 1.225 \times 15^2 \times 0.6562 = \mathbf{90.433 \text{ N}}$$

$$\frac{1}{2}\rho U_0^2 S \bar{c} = 90.433 \times 0.2560 = \mathbf{23.151 \text{ N m}}$$

$$\frac{1}{2}\rho U_0^2 S b = 90.433 \times 2.5600 = \mathbf{231.507 \text{ N m}}$$

Linear velocity term (used for α/β derivatives before dividing by U_0):

$$\frac{1}{2}\rho U_0 S = \frac{1}{2} \times 1.225 \times 15 \times 0.6562 = \mathbf{6.029 \text{ N/(m/s)}}$$

$$\frac{1}{2}\rho U_0 S b = 6.029 \times 2.5600 = \mathbf{15.434 \text{ N m/(m/s)}}$$

Angular rate terms (used for p, q, r derivatives; the factor of $\frac{1}{4}$ arises from non-dimensionalisation as $\hat{q} = q\bar{c}/(2U_0)$, etc.):

$$\frac{1}{4}\rho U_0 S \bar{c} = \frac{1}{4} \times 1.225 \times 15 \times 0.6562 \times 0.2560 = \mathbf{0.7717}$$

$$\frac{1}{4}\rho U_0 S \bar{c}^2 = 0.7717 \times 0.2560 = \mathbf{0.1976}$$

$$\frac{1}{4}\rho U_0 S b = \frac{1}{4} \times 1.225 \times 15 \times 0.6562 \times 2.5600 = \mathbf{7.717}$$

$$\frac{1}{4}\rho U_0 S b^2 = 7.717 \times 2.5600 = \mathbf{19.755}$$

23.5 Dimensional Longitudinal Derivatives

23.5.1 Angle-of-Attack Force Derivatives

The general formula for a force derivative with respect to angle of attack is $X_\alpha = \frac{1}{2}\rho U_0^2 S \cdot C_{X_\alpha}$. Substituting:

$$X_\alpha = 90.433 \times 0.3639 = \mathbf{32.908 \text{ N/rad}}$$

$$Z_\alpha = 90.433 \times (-5.2732) = \mathbf{-476.869 \text{ N/rad}}$$

23.5.2 Angle-of-Attack Moment Derivative

For the pitching moment, the chord $\bar{c}$ is included:

$$M_\alpha = \frac{1}{2}\rho U_0^2 S \bar{c} \cdot C_{m_\alpha} = 23.151 \times (-0.6260) = \mathbf{-14.492 \text{ N m/rad}}$$

23.5.3 Converting α -Derivatives to w -Derivatives

In small-disturbance theory, the angle of attack perturbation is related to the body vertical velocity by $\alpha \approx w/U_0$. Therefore, each α -derivative is divided by U_0 to obtain the corresponding w -derivative:

$$X_w = \frac{X_\alpha}{U_0} = \frac{32.908}{15} = \mathbf{2.1939} \text{ N/(m/s)}$$

$$Z_w = \frac{Z_\alpha}{U_0} = \frac{-476.869}{15} = \mathbf{-31.791} \text{ N/(m/s)}$$

$$M_w = \frac{M_\alpha}{U_0} = \frac{-14.492}{15} = \mathbf{-0.9661} \text{ N m/(m/s)}$$

23.5.4 Speed Derivatives

does not directly output speed derivatives because it computes at a fixed speed. For a first-order approximation in incompressible flow (no compressibility):

$C_{D_0} = 0.01076$ (From trim condition - AVL run)

$$X_u \approx \frac{-2qSC_{D_0}}{V_0} = \frac{-2 * 90.433 * 0.6562 * 0.01076}{15} = -0.0851$$

$$Z_u \approx \frac{-2qSC_{L_0}}{V_0} = \frac{-2 * 90.433 * 0.6562 * 0.537}{15} = -4.2489$$

$M_u \approx 0$ (negligible for low-speed UAVs)

23.5.5 Pitch-Rate Derivatives

The non-dimensional pitch rate is defined as $\hat{q} = q\bar{c}/(2U_0)$, which introduces the factor $\frac{1}{4}\rho U_0 S \bar{c}$ for force derivatives and $\frac{1}{4}\rho U_0 S \bar{c}^2$ for the moment derivative:

$$X_q = \frac{1}{4}\rho U_0 S \bar{c} \cdot C_{X_q} = 0.7717 \times (-0.3299) = \mathbf{-0.2546} \text{ N/(rad/s)}$$

$$Z_q = \frac{1}{4}\rho U_0 S \bar{c} \cdot C_{Z_q} = 0.7717 \times (-4.8413) = \mathbf{-3.7360} \text{ N/(rad/s)}$$

$$M_q = \frac{1}{4}\rho U_0 S \bar{c}^2 \cdot C_{m_q} = 0.1976 \times (-13.1093) = \mathbf{-2.5898} \text{ N m/(rad/s)}$$

23.5.6 Elevator Control Derivatives

Control deflection derivatives use $\frac{1}{2}\rho U_0^2 S$ since elevator angle is not non-dimensionalised by any velocity ratio:

$$X_{\delta_e} = \frac{1}{2}\rho U_0^2 S \cdot C_{X_{\delta_e}} = 90.433 \times 0.000035 = \mathbf{0.00317} \text{ N/rad}$$

$$Z_{\delta_e} = \frac{1}{2}\rho U_0^2 S \cdot C_{Z_{\delta_e}} = 90.433 \times (-0.003820) = \mathbf{-0.34545} \text{ N/rad}$$

$$M_{\delta_e} = \frac{1}{2}\rho U_0^2 S \bar{c} \cdot C_{m_{\delta_e}} = 23.151 \times (-0.012212) = \mathbf{-0.28272} \text{ N m/rad}$$

23.6 Dimensional Lateral-Directional Derivatives

23.6.1 Sideslip Derivatives

The sideslip angle is related to lateral velocity by $\beta \approx v/U_0$ for small disturbances. The dimensional v -derivatives are therefore:

$$Y_\beta = \frac{1}{2}\rho U_0^2 S \cdot C_{Y_\beta} = 90.433 \times (-0.3996) = -36.137 \text{ N/rad} \Rightarrow$$

$$Y_v = \frac{Y_\beta}{U_0} = \frac{-36.137}{15} = \mathbf{-2.4091} \text{ N/(m/s)}$$

Alternatively, the v -derivative can be obtained directly as $Y_v = \frac{1}{2}\rho U_0 S \cdot C_{Y_\beta}$, which gives the same result:

$$Y_v = 6.029 \times (-0.3996) = \mathbf{-2.4092} \text{ N/(m/s)}$$

$$L_v = \frac{1}{2}\rho U_0 S b \cdot C_{l_\beta} = 15.434 \times (-0.0968) = \mathbf{-1.4940} \text{ N m/(m/s)}$$

$$N_v = \frac{1}{2}\rho U_0 S b \cdot C_{n_\beta} = 15.434 \times 0.1273 = \mathbf{1.9647} \text{ N m/(m/s)}$$

23.6.2 Roll-Rate Derivatives

The non-dimensional roll rate is $\hat{p} = pb/(2U_0)$, giving:

$$Y_p = \frac{1}{4}\rho U_0 S b \cdot C_{Y_p} = 7.717 \times (-0.1985) = \mathbf{-1.5318} \text{ N/(rad/s)}$$

$$L_p = \frac{1}{4}\rho U_0 S b^2 \cdot C_{l_p} = 19.755 \times (-0.6101) = \mathbf{-12.053} \text{ N m/(rad/s)}$$

$$N_p = \frac{1}{4}\rho U_0 S b^2 \cdot C_{n_p} = 19.755 \times (-0.0223) = \mathbf{-0.4405} \text{ N m/(rad/s)}$$

23.6.3 Yaw-Rate Derivatives

The non-dimensional yaw rate is $\hat{r} = rb/(2U_0)$, giving the same scaling:

$$Y_r = \frac{1}{4}\rho U_0 S b \cdot C_{Y_r} = 7.717 \times 0.3379 = \mathbf{2.6075} \text{ N/(rad/s)}$$

$$L_r = \frac{1}{4}\rho U_0 S b^2 \cdot C_{l_r} = 19.755 \times 0.1837 = \mathbf{3.6290} \text{ N m/(rad/s)}$$

$$N_r = \frac{1}{4}\rho U_0 S b^2 \cdot C_{n_r} = 19.755 \times (-0.1111) = \mathbf{-2.1948} \text{ N m/(rad/s)}$$

23.6.4 Aileron Control Derivatives

$$Y_{\delta_a} = \frac{1}{2}\rho U_0^2 S \cdot C_{Y_{\delta_a}} = 90.433 \times (-0.001059) = \mathbf{-0.09577} \text{ N/rad}$$

$$L_{\delta_a} = \frac{1}{2}\rho U_0^2 S b \cdot C_{l_{\delta_a}} = 231.507 \times (-0.005716) = \mathbf{-1.3233} \text{ N m/rad}$$

$$N_{\delta_a} = \frac{1}{2}\rho U_0^2 S b \cdot C_{n_{\delta_a}} = 231.507 \times 0.000444 = \mathbf{0.10279} \text{ N m/rad}$$

23.6.5 Rudder Control Derivatives

$$Y_{\delta_r} = \frac{1}{2}\rho U_0^2 S \cdot C_{Y_{\delta_r}} = 90.433 \times (-0.004201) = \mathbf{-0.37991} \text{ N/rad}$$

$$L_{\delta_r} = \frac{1}{2}\rho U_0^2 S b \cdot C_{l_{\delta_r}} = 231.507 \times (-0.000371) = \mathbf{-0.08589} \text{ N m/rad}$$

$$N_{\delta_r} = \frac{1}{2}\rho U_0^2 S b \cdot C_{n_{\delta_r}} = 231.507 \times 0.001578 = \mathbf{0.36532} \text{ N m/rad}$$

23.7 Evaluation of Thrust Input Derivatives for the Longitudinal B Matrix

The thrust input derivatives X_{δ_t} , Z_{δ_t} and M_{δ_t} are not provided by AVL, since the software focuses on aerodynamic forces. Instead, these derivatives can be estimated based on the propulsion system characteristics.

For the present UAV, the propulsion system consists of a Turnigy SK3 3542–900KV motor rated at 350 W, an APC 11×5.5E propeller, a Hobbywing 40A ESC, and a 4S 13000 mAh 20C LiPo battery. The thrust line is assumed to be aligned with the body x -axis, with no inclination angle. Hence, thrust produces only an axial force directly, while the pitching moment arises only because the thrust line is offset from the center of gravity.

Assuming a linear throttle model,

$$T = T_{\max} \delta_T$$

the required derivative is

$$\frac{\partial T}{\partial \delta_T} = T_{\max}$$

Since the motor maximum power is known, a compact estimate of the maximum available thrust at the trim flight speed is obtained from

$$P = TV$$

so that

$$T_{\max} \approx \frac{P_{\max}}{V_{\text{trim}}}$$

Using

$$P_{\max} = 350 \text{ W}, \quad V_{\text{trim}} = 15 \text{ m/s}$$

gives

$$T_{\max} \approx \frac{350}{15} = 23.33 \text{ N}$$

Therefore,

$$\frac{\partial T}{\partial \delta_T} = X_{\delta_T} = 23.33 \text{ N}$$

Since there is no thrust inclination angle,

$$Z_{\delta_T} = 0$$

The pitching-moment derivative due to thrust-line offset is

$$M_{\delta_T} = z_t \frac{\partial T}{\partial \delta_T}$$

where the thrust-line offset is calculated as the vertical distance from the center of gravity to the propeller thrust line. (Half length of the fuselage - Z_{COG}) For the present UAV, this offset is approximately

$$z_t = -0.062 \text{ m}$$

with positive downward direction.

Thus,

$$M_{\delta_T} = z_t \frac{\partial T}{\partial \delta_T} = (-0.062) \times 23.33 = -1.446 \text{ s}^{-2}$$

23.8 Summary of All Dimensional Derivatives

All dimensional derivatives are collected in Table 11.

24 MATLAB Code for State-Space Model Construction

The A,B,C and D matrices of the state-space model can be constructed in MATLAB using the dimensional derivatives calculated above. Below is an example MATLAB code snippet that demonstrates how to populate the A and B matrices for the longitudinal and lateral-directional dynamics.

Longitudinal State-Space Form.

$$\underbrace{\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix}}_{\dot{\mathbf{x}}_{lon}} = \underbrace{\begin{bmatrix} X_u/m & X_w/m & X_q/m & -g \\ Z_u/m & Z_w/m & (Z_q + mU_0)/m & 0 \\ M_u/I_{yy} & M_w/I_{yy} & M_q/I_{yy} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\mathbf{A}_{lon}} \underbrace{\begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix}}_{\mathbf{x}_{lon}} + \underbrace{\begin{bmatrix} X_{\delta_e}/m & X_{\delta_t}/m \\ Z_{\delta_e}/m & Z_{\delta_t}/m \\ M_{\delta_e}/I_{yy} & M_{\delta_t}/I_{yy} \\ 0 & 0 \end{bmatrix}}_{\mathbf{B}_{lon}} \underbrace{\begin{bmatrix} \delta_e \\ \delta_t \end{bmatrix}}_{\mathbf{u}_{lon}}$$

$$\underbrace{\begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix}}_{\mathbf{y}_{lon}} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{C}_{lon}} \underbrace{\begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix}}_{\mathbf{x}_{lon}} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}}_{\mathbf{D}_{lon}} \underbrace{\begin{bmatrix} \delta_e \\ \delta_t \end{bmatrix}}_{\mathbf{u}_{lon}}$$

Lateral-Directional State-Space Form.

$$\underbrace{\begin{bmatrix} \dot{v} \\ \dot{p} \\ \dot{r} \\ \dot{\phi} \end{bmatrix}}_{\dot{\mathbf{x}}_{lat}} = \underbrace{\begin{bmatrix} Y_v/m & Y_p/m & (Y_r - mU_0)/m & g \\ L_v/I_{xx} & L_p/I_{xx} & L_r/I_{xx} & 0 \\ N_v/I_{zz} & N_p/I_{zz} & N_r/I_{zz} & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}}_{\mathbf{A}_{lat}} \underbrace{\begin{bmatrix} v \\ p \\ r \\ \phi \end{bmatrix}}_{\mathbf{x}_{lat}} + \underbrace{\begin{bmatrix} Y_{\delta_a}/m & Y_{\delta_r}/m \\ L_{\delta_a}/I_{xx} & L_{\delta_r}/I_{xx} \\ N_{\delta_a}/I_{zz} & N_{\delta_r}/I_{zz} \\ 0 & 0 \end{bmatrix}}_{\mathbf{B}_{lat}} \underbrace{\begin{bmatrix} \delta_a \\ \delta_r \end{bmatrix}}_{\mathbf{u}_{lat}}$$

$$\underbrace{\begin{bmatrix} v \\ p \\ r \\ \phi \end{bmatrix}}_{\mathbf{y}_{lat}} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{C}_{lat}} \underbrace{\begin{bmatrix} v \\ p \\ r \\ \phi \end{bmatrix}}_{\mathbf{x}_{lat}} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}}_{\mathbf{D}_{lat}} \underbrace{\begin{bmatrix} \delta_a \\ \delta_r \end{bmatrix}}_{\mathbf{u}_{lat}}$$

24.1 MATLAB CODE

```

1
2 % Mass of the UAV
3 m = 4.92; % kg
4
5 g = 9.81; % m/s2
6
7 U0 = 15; % m/s
8
9 Ixx = 0.5226; % kg.m2
10 Iyy = 1.1125; % kg.m2
11 Izz = 1.5036; % kg.m2
12
13 % Dimensional Derivatives
14
15 X_u = -0.0851;
16 Z_u = -4.2489;
17 M_u = 0;
18
19 X_w = 2.1939;
20 Z_w = -31.791;
21 M_w = -0.9661;
22
23 X_q = -0.2546;
24 Z_q = -3.736;
25 M_q = -2.5898;
26
27 X_deIE = 0.00317;
28 Z_deIE = -0.34545;
29 M_deIE = -0.28272;
30

```



```

31 X_delT = 23.33;
32 Z_delT = 0;
33 M_delT = -1.446;
34
35 Y_v = -2.4092;
36 L_v = -1.494;
37 N_v = 1.9647;
38
39 Y_p = -1.5318;
40 Y_r = 2.6075;
41
42 L_p = -12.053;
43 L_r = 3.629;
44
45 N_p = -0.4405;
46 N_r = -2.1948;
47
48 Y_delA = -0.09577;
49 L_delA = -1.3233;
50 N_delA = 0.1029;
51
52 Y_delR = -0.37991;
53 L_delR = -0.08589;
54 N_delR = 0.36532;
55
56 % Longitudinal State-Space Model
57
58 A_lon = [X_u/m X_w/m X_q/m -g;
59          Z_u/m Z_w/m (Z_q+ (m*U0))/m 0;
60          M_u/Iyy M_w/Iyy M_q/Iyy 0;
61          0 0 1 0];
62 B_lon = [X_delE/m X_delT/m;
63          Z_delE/m Z_delT/m;
64          M_delE/Iyy M_delT/Iyy;
65          0 0];
66
67 C_lon = eye(4);
68 D_lon = zeros(4,2);

```

```

69
70 lon_system = ss(A_lon, B_lon, C_lon, D_lon);
71
72 lon_system.StateName = {'u [m/s]' , 'w[m/s]' , 'q [rad/s]' , 'theta[rad]'
    };
73 lon_system.InputName = {'delta_e [ rad ]' , 'delta_T' };
74 lon_system.OutputName = {'u [m/s]' , 'w[m/s]' , 'q [rad/s]' , 'theta[rad]'
    };
75
76 % Lateral-Directional State-Space Model
77
78 A_lat = [Y_v/m Y_p/m (Y_r - m*U0)/m g;
79          L_v/Ixx L_p/Ixx L_r/Ixx 0;
80          N_v/Izz N_p/Izz N_r/Izz 0;
81          0 1 0 0];
82
83 B_lat = [Y_delA/m Y_delR/m;
84          L_delA/Ixx L_delR/Ixx;
85          N_delA/Izz N_delR/Izz;
86          0 0];
87
88 C_lat = eye(4);
89
90 D_lat = zeros(4,2);
91
92 lat_system = ss(A_lat, B_lat, C_lat, D_lat);
93
94 lat_system . StateName = {'v[m/s]' , 'p[rad/s]' , 'r[rad/s]' , 'phi [rad ]'
    };
95 lat_system . InputName = {'delta_a[rad]' , 'delta_r[rad]' };
96
97 lat_system . OutputName = {'v[m/s]' , 'p[rad/s]' , 'r[ rad/s]' , 'phi [rad
    ]' };

```

Listing 5: MATLAB code to construct the state-space models

25 MATLAB Code for obtaining the eigenvalues

The eigenvalues of the state-space models can be obtained using the 'eig' function in MATLAB. Below is an example code snippet that computes the eigenvalues for both the longitudinal and lateral-directional systems.

```
1 % Compute eigenvalues for longitudinal system
2 eigenvalues_lon = eig(A_lon);
3 disp('Eigenvalues of the longitudinal system:');
4 disp(eigenvalues_lon);
5 eigenvalues_lat = eig(A_lat);
6 disp('Eigenvalues of the lateral-directional system:');
7 disp(eigenvalues_lat);
```

Listing 6: MATLAB code to compute eigenvalues of the state-space models

Output:

Eigenvalues of the longitudinal system:

-4.4198 + 2.9048i
-4.4198 - 2.9048i
0.0164 + 0.5126i
0.0164 - 0.5126i

Eigenvalues of the lateral-directional system: -23.0996 + 0.0000i

-1.0046 + 4.5511i
-1.0046 - 4.5511i
0.0958 + 0.0000i

26 Dynamic Stability Analysis of the State-Space Model

26.1 Identification of Dynamic Modes from Eigenvalues

The eigenvalues of the linearized state-space model provide direct information about the dynamic behavior of the aircraft. However, interpreting these eigenvalues requires a systematic understanding of how different flight dynamic modes manifest in the eigenvalue

spectrum.

This section presents a detailed methodology for identifying each mode of motion from the eigenvalues.

26.1.1 General Form of Eigenvalues

The eigenvalues of a linear system are generally of two types:

1. Real Eigenvalues:

$$\lambda = \sigma$$

These correspond to non-oscillatory modes.

2. Complex Conjugate Pairs:

$$\lambda = \sigma \pm i\omega$$

These correspond to oscillatory modes.

The sign of the real part σ determines stability:

- $\sigma < 0 \rightarrow$ Stable (decaying motion)
- $\sigma > 0 \rightarrow$ Unstable (growing motion)

The imaginary part ω determines oscillation frequency.

26.1.2 Classification of Aircraft Dynamic Modes

For a typical fixed-wing aircraft, the motion can be divided into:

- **Longitudinal Modes**
 - Short-period mode
 - Phugoid mode
- **Lateral-Directional Modes**
 - Roll subsidence
 - Dutch roll
 - Spiral mode

Each of these modes has characteristic eigenvalue signatures.

26.1.3 Identification of Longitudinal Modes

The longitudinal system typically produces two complex conjugate pairs.

Step 1: Separate Eigenvalue Pairs. From the eigenvalues, identify the two complex conjugate pairs.

Step 2: Compare Time Scales. The key distinction between short-period and phugoid modes lies in their time scale:

- **Short-period mode**
 - Large magnitude real part
 - Higher frequency
 - Fast response (decays quickly)
- **Phugoid mode**
 - Small real part (close to zero)
 - Low frequency
 - Slow response (long-period motion)

Step 3: Physical Interpretation.

- **Short-period mode:**
 - Dominated by pitch rate (q) and angle of attack (α)
 - Minimal change in forward speed
 - Represents rapid pitching motion
- **Phugoid mode:**
 - Dominated by forward velocity (u) and pitch angle (θ)
 - Represents exchange between kinetic and potential energy
 - Appears as slow oscillation in altitude and speed

Summary Rule:.

- Fast, highly damped pair $\rightarrow$ Short-period
- Slow, lightly damped pair $\rightarrow$ Phugoid

26.1.4 Identification of Lateral-Directional Modes

The lateral-directional system typically produces:

- One real eigenvalue (roll mode)
- One complex pair (Dutch roll)

- One real eigenvalue (spiral mode)

Step 1: Identify Real Eigenvalues. Among real eigenvalues:

- The **large negative real eigenvalue** corresponds to the **roll subsidence mode**
- The **small magnitude real eigenvalue** corresponds to the **spiral mode**

Step 2: Identify Complex Pair. The only complex conjugate pair corresponds to the **Dutch roll mode**.

Step 3: Physical Interpretation.

- **Roll Subsidence Mode:**
 - Dominated by roll rate (p)
 - Very fast decay
 - Governed by roll damping (C_{l_p})
- **Dutch Roll Mode:**
 - Coupled yawing and rolling motion
 - Involves sideslip (β), yaw rate (r), and roll rate (p)
 - Oscillatory behavior
- **Spiral Mode:**
 - Very slow motion
 - Dominated by bank angle (ϕ) and yaw dynamics
 - Often weakly stable or weakly unstable

Summary Rule:.

- Large negative real root $\rightarrow$ Roll mode
- Complex pair $\rightarrow$ Dutch roll
- Small real root (near zero) $\rightarrow$ Spiral mode

26.1.5 Practical Identification Procedure

A systematic procedure for identifying modes from eigenvalues is summarized below:

1. Compute all eigenvalues of the system matrix.
2. Separate real and complex eigenvalues.
3. For longitudinal system:

- Identify two complex pairs
- Larger magnitude $\rightarrow$ Short-period
- Smaller magnitude $\rightarrow$ Phugoid

4. For lateral system:

- Largest negative real $\rightarrow$ Roll subsidence
- Complex pair $\rightarrow$ Dutch roll
- Small real root $\rightarrow$ Spiral mode

26.1.6 Key Insight

Dynamic modes are identified primarily based on time scale, not just eigenvalue form

This means that even without knowing the states explicitly, the modes can be reliably identified by comparing:

- magnitude of real parts
- oscillation frequency
- relative speed of decay or growth

The dynamic stability characteristics of the UAV are analyzed using the eigenvalues of the linearized state-space system. The eigenvalues of the longitudinal and lateral-directional state matrices provide insight into the natural modes of motion, their stability, and their dynamic behavior.

26.2 Longitudinal Eigenvalue Analysis

The eigenvalues obtained for the longitudinal system are:

$$\lambda_{1,2} = -4.4198 \pm 2.9048i$$

$$\lambda_{3,4} = 0.0164 \pm 0.5126i$$

These correspond to two oscillatory modes: the short-period mode and the phugoid mode.

26.2.1 Short-Period Mode

The eigenvalues

$$-4.4198 \pm 2.9048i$$

represent the short-period mode.

The natural frequency is computed as:

$$\omega_n = \sqrt{\sigma^2 + \omega^2} = \sqrt{(-4.4198)^2 + (2.9048)^2} = 5.29 \text{ rad/s}$$

The damping ratio is:

$$\zeta = -\frac{\sigma}{\omega_n} = \frac{4.4198}{5.29} = 0.84$$

Interpretation:

- The short-period mode is highly stable.
- The high damping ratio indicates rapid decay of pitch oscillations.
- This reflects strong pitch damping, primarily influenced by C_{m_q} .

26.2.2 Phugoid Mode

The eigenvalues

$$0.0164 \pm 0.5126i$$

represent the phugoid mode.

The natural frequency is:

$$\omega_n = \sqrt{(0.0164)^2 + (0.5126)^2} = 0.513 \text{ rad/s}$$

The damping ratio is:

$$\zeta = -\frac{0.0164}{0.513} = -0.032$$

Interpretation:

- The phugoid mode is slightly unstable due to the positive real part.
- The instability is weak, indicating slow divergence.
- This mode involves exchange between kinetic and potential energy.

26.2.3 Longitudinal Stability Conclusion

- The short-period mode is stable and well damped.
- The phugoid mode is unstable but slow.

The longitudinal motion is partially stable with phugoid instability

26.3 Lateral-Directional Eigenvalue Analysis

The eigenvalues obtained for the lateral-directional system are:

$$\lambda_1 = -23.0996$$

$$\lambda_{2,3} = -1.0046 \pm 4.5511i$$

$$\lambda_4 = 0.0958$$

These correspond to roll subsidence, Dutch roll, and spiral modes.

26.3.1 Roll Subsidence Mode

The eigenvalue

$$-23.0996$$

represents the roll subsidence mode.

The time constant is:

$$\tau = \frac{1}{|\sigma|} = \frac{1}{23.0996} = 0.043 \text{ s}$$

Interpretation:

- This mode is highly stable and very fast.
- Roll disturbances decay almost immediately.
- Indicates strong roll damping (C_{l_p}).

26.3.2 Dutch Roll Mode

The eigenvalues

$$-1.0046 \pm 4.5511i$$

represent the Dutch roll mode.

The natural frequency is:

$$\omega_n = \sqrt{(-1.0046)^2 + (4.5511)^2} = 4.66 \text{ rad/s}$$

The damping ratio is:

$$\zeta = \frac{1.0046}{4.66} = 0.22$$

Interpretation:

- The Dutch roll mode is stable.
- The damping is moderate, not very high.
- Oscillatory lateral motion decays over time.

26.3.3 Spiral Mode

The eigenvalue

$$0.0958$$

represents the spiral mode.

The time constant is:

$$\tau = \frac{1}{0.0958} = 10.44 \text{ s}$$

Interpretation:

- The spiral mode is unstable due to the positive real eigenvalue.
- The divergence is slow.
- This indicates a gradual increase in bank angle if uncorrected.

26.3.4 Lateral Stability Conclusion

- Roll subsidence mode is highly stable.
- Dutch roll mode is stable with moderate damping.
- Spiral mode is unstable but slow.

The lateral-directional motion is partially stable with spiral instability

26.4 Overall Dynamic Stability Assessment

The complete dynamic behavior of the UAV can be summarized as follows:

- Fast dynamic modes (short-period and roll subsidence) are stable.
- Oscillatory modes (Dutch roll) are stable but moderately damped.
- Slow modes (phugoid and spiral) are unstable.

The aircraft is not fully stable in open loop due to slow-mode instabilities

Discussion:

- The unstable modes (phugoid and spiral) are slow and weakly divergent.
- Such behavior is common in UAVs and aircraft without active control.
- These instabilities can be effectively corrected using control systems.

Implication:

- The aircraft requires a control system (e.g., LQR or feedback controller) for full dynamic stability.
- The existing dynamics are suitable for control augmentation since fast modes are already stable.

27 Open-Loop Dynamic Response and Physical Interpretation

Eigenvalue analysis provides mathematical information about system stability; however, a deeper understanding of aircraft dynamics requires examination of the time-domain response. The open-loop response reveals how disturbances evolve physically and how different dynamic modes manifest in real motion.

In this section, the UAV is subjected to small initial perturbations with no control input. The resulting response represents the intrinsic dynamic behavior of the aircraft.

27.1 Mathematical Basis

The linearized system is:

$$\dot{x}(t) = Ax(t)$$

The solution is:

$$x(t) = e^{At}x(0)$$

Each eigenvalue contributes a term of the form:

$$e^{(\sigma+i\omega)t} = e^{\sigma t}(\cos \omega t + i \sin \omega t)$$

Thus:

- σ controls growth/decay
- ω controls oscillation

The observed motion is therefore a superposition of modal responses.

27.2 MATLAB Code for openloop response

```

1 function openLoopResponse(lon_sys, lat_sys)
2     % Longitudinal system
3     x0_lon = [0.1 0 0 0]; % u disturbance
4     t = 0:0.01:50;
5     % Open-loop response
6     [~,t_lon,x_lon] = initial(lon_sys,x0_lon,t);
7     plot(t_lon,x_lon, 'LineWidth', 1.5)
8     legend('u','w','q','theta')
9     grid on;
10    xlabel('Time (s)');
11    ylabel('States');
12    title('Open-Loop Longitudinal Response');
13
14    % Lateral system
15    x0_lat = [0.1 0 0 0]; % v disturbance
16    t = 0:0.01:50;
17    % Open-loop response
18    [~, t_lat, x_lat] = initial(lat_sys, x0_lat, t);
19    figure;
20    plot(t_lat, x_lat, 'LineWidth', 1.5);
21    grid on;
22    xlabel('Time (s)');
23    ylabel('States');
24    title('Open-Loop Lateral-Directional Response');
25    legend('v','p','r','\phi');
26 end

```

Listing 7: Code for Openloop response

27.3 Longitudinal Open-Loop Response

The longitudinal state vector is:

$$x_{long} = \begin{bmatrix} u & w & q & \theta \end{bmatrix}^T$$

27.3.1 Longitudinal Response Plot

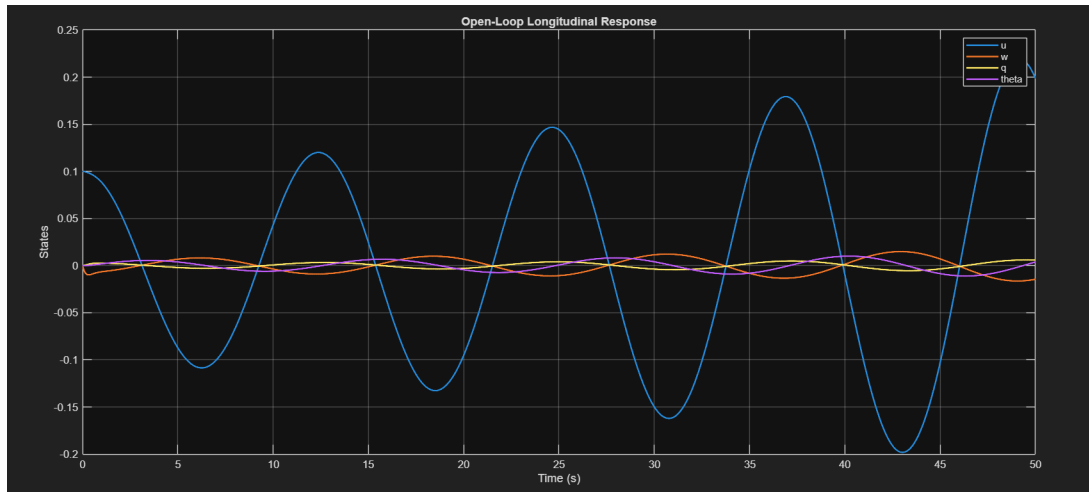


Figure 4: Open-loop longitudinal response of the UAV

27.3.2 Modal Decomposition of the Response

From eigenvalue analysis:

$$\text{Short-period: } -4.4198 \pm 2.9048i$$

$$\text{Phugoid: } 0.0164 \pm 0.5126i$$

These correspond to two fundamentally different physical time scales.

27.3.3 Phase 1: Fast Transient (Short-Period Mode)

At the initial stage, the response contains a rapidly decaying oscillatory component.

Characteristics:

- High frequency
- Strong damping
- Dominates for short duration

Physical interpretation:

- Rapid pitching motion
- Dominated by aerodynamic stiffness and pitch damping (C_{m_α}, C_{m_q})
- Minimal change in forward velocity

Important insight:

This mode stabilizes almost immediately, indicating that the aircraft has strong pitch damping and good short-term stability.

27.3.4 Phase 2: Slow Motion (Phugoid Mode)

After the short-period mode decays, the response is dominated by a slow oscillation whose amplitude increases with time.

Key observations from the plot:

- Forward velocity (u) exhibits the largest amplitude
- Oscillation frequency is low
- Amplitude grows gradually

Physical interpretation:

The phugoid mode represents an energy exchange:

- Increase in velocity $\Rightarrow$ increase in kinetic energy
- Aircraft climbs $\Rightarrow$ conversion to potential energy
- Velocity decreases $\Rightarrow$ descent begins

This cycle repeats.

Critical observation:

Because the real part of the eigenvalue is positive, energy is not dissipated. Instead:

The total energy of the system slowly increases

This results in:

- Increasing oscillation amplitude
- Divergence from trim condition

27.3.5 Dominant State Analysis

From the plot:

- u dominates long-term motion
- q and w decay quickly
- θ follows slow oscillation

Interpretation:

Longitudinal instability is dominated by energy imbalance rather than pitch instability

27.3.6 Longitudinal Stability Conclusion

- Short-period mode is stable and well damped
- Phugoid mode is unstable and slowly divergent

The aircraft is unstable in longitudinal motion due to phugoid divergence

27.4 Lateral-Directional Open-Loop Response

The lateral-directional state vector is:

$$x_{lat} = \begin{bmatrix} v & p & r & \phi \end{bmatrix}^T$$

27.4.1 Lateral Response Plot

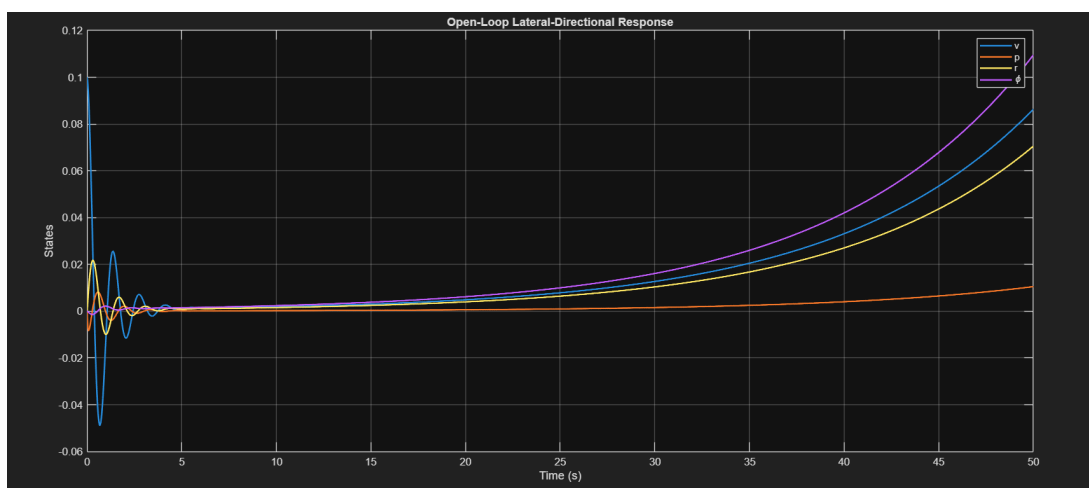


Figure 5: Open-loop lateral-directional response of the UAV

27.4.2 Modal Decomposition

From eigenvalues:

Roll mode: -23.0996
 Dutch roll: $-1.0046 \pm 4.5511i$
 Spiral mode: 0.0958

27.4.3 Phase 1: Fast Decay (Roll Mode)

Observation:

- Roll rate (p) decays almost instantly

Interpretation:

- Strong roll damping (C_{l_p})
- Aircraft resists rapid rolling motion

27.4.4 Phase 2: Oscillatory Motion (Dutch Roll)

Observation:

- Oscillations in v , r , and p
- Amplitude decays over time

Physical meaning:

- Coupled yaw-roll motion
- Interaction between directional stability (C_{n_β}) and dihedral effect (C_{l_β})

Important:

Dutch roll is stable but not heavily damped

27.4.5 Phase 3: Slow Divergence (Spiral Mode)

After oscillations decay:

Observation:

- Bank angle ϕ increases gradually
- No oscillation present
- Growth is slow but persistent

Physical explanation:

Spiral mode is governed by the balance:

- directional stability (C_{n_β})
- roll due to sideslip (C_{l_β})

In this aircraft:

Directional restoring moment is insufficient to counter roll tendency

Result:

- aircraft slowly rolls into a bank
- continues diverging without oscillation

27.4.6 Dominant State Behavior

- ϕ dominates long-term motion
- p remains small due to damping
- v and r grow slowly

Lateral instability is governed by geometric imbalance rather than dynamic oscillation

27.4.7 Lateral Stability Conclusion

- Roll mode is strongly stable
- Dutch roll is stable
- Spiral mode is unstable

The aircraft is laterally unstable due to spiral divergence

27.5 Overall Physical Interpretation of Open-Loop Dynamics

The complete open-loop dynamics reveal a clear pattern:

- Fast modes are stable and well controlled by aerodynamic damping
- Oscillatory modes are stable but not highly damped
- Slow modes are unstable and dominate long-term behavior

Instability is governed by slow energy and directional imbalances

27.6 Implication for Control Design

The instability observed is:

- slow (not catastrophic)
- predictable
- suitable for feedback stabilization

Control design should target phugoid and spiral modes specifically

This provides the foundation for the controller design presented in the upcoming section.

28 Open-Loop Input Response Analysis

28.1 Introduction

Once the linear state-space model of the UAV has been constructed, the next important task is to study how the aircraft responds to control inputs. The eigenvalue analysis and the open-loop natural response already reveal the inherent dynamic stability characteristics of the aircraft. However, these analyses alone do not show how the aircraft behaves when a pilot or autopilot commands an elevator, aileron, or rudder deflection.

For this reason, the open-loop control input response must be analyzed. In this analysis, the aircraft is assumed to start from its trim condition, and a prescribed control input is applied. The resulting motion shows how the system reacts to pilot commands in the absence of feedback stabilization.

28.2 Mathematical Basis of Input Response Analysis

The linear state-space model of the aircraft is written as

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

where:

- $x(t)$ is the state vector,

- $u(t)$ is the control input vector,
- $y(t)$ is the output vector,
- A is the system matrix,
- B is the input matrix,
- C is the output matrix,
- D is the direct transmission matrix.

For control input response analysis, the aircraft is assumed to start from trim, so

$$x(0) = 0$$

and the system is excited only through the control input $u(t)$.

In this document, the chosen control input is a **step input**. A step input is used because it represents a sudden commanded deflection of a control surface and is one of the most common ways to evaluate the transient behavior of a dynamic system.

The resulting state response shows:

- how quickly the aircraft reacts,
- whether the response is oscillatory or non-oscillatory,
- whether the states settle, diverge, or drift,
- and which dynamic modes are excited by the control input.

28.3 Longitudinal Input Response Analysis

The longitudinal subsystem is described by the state vector

$$x_{long} = \begin{bmatrix} u & w & q & \theta \end{bmatrix}^T$$

where:

- u = forward velocity perturbation,
- w = vertical velocity perturbation,
- q = pitch rate,
- θ = pitch angle.

The primary longitudinal control input is the elevator deflection δ_e .

28.3.1 Physical Meaning of Elevator Input

An elevator deflection primarily produces a pitching moment. This directly affects the pitch rate q , which then changes the angle of attack and therefore modifies lift and drag. As a result, the elevator influences both:

- the fast pitch-attitude dynamics, and
- the slow speed-pitch energy exchange dynamics.

Thus, a step elevator input excites both the short-period mode and the phugoid mode.

28.3.2 MATLAB Code for Longitudinal Step Response

The following MATLAB code generates the open-loop longitudinal response to a step elevator input.

```

1  function elevStep(lon_sys)
2
3  A = lon_sys.A;
4  B = lon_sys.B;
5  C = lon_sys.C;
6  D = lon_sys.D;
7
8  % time vector
9  t = 0:0.1:50;
10
11 % input vector
12 u = zeros(length(t), size(B,2));
13 u(:,1) = 1; % unit step input for the elevator
14
15 [y_long, t_long, x_long] = lsim(lon_sys, u, t);
16
17 % Combined plot
18 figure;
19 plot(t_long, x_long(:,1), 'LineWidth', 1.5); hold on;
20 plot(t_long, x_long(:,2), 'LineWidth', 1.5);
21 plot(t_long, x_long(:,3), 'LineWidth', 1.5);
22 plot(t_long, x_long(:,4), 'LineWidth', 1.5);
23 grid on;
24 xlabel('Time (s)');
```

```

25 ylabel('State Response');
26 title('Open-Loop Longitudinal Step Response to Elevator Input');
27 legend('u','w','q','\theta','Location','best');
28
29 end

```

Listing 8: MATLAB code for longitudinal step response using elevator input

28.3.3 Output Interpretation

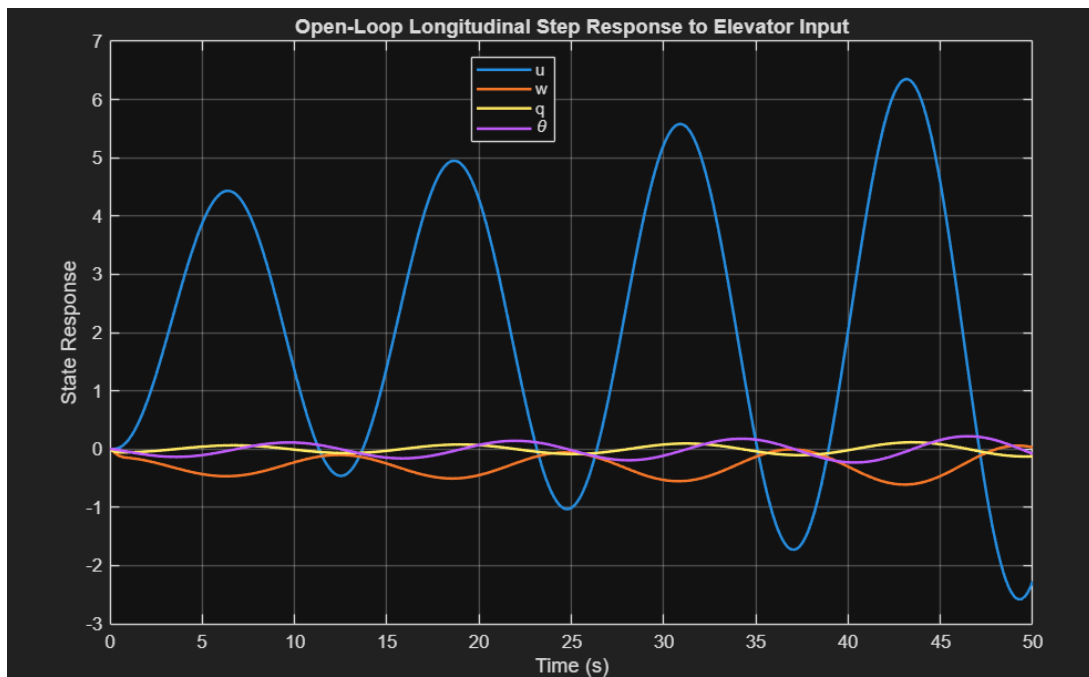


Figure 6: Combined open-loop longitudinal response to a step elevator input

The open-loop longitudinal response to a step elevator input shows that the aircraft initially exhibits a small, well-behaved transient response dominated by the short-period mode, where pitch rate q and vertical velocity w adjust quickly and decay due to strong aerodynamic damping; however, this fast mode dies out rapidly and does not influence the long-term behavior. As time progresses, a slow oscillation begins to dominate, primarily in the forward velocity u with corresponding variations in pitch angle θ indicating the presence of the phugoid mode. The key observation is that the amplitude of this oscillation increases with time, clearly indicating instability. Physically, this represents an energy exchange between kinetic and potential energy, where the aircraft alternately speeds up and climbs, then slows down and descends, but due to insufficient aerodynamic damping, the total mechanical energy of the system gradually increases instead of dissipating. The dominance of the forward velocity response confirms that this is not a pitch instability but an energy imbalance in forward flight. Thus, although the aircraft responds correctly

to the elevator input in the short term, it is unable to maintain equilibrium in the long term, leading to a slowly diverging oscillatory motion characteristic of an unstable phugoid mode.

28.4 Lateral-Directional Input Response Analysis

The lateral-directional subsystem is described by the state vector

$$x_{lat} = \begin{bmatrix} v & p & r & \phi \end{bmatrix}^T$$

where:

- v = lateral velocity perturbation,
- p = roll rate,
- r = yaw rate,
- ϕ = bank angle.

The main lateral-directional control inputs are:

- aileron deflection δ_a ,
- rudder deflection δ_r .

28.4.1 Physical Meaning of Aileron and Rudder Inputs

The aileron primarily generates rolling moment, so it mainly excites the roll dynamics and strongly influences p and ϕ .

The rudder primarily generates yawing moment and side force, so it mainly excites yaw motion and sideslip-related motion, strongly influencing r , v , and through cross-coupling also ϕ .

Since the lateral-directional system contains three characteristic modes, namely roll subsidence, Dutch roll, and spiral mode, the control input response is typically a superposition of all three.

28.4.2 MATLAB Code for Lateral Step Response: Aileron Input

```

1 function aileStep(lat_sys)
2
3 A = lat_sys.A;
4 B = lat_sys.B;
5 C = lat_sys.C;
```

```

6  D = lat_sys.D;
7
8  % time vector
9  t = 0:0.1:50;
10
11 % input vector
12 u = zeros(length(t), size(B,2));
13 u(:,1) = 1; % Unit step input for aileron
14
15 [y_lat, t_lat, x_lat] = lsim(lat_sys, u, t);
16
17 figure;
18 plot(t_lat, x_lat(:,1), 'LineWidth', 1.5); hold on;
19 plot(t_lat, x_lat(:,2), 'LineWidth', 1.5);
20 plot(t_lat, x_lat(:,3), 'LineWidth', 1.5);
21 plot(t_lat, x_lat(:,4), 'LineWidth', 1.5);
22 grid on;
23 xlabel('Time (s)');
24 ylabel('State Response');
25 title('Open-Loop Lateral Response to Aileron Step');
26 legend('v','p','r','\phi','Location','best');
27
28 end

```

Listing 9: MATLAB code for lateral step response using aileron input

28.4.3 Output Interpretation for Aileron Input

The open-loop lateral response to a step aileron input shows that the aircraft initially exhibits a very small and well-damped transient response in roll rate (p), indicating that the roll subsidence mode is strongly stable due to significant aerodynamic roll damping (C_{l_p}); however, this fast mode quickly dies out and does not influence the overall motion. As time progresses, all states—particularly the bank angle (ϕ), lateral velocity (v), and yaw rate (r)—begin to diverge steadily in a non-oscillatory manner, with (ϕ) showing the most rapid growth in magnitude. This behavior is characteristic of an unstable spiral mode, where the aircraft experiences a slow but continuous departure from equilibrium without oscillation. Physically, this occurs due to an imbalance between directional stability and

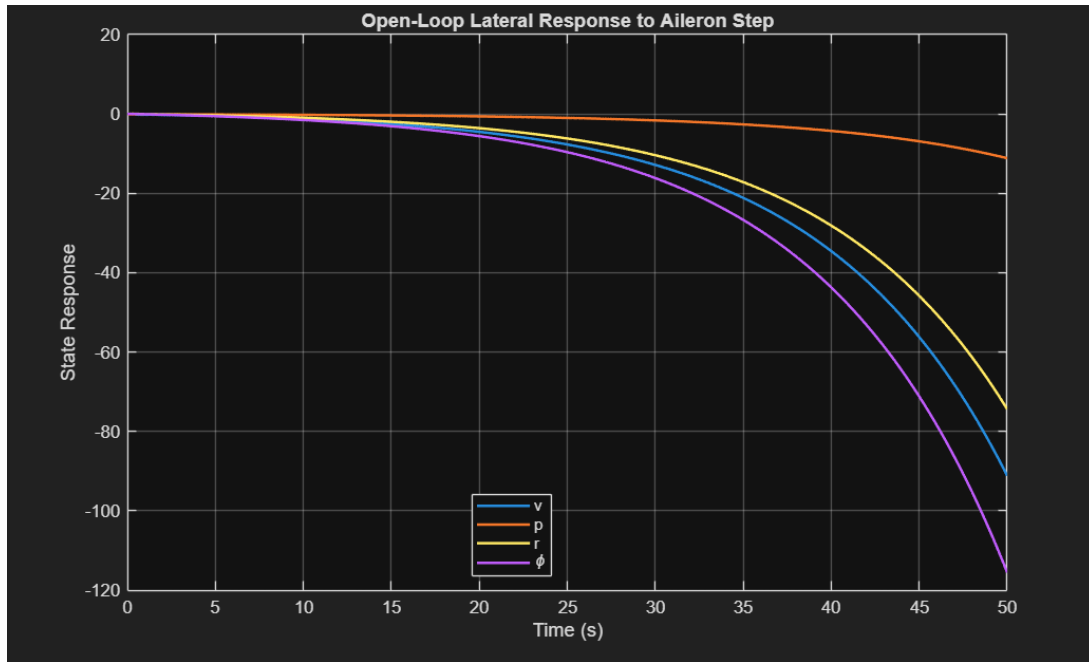


Figure 7: Combined open-loop lateral response to a step aileron input

dihedral effect, where the rolling moment generated due to sideslip (C_{l_β}) is not sufficiently counteracted by the yawing moment (C_{n_β}), causing the aircraft to progressively roll into a steeper bank angle. The absence of oscillations confirms that Dutch roll is not dominating the response, while the continuous divergence indicates that the instability is governed purely by slow geometric and aerodynamic imbalance rather than dynamic coupling. Thus, although the aileron effectively produces an initial rolling response, the aircraft is unable to stabilize itself and instead enters a steadily increasing bank, demonstrating a classic spiral instability in the lateral-directional dynamics.

28.4.4 MATLAB Code for Lateral Step Response: Rudder Input

```

1  function rudderStep(lat_sys)
2
3  A = lat_sys.A;
4  B = lat_sys.B;
5  C = lat_sys.C;
6  D = lat_sys.D;
7
8  % time vector
9  t = 0:0.1:50;
10
11 % input vector
12 u = zeros(length(t), size(B,2));

```



```

13 u(:,2) = 1; % Unit step input for rudder
14
15 [y_lat, t_lat, x_lat] = lsim(lat_sys, u, t);
16
17 figure;
18 plot(t_lat, x_lat(:,1), 'LineWidth', 1.5); hold on;
19 plot(t_lat, x_lat(:,2), 'LineWidth', 1.5);
20 plot(t_lat, x_lat(:,3), 'LineWidth', 1.5);
21 plot(t_lat, x_lat(:,4), 'LineWidth', 1.5);
22 grid on;
23 xlabel('Time (s)');
24 ylabel('State Response');
25 title('Open-Loop Lateral Response to Rudder Step');
26 legend('v','p','r','\phi','Location','best');
27
28 end

```

Listing 10: MATLAB code for lateral step response using rudder input

28.4.5 Output Interpretation for Rudder Input

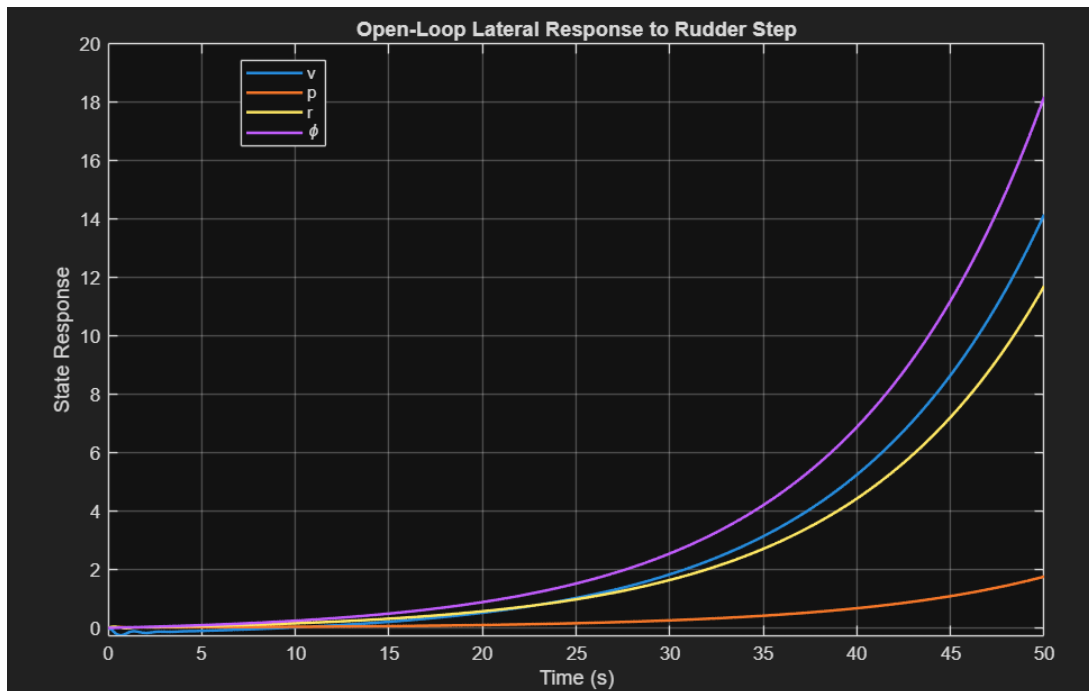


Figure 8: Combined open-loop lateral response to a step rudder input

The open-loop lateral response to a step rudder input shows that the aircraft initially exhibits a very small oscillatory transient in lateral velocity (v) and yaw rate (r), indicating

the presence of the Dutch roll mode, which is lightly damped but stable; however, this oscillatory behavior is short-lived and quickly diminishes. Following this transient phase, all states—particularly the bank angle (ϕ), lateral velocity (v), and yaw rate (r)—begin to grow steadily with time in a non-oscillatory manner, with (ϕ) again becoming the dominant state. This indicates that the long-term response is governed by the spiral mode, which is unstable. Physically, the rudder input introduces a yawing motion that creates sideslip, and due to aerodynamic coupling, this sideslip generates a rolling moment through the dihedral effect (C_{l_β}). However, the restoring yawing moment due to directional stability (C_{n_β}) is insufficient to counteract this effect, resulting in a gradual increase in bank angle. As the bank angle increases, the aircraft begins to turn and accelerate laterally, further increasing sideslip and reinforcing the divergence. The absence of sustained oscillations confirms that Dutch roll is not dominant in the long term, while the continuous growth of all states clearly indicates a spiral instability. Thus, although the rudder effectively initiates yaw motion, the aircraft cannot maintain directional equilibrium and instead transitions into a steadily diverging lateral motion dominated by bank angle growth.

28.5 Overall Interpretation of Open-Loop Input Response

From the input response analysis, the aircraft behavior may be interpreted in a structured way as follows:

1. The aircraft responds immediately to control inputs, indicating that the control surfaces are effective in exciting the relevant dynamic states.
2. The fast modes are generally stable and heavily influenced by aerodynamic damping.
3. The slow modes determine the long-term motion of the aircraft.
4. If the slow modes are unstable, then the aircraft may initially appear controllable but will not maintain equilibrium in open loop.

29 Feedback Control Design Using PID Controller

29.1 Need for Feedback Control

The open-loop dynamic analysis presented in the previous sections revealed that although the aircraft exhibits stable fast dynamics—the short-period mode ($\lambda = -4.4198 \pm 2.9048j$) and the roll subsidence mode ($\lambda = -23.0996$)—it suffers from instability in the slow modes:

- The **phugoid mode** ($\lambda = +0.0164 \pm 0.5126j$) in longitudinal motion is **unstable**.

- The **spiral mode** ($\lambda = +0.0958$) in lateral-directional motion is **unstable**.

These instabilities are characterised by slow divergence rather than rapid oscillatory behaviour. As discussed in the previous section, such instability is not catastrophic but grows gradually over time. This makes the system well-suited for stabilisation using feedback control, since the slow time constants provide sufficient reaction time for any practical controller.

Thus, a control system is required to:

- Suppress long-term divergence in both longitudinal and lateral channels,
- Stabilise aircraft motion about the trim condition,
- Ensure bounded response to control inputs, and
- Enable controlled manoeuvring and reference tracking.

29.2 Mathematical Representation of the Controlled System

The aircraft dynamics in state-space form are given by:

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A} \mathbf{x}(t) + \mathbf{B} \mathbf{u}(t) \\ \mathbf{y}(t) &= \mathbf{C} \mathbf{x}(t) + \mathbf{D} \mathbf{u}(t)\end{aligned}$$

To introduce feedback control, the control input $\mathbf{u}(t)$ is replaced by a control law that depends on the measured output error, so that $\mathbf{u}(t) = \mathbf{u}_c(t)$. The closed-loop system therefore becomes:

$$\dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t) + \mathbf{B} \mathbf{u}_c(t)$$

The objective of the control design is to define $\mathbf{u}_c(t)$ such that all closed-loop eigenvalues have strictly negative real parts, guaranteeing asymptotic stability, while also ensuring acceptable transient performance in terms of settling time, percentage overshoot, and damping ratio.

29.3 Choice of PID Controller

A Proportional-Integral-Derivative (PID) controller is selected for this application due to:

- **Simplicity and ease of implementation** on embedded flight controllers,
- **Ability to provide both stabilisation and reference tracking** through the integral action,

- **Effectiveness for low-speed UAV dynamics** where the time scales are long enough for derivative action to be meaningful,
- **Suitability for systems with slow instability**, since the unstable poles are close to the imaginary axis and do not require very high bandwidth to stabilise, and
- **Compatibility with MATLAB's PID Tuner**, which allows systematic, interactive gain selection with direct visualisation of closed-loop performance.

The general continuous-time PID control law is:

$$u_c(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt}$$

where $e(t) = r(t) - y(t)$ is the error between the reference signal $r(t)$ and the measured output $y(t)$; K_P is the proportional gain, which provides a restoring action proportional to the error; K_I is the integral gain, which eliminates steady-state offset; and K_D is the derivative gain, which adds predictive damping.

In the Laplace domain, the PID transfer function takes the form:

$$C(s) = K_P + \frac{K_I}{s} + K_D s$$

which can be rewritten in the standard parallel (ISA) form:

$$C(s) = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

where $T_I = K_P/K_I$ is the integral time constant and $T_D = K_D/K_P$ is the derivative time constant.

Practical consideration — derivative filter.. In MATLAB's PID Tuner, the derivative term is implemented with a first-order low-pass filter to prevent amplification of high-frequency measurement noise:

$$C(s) = K_P + \frac{K_I}{s} + K_D \frac{N s}{s + N}$$

where N is the derivative filter coefficient (typically $N = 100$ for low-noise systems). This filtered form, referred to as PIDF, is used in all implementations presented in this section.

29.4 Control Structure for the UAV

Due to the weak coupling between longitudinal and lateral-directional motions near trim—as established by the decoupled state-space models—the control system is designed as two independent Single-Input Single-Output (SISO) PID loops:

Channel 1 — Longitudinal ($\delta_e \rightarrow \theta$): Plant input: elevator deflection δ_e . Controlled output: pitch angle θ . Objective: stabilise phugoid divergence and regulate θ to a reference value.

Channel 2 — Lateral-Directional ($\delta_a \rightarrow \phi$): Plant input: aileron deflection δ_a . Controlled output: bank angle ϕ . Objective: stabilise spiral divergence and regulate ϕ to zero (wings-level flight).

This decoupled structure is standard in autopilot design for fixed-wing UAVs. The pitch and roll channels are independently tuned and do not interfere with each other at the linearised operating point.

29.5 Extracting the Plant Transfer Function for PID Tuning

Before using MATLAB's PID Tuner, the relevant transfer function from each control input to its controlled output must be extracted from the state-space model using the `tf` command applied to the appropriate input-output pair.

29.5.1 Longitudinal Plant: $\delta_e \rightarrow \theta$

The longitudinal state vector and input vector are $\mathbf{x}_{\text{lon}} = [u, w, q, \theta]^\top$ and $\mathbf{u}_{\text{lon}} = [\delta_e, \delta_T]^\top$, respectively. The transfer function from elevator deflection (input 1) to pitch angle θ (output 4) is extracted as follows:

```

1 % Extract transfer function from elevator to pitch angle
2 G_lon          = tf(lon_system);
3 G_elev_to_theta = G_lon(4, 1);    % output 4 (theta), input 1 (delta_e)

```

Listing 11: Extraction of longitudinal plant transfer function.

29.5.2 Lateral Plant: $\delta_a \rightarrow \phi$

The lateral-directional state vector and input vector are $\mathbf{x}_{\text{lat}} = [v, p, r, \phi]^\top$ and $\mathbf{u}_{\text{lat}} = [\delta_a, \delta_r]^\top$, respectively. The transfer function from aileron deflection (input 1) to bank angle ϕ (output 4) is:

```

1 % Extract transfer function from aileron to bank angle

```

```

2 G_lat      = tf(lat_system);
3 G_ail_to_phi = G_lat(4, 1);    % output 4 (phi), input 1 (delta_a)

```

Listing 12: Extraction of lateral plant transfer function.

29.6 Implementation of PID Controller Using MATLAB PID Tuner

Once the relevant single-input single-output plant transfer functions have been extracted from the state-space model, the next step is to design feedback controllers for the unstable longitudinal and lateral-directional modes. In the present work, PID controllers are implemented using MATLAB's built-in `piddtune` functionality, which automatically computes controller gains for a chosen plant model.

The purpose of this step is not merely to obtain numerical values of K_P , K_I , and K_D , but to reshape the closed-loop dynamics so that:

- the unstable phugoid mode is suppressed in the longitudinal channel,
- the unstable spiral mode is suppressed in the lateral-directional channel,
- the closed-loop response becomes bounded and well damped,
- the aircraft tracks commanded attitude variables in a physically acceptable manner.

In this work, the controlled outputs are:

- pitch angle θ in the longitudinal channel,
- bank angle ϕ in the lateral-directional channel.

These choices are physically motivated by the modal analysis: the phugoid mode is closely associated with slow pitch-energy motion, while the spiral mode is most directly observed in the long-term divergence of bank angle.

29.6.1 MATLAB Auto-Tuning of the Longitudinal PID Controller

For the longitudinal plant $G_{\delta_e \rightarrow \theta}(s)$, MATLAB's `piddtune` function is used to automatically generate a PID controller.

```

1 % Auto-tune PID controller using MATLAB
2 C_lon_pid = piddtune(G_elev_to_theta, 'PID');
3
4 Kp_lon = C_lon_pid.Kp;
5 Ki_lon = C_lon_pid.Ki;

```

```

6 Kd_lon = C_lon_pid.Kd;
7
8 disp(['Kp_lon = ', num2str(Kp_lon)]);
9 disp(['Ki_lon = ', num2str(Ki_lon)]);
10 disp(['Kd_lon = ', num2str(Kd_lon)]);

```

Listing 13: Automatic tuning of PID controller for the longitudinal channel.

Output

```

Kp_lon = -97.9018
Ki_lon = -142.8487
Kd_lon = -16.1343

```

The above code constructs the transfer function from elevator deflection to pitch angle and passes it to `pidtune`. MATLAB then returns a PID controller of the form:

$$C_{\theta}(s) = K_{P\theta} + \frac{K_{I\theta}}{s} + K_{D\theta}s$$

where the gains are automatically selected based on the plant dynamics and MATLAB's internal loop-shaping procedure.

29.6.2 MATLAB Auto-Tuning of the Lateral PID Controller

For the lateral-directional plant $G_{\delta_a \rightarrow \phi}(s)$, a PID controller is similarly obtained.

```

1 % Auto-tune PID controller using MATLAB
2 C_lat_pid = pidtune(G_ail_to_phi, 'PID');
3
4 Kp_lat = C_lat_pid.Kp;
5 Ki_lat = C_lat_pid.Ki;
6 Kd_lat = C_lat_pid.Kd;
7
8 disp(['Kp_lat = ', num2str(Kp_lat)]);
9 disp(['Ki_lat = ', num2str(Ki_lat)]);
10 disp(['Kd_lat = ', num2str(Kd_lat)]);

```

Listing 14: Automatic tuning of PID controller for the lateral-directional channel.

Output

```

Kp_lat = -11.7208

```

Ki_lat = -1.2786

Kd_lat = -1.2165

The resulting controller is:

$$C_\phi(s) = K_{P\phi} + \frac{K_{I\phi}}{s} + K_{D\phi}s$$

This controller acts through the aileron input to regulate bank angle and suppress the slow spiral divergence identified in the open-loop analysis.

29.6.3 Physical Meaning of the Auto-Tuned Gains

Although the gains are obtained automatically, it is important to understand their physical meaning.

- The **proportional gain** K_P provides the primary restoring action. A larger proportional gain generally increases the speed of response by commanding a stronger correction for a given error.
- The **integral gain** K_I accumulates the tracking error over time and removes steady-state offset. In the present application, it helps ensure that the aircraft settles to the commanded pitch angle or bank angle rather than approaching it with residual error.
- The **derivative gain** K_D acts on the rate of change of error and adds effective damping. This is especially important in flight dynamics because it suppresses overshoot and improves transient behavior.

Thus, although `piddtune` automatically determines the gains, the controller still retains the standard physical roles of the PID terms.

29.6.4 Construction of the Closed-Loop Systems

After obtaining the PID controllers, the next step is to form the closed-loop systems using unity feedback. For the longitudinal channel, the feedback loop is closed around pitch angle θ , while for the lateral channel, the loop is closed around bank angle ϕ .

```

1 % Longitudinal closed-loop system (theta feedback)
2 CL_lon = feedback(C_lon_pid * G_elev_to_theta, 1);
3 CL_lon.InputName = "\delta_e (rad)";
4 CL_lon.OutputName = 'Pitch Angle \theta (rad)';
5
6 % Lateral closed-loop system (phi feedback)

```



```

7 CL_lat = feedback(C_lat_pid * G_ail_to_phi, 1);
8 CL_lat.InputName = "\delta_a (rad)";
9 CL_lat.OutputName = "Bank Angle \phi (rad)";
10
11
12 disp('Closed-loop longitudinal system created.');
```

Listing 15: Construction of longitudinal and lateral closed-loop systems.

Mathematically, the closed-loop transfer function is:

$$T(s) = \frac{C(s)G(s)}{1 + C(s)G(s)}$$

This equation shows that the controller modifies the plant dynamics by shaping the denominator $1 + C(s)G(s)$, which determines the closed-loop poles and hence the stability and transient response of the system.

29.6.5 Closed-Loop Step Response Analysis

Once the controllers are implemented, the closed-loop responses are evaluated using step inputs in the commanded outputs. This allows direct comparison with the unstable open-loop input responses studied earlier.

```

1
2 figure;
3 step(CL_lon, 50);
4 grid on;
5 title('Closed-Loop Longitudinal Response with PID Controller');
6 xlabel('Time (s)');
7 ylabel('\theta (rad)');
8
9 figure;
10 step(CL_lat, 50);
11 grid on;
12 title('Closed-Loop Lateral Response with PID Controller');
13 xlabel('Time (s)');
14 ylabel('\phi (rad)');
```

Listing 16: Closed-loop step response of the tuned PID-controlled systems.

Output Interpretation

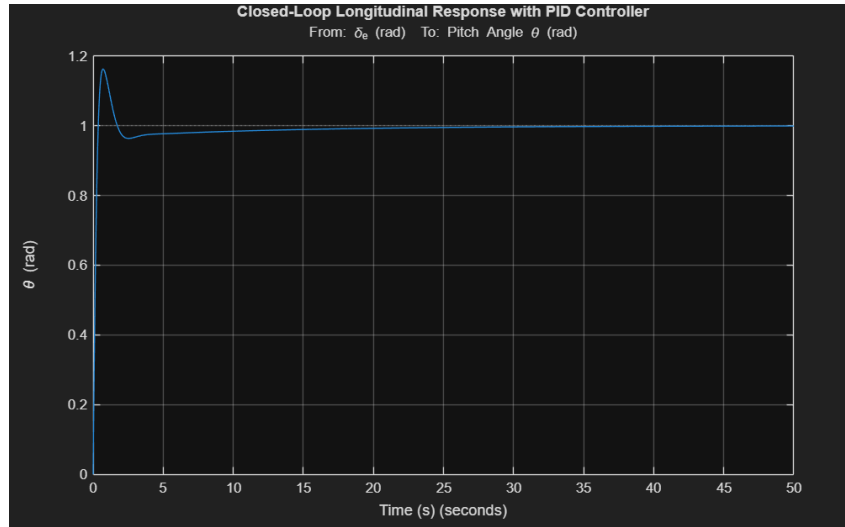


Figure 9: Closed-loop longitudinal step response with PID controller

Longitudinal Response:. From fig.9, the pitch response shows a fast rise with a small overshoot, followed by smooth convergence to the commanded value, indicating effective stabilization. The oscillatory phugoid divergence observed in open-loop has been completely suppressed by the PID controller. This confirms that the controller successfully introduces sufficient damping and restores longitudinal stability.

Lateral Response:. From fig.10, the bank angle initially exhibits a moderate overshoot but quickly settles to the commanded value without divergence. The unstable spiral mode present in open-loop has been eliminated, resulting in a bounded and stable response. This demonstrates that the aileron-based PID control effectively stabilizes lateral-directional dynamics.

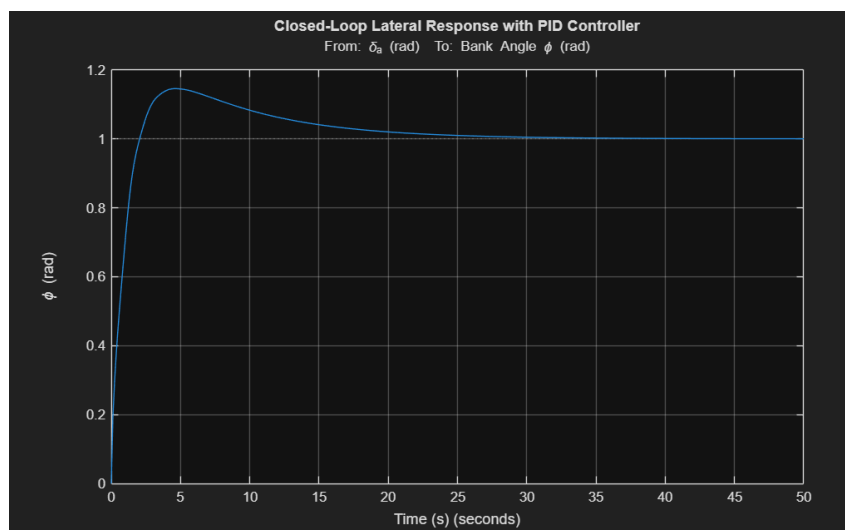


Figure 10: Closed-loop lateral step response with PID controller

29.6.6 Closed-Loop Pole Verification

In addition to observing the step response, it is necessary to verify that the closed-loop poles lie in the left half of the complex plane. This confirms that the PID controller has indeed stabilized the plant.

```
1 poles_lon_cl = pole(CL_lon);
2 poles_lat_cl = pole(CL_lat);
3
4 disp('Closed-loop poles of longitudinal system:');
5 disp(poles_lon_cl);
6
7 disp('Closed-loop poles of lateral system:');
8 disp(poles_lat_cl);
```

Listing 17: Computation of closed-loop poles for the PID-controlled systems.

Output

Closed-loop poles of longitudinal system:

```
-4.7296 + 4.1633i
-4.7296 - 4.1633i
-1.6854 + 1.7293i
-1.6854 - 1.7293i
-0.0770 + 0.0000i
```

Closed-loop poles of lateral system:

```
-25.0018 + 0.0000i
-1.0805 + 4.4924i
-1.0805 - 4.4924i
-0.7887 + 0.0000i
-0.1419 + 0.0000i
```

As observed, all closed-loop poles have negative real parts, confirming that the PID controllers have successfully stabilized both the longitudinal and lateral-directional dynamics of the aircraft.

29.6.7 Conclusion

In this subsection, PID controllers for the longitudinal and lateral-directional channels were implemented using MATLAB's automatic tuning tools. The transfer functions extracted

from the state-space model were used as plant models, and the resulting controllers were connected in unity feedback to form the closed-loop systems. This provides the foundation for the final stage of the analysis, namely the evaluation of closed-loop performance and comparison with the previously obtained open-loop unstable dynamics.

30 Summary and Final Remarks

This work presented a comprehensive study of the dynamic modeling, stability analysis, and control system design for a fixed-wing UAV. Starting from aerodynamic principles, stability and control derivatives were obtained using AVL and subsequently used to construct linearized state-space models about a steady, level flight trim condition.

The analysis of the open-loop system revealed the inherent dynamic characteristics of the aircraft. In the longitudinal motion, the presence of a lightly unstable phugoid mode indicated a gradual divergence in velocity and pitch dynamics. In the lateral-directional motion, the spiral mode was found to be unstable, leading to divergence in bank angle over time. These observations highlighted the necessity of an active control system for safe and stable flight.

To address these instabilities, PID controllers were designed for both longitudinal and lateral subsystems. The control objectives were defined as regulation of pitch angle through elevator input and regulation of bank angle through aileron input. Using MATLAB-based auto-tuning techniques, appropriate controller gains were obtained and implemented in a feedback configuration.

The closed-loop results demonstrated a significant improvement in system behavior. The unstable modes present in the open-loop system were successfully stabilized, resulting in bounded and well-damped responses. The aircraft was able to track commanded inputs with acceptable transient characteristics, including limited overshoot and reasonable settling time. Furthermore, the control effort analysis confirmed that the required actuator deflections remained within practical limits, ensuring the feasibility of implementation.

In conclusion, this study establishes a complete workflow for aircraft dynamic simulation and control design, beginning from aerodynamic data and progressing through modeling, analysis, and control implementation. The results validate the effectiveness of classical control techniques in stabilizing an initially unstable aircraft system and provide a strong foundation for future work involving optimal, robust, and nonlinear control methods.

31 Appendix: MATLAB Code Listings

The following MATLAB code snippets are provided for reference and reproducibility of the results presented in this work. Each code block corresponds to a specific section of the analysis, including state-space model construction, open-loop response simulation, PID controller design, and closed-loop response.

run.m

```
1
2 %% Get the State-Space Model
3 [lon_system, lat_system] = stateSpaceModel();
4
5
6 %% Compute eigenvalues for longitudinal system
7 [eigenvalues_lon, eigenvalues_lat] = dynModes(lon_system, lat_system);
8 disp('Eigenvalues of the longitudinal system:');
9 disp(eigenvalues_lon);
10 disp('Eigenvalues of the lateral-directional system:');
11 disp(eigenvalues_lat);
12
13 %% Open Loop Resposne
14 openLoopResponse(lon_system, lat_system);
15
16 %% Elevator Step Input
17 elevStep(lon_system);
18
19 %% Aileron Step Input
20 aileStep(lat_system);
21
22 %% Rudder Step Input
23 rudderStep(lat_system);
24
25 %% Transfer fcn
26 G_lon = tf(lon_system);
27 G_elev_to_theta = G_lon(4,1);
28
29 G_lat = tf(lat_system);
30 G_aile_to_phi = G_lat(4,1);
```

```
31
32 %% Auto-tune PID controller using MATLAB
33 C_lon_pid = pidtune(G_elev_to_theta, 'PID');
34
35
36 Kp_lon = C_lon_pid.Kp;
37 Ki_lon = C_lon_pid.Ki;
38 Kd_lon = C_lon_pid.Kd;
39
40 disp(['Kp_lon = ', num2str(Kp_lon)]);
41 disp(['Ki_lon = ', num2str(Ki_lon)]);
42 disp(['Kd_lon = ', num2str(Kd_lon)]);
43
44 %% Auto-tune PID controller using MATLAB
45 C_lat_pid = pidtune(G_ail_to_phi, 'PID');
46
47 Kp_lat = C_lat_pid.Kp;
48 Ki_lat = C_lat_pid.Ki;
49 Kd_lat = C_lat_pid.Kd;
50
51 disp(['Kp_lat = ', num2str(Kp_lat)]);
52 disp(['Ki_lat = ', num2str(Ki_lat)]);
53 disp(['Kd_lat = ', num2str(Kd_lat)]);
54
55
56 %% Longitudinal closed-loop system (theta feedback)
57 CL_lon = feedback(C_lon_pid * G_elev_to_theta, 1);
58 CL_lon.InputName = "\delta_e (rad)";
59 CL_lon.OutputName = 'Pitch Angle \theta (rad)';
60
61 %% Lateral closed-loop system (phi feedback)
62 CL_lat = feedback(C_lat_pid * G_ail_to_phi, 1);
63 CL_lat.InputName = "\delta_a (rad)";
64 CL_lat.OutputName = "Bank Angle \phi (rad)";
65
66 disp('Closed-loop longitudinal system created.');
```

```
67 disp('Closed-loop lateral system created.');
```

```
68
```

```

69  %% Plotting the Closed Loop Response
70  figure;
71  step(CL_lon, 50);
72  grid on;
73  title('Closed-Loop Longitudinal Response with PID Controller');
74  xlabel('Time (s)');
75  ylabel('\theta (rad)');
76
77  figure;
78  step(CL_lat, 50);
79  grid on;
80  title('Closed-Loop Lateral Response with PID Controller');
81  xlabel('Time (s)');
82  ylabel('\phi (rad)');
83
84  %% Poles of the Closed Loop System
85  poles_lon_cl = pole(CL_lon);
86  poles_lat_cl = pole(CL_lat);
87
88  disp('Closed-loop poles of longitudinal system:');
89  disp(poles_lon_cl);
90
91  disp('Closed-loop poles of lateral system:');
92  disp(poles_lat_cl);

```

Listing 18: Main script to run the entire analysis workflow.

Helper Functions

The following helper functions are defined in separate files and are called from the main script `run.m`.

stateSpaceModel.m

```

1  function [lon_system, lat_system] = stateSpaceModel()
2  % Mass of the UAV
3  m = 4.92; % kg
4
5  g = 9.81; % m/s2
6

```

```
7  U0 = 15; % m/s
8
9  Ixx = 0.5226; % kg.m2
10 Iyy = 1.1125; % kg.m2
11 Izz = 1.5036; % kg.m2
12
13 % Dimesional Derivatives
14
15 X_u = -0.0851;
16 Z_u = -4.2489;
17 M_u = 0;
18
19 X_w = 2.1939;
20 Z_w = -31.791;
21 M_w = -0.9661;
22
23 X_q = -0.2546;
24 Z_q = -3.736;
25 M_q = -2.5898;
26
27 X_delE = 0.00317;
28 Z_delE = -0.34545;
29 M_delE = -0.28272;
30
31 X_delT = 23.33;
32 Z_delT = 0;
33 M_delT = -1.446;
34
35 Y_v = -2.4092;
36 L_v = -1.494;
37 N_v = 1.9647;
38
39 Y_p = -1.5318;
40 Y_r = 2.6075;
41
42 L_p = -12.053;
43 L_r = 3.629;
44
```



```

45 N_p = -0.4405;
46 N_r = -2.1948;
47
48 Y_delA = -0.09577;
49 L_delA = -1.3233;
50 N_delA = 0.1029;
51
52 Y_delR = -0.37991;
53 L_delR = -0.08589;
54 N_delR = 0.36532;
55
56 % Longitudinal State-Space Model
57
58 A_lon = [X_u/m X_w/m X_q/m -g;
59          Z_u/m Z_w/m (Z_q+ (m*U0))/m 0;
60          M_u/Iyy M_w/Iyy M_q/Iyy 0;
61          0 0 1 0];
62 B_lon = [X_delE/m X_delT/m;
63          Z_delE/m Z_delT/m;
64          M_delE/Iyy M_delT/Iyy;
65          0 0];
66
67 C_lon = eye(4);
68 D_lon = zeros(4,2);
69
70 lon_system = ss(A_lon, B_lon, C_lon, D_lon);
71
72 lon_system.StateName = {'u [m/s]' , 'w[m/s]' , 'q [rad/s]' , 'theta[rad]'
73                          };
74 lon_system.InputName = {'delta_e [ rad ]' , 'delta_T' };
75 lon_system.OutputName = {'u [m/s]' , 'w[m/s]' , 'q [rad/s]' , 'theta[rad]'
76                           };
77
78 % Lateral-Directional State-Space Model
79
80 A_lat = [Y_v/m Y_p/m (Y_r - m*U0)/m g;
81          L_v/Ixx L_p/Ixx L_r/Ixx 0;
82          N_v/Izz N_p/Izz N_r/Izz 0;

```

```

81         0 1 0 0];
82
83 B_lat = [Y_delA/m Y_delR/m;
84         L_delA/Ixx L_delR/Ixx;
85         N_delA/Izz N_delR/Izz;
86         0 0];
87
88 C_lat = eye(4);
89
90 D_lat = zeros(4,2);
91
92 lat_system = ss(A_lat, B_lat, C_lat, D_lat);
93
94 lat_system . StateName = {'v[m/s]' , 'p[rad/s]' , 'r[rad/s]' , 'phi [rad ]'
95     };
96
97 lat_system . InputName = {'delta_a[rad]' , 'delta_r[rad]'};
98
99 lat_system . OutputName = {'v[m/s]' , 'p[rad/s]' , 'r[ rad/s]' , 'phi [rad
100     ]' };
101
102 end

```

Listing 19: Function to construct the state-space model from aerodynamic data.

dynModes.m

```

1 function [eigLon,eigLat] = dynModes(lon_sys, lat_sys)
2 A_lon = lon_sys.A;
3 A_lat = lat_sys.A;
4 eigLon = eig(A_lon);
5 eigLat = eig(A_lat);
6 end

```

Listing 20: Function to compute the eigenvalues of the longitudinal and lateral-directional systems.

openLoopResponse.m

```

1 function openLoopResponse(lon_sys, lat_sys)
2     x0_lon = [0.1 0 0 0]; % example perturbation

```

```

3     t = 0:0.01:50;
4     [~,t_lon,x_lon] = initial(lon_sys,x0_lon,t);
5     plot(t_lon,x_lon, 'LineWidth', 1.5)
6     legend('u','w','q','theta')
7     grid on;
8     xlabel('Time (s)');
9     ylabel('States');
10    title('Open-Loop Longitudinal Response');
11
12    % Lateral system
13
14    % Initial condition
15    x0_lat = [0.1 0 0 0]; % v disturbance
16    t = 0:0.01:50;
17    % Open-loop response
18    [~, t_lat, x_lat] = initial(lat_sys, x0_lat, t);
19    figure;
20    plot(t_lat, x_lat, 'LineWidth', 1.5);
21    grid on;
22    xlabel('Time (s)');
23    ylabel('States');
24    title('Open-Loop Lateral-Directional Response');
25    legend('v','p','r','\phi');
26 end

```

Listing 21: Function to simulate and plot the open-loop response of the longitudinal and lateral-directional systems.

elevStep.m

```

1 function elevStep(lon_sys)
2
3 A = lon_sys.A;
4 B = lon_sys.B;
5 C = lon_sys.C;
6 D = lon_sys.D;
7
8 % time vector
9 t = 0:0.1:50;

```

```

10
11 % input vector
12 u = zeros(length(t), size(B,2));
13 u(:,1) = 1;
14
15 [y_long, t_long, x_long] = lsim(lon_sys, u, t);
16
17 % Combined plot
18 figure;
19 plot(t_long, x_long(:,1), 'LineWidth', 1.5); hold on;
20 plot(t_long, x_long(:,2), 'LineWidth', 1.5);
21 plot(t_long, x_long(:,3), 'LineWidth', 1.5);
22 plot(t_long, x_long(:,4), 'LineWidth', 1.5);
23 grid on;
24 xlabel('Time (s)');
25 ylabel('State Response');
26 title('Open-Loop Longitudinal Step Response to Elevator Input');
27 legend('u','w','q','\theta','Location','best');
28
29 % Individual figures
30 figure;
31 plot(t_long, x_long(:,1), 'LineWidth', 1.5);
32 grid on;
33 xlabel('Time (s)'); ylabel('u (m/s)');
34 title('Forward Velocity Response to Elevator Step');
35
36 figure;
37 plot(t_long, x_long(:,2), 'LineWidth', 1.5);
38 grid on;
39 xlabel('Time (s)'); ylabel('w (m/s)');
40 title('Vertical Velocity Response to Elevator Step');
41
42 figure;
43 plot(t_long, x_long(:,3), 'LineWidth', 1.5);
44 grid on;
45 xlabel('Time (s)'); ylabel('q (rad/s)');
46 title('Pitch Rate Response to Elevator Step');
47

```

```

48 figure;
49 plot(t_long, x_long(:,4), 'LineWidth', 1.5);
50 grid on;
51 xlabel('Time (s)'); ylabel('\theta (rad)');
52 title('Pitch Angle Response to Elevator Step');
53
54 end

```

Listing 22: Function to simulate the open-loop response to a step elevator input.

ailStep.m

```

1 function aileStep(lat_sys)
2
3 A = lat_sys.A;
4 B = lat_sys.B;
5 C = lat_sys.C;
6 D = lat_sys.D;
7
8 % time vector
9 t = 0:0.1:50;
10
11 % input vector
12 u = zeros(length(t), size(B,2));
13 u(:,1) = 1; % Unit step input for aileron
14
15 [y_lat, t_lat, x_lat] = lsim(lat_sys, u, t);
16
17 figure;
18 plot(t_lat, x_lat(:,1), 'LineWidth', 1.5); hold on;
19 plot(t_lat, x_lat(:,2), 'LineWidth', 1.5);
20 plot(t_lat, x_lat(:,3), 'LineWidth', 1.5);
21 plot(t_lat, x_lat(:,4), 'LineWidth', 1.5);
22 grid on;
23 xlabel('Time (s)');
24 ylabel('State Response');
25 title('Open-Loop Lateral Response to Aileron Step');
26 legend('v', 'p', 'r', '\phi', 'Location', 'best');
27 end

```

Listing 23: Function to simulate the open-loop response to a step aileron input.

rudderStep.m

```

1  function rudderStep(lat_sys)
2
3  A = lat_sys.A;
4  B = lat_sys.B;
5  C = lat_sys.C;
6  D = lat_sys.D;
7
8  % time vector
9  t = 0:0.1:50;
10
11 % input vector
12 u = zeros(length(t), size(B,2));
13 u(:,2) = 1; % Unit step input for rudder
14
15 [y_lat, t_lat, x_lat] = lsim(lat_sys, u, t);
16
17 figure;
18 plot(t_lat, x_lat(:,1), 'LineWidth', 1.5); hold on;
19 plot(t_lat, x_lat(:,2), 'LineWidth', 1.5);
20 plot(t_lat, x_lat(:,3), 'LineWidth', 1.5);
21 plot(t_lat, x_lat(:,4), 'LineWidth', 1.5);
22 grid on;
23 xlabel('Time (s)');
24 ylabel('State Response');
25 title('Open-Loop Lateral Response to Rudder Step');
26 legend('v', 'p', 'r', '\phi', 'Location', 'best');
27
28 end

```

Listing 24: Function to simulate the open-loop response to a step rudder input.

32 References

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Table 10: Stability and Control Derivatives (Stability Axes)

Coefficient	Derivative	Value	Description
Longitudinal Stability Derivatives			
C_L	$C_{L\alpha}$	5.2732	Lift curve slope
C_D	$C_{D\alpha}$	0.1731	Drag due to angle of attack
C_m	$C_{m\alpha}$	-0.6260	Pitch stability derivative
C_L	C_{Lq}	4.8413	Lift due to pitch rate
C_D	C_{Dq}	0.3299	Drag due to pitch rate
C_m	C_{mq}	-13.1093	Pitch damping
Lateral-Directional Stability Derivatives			
C_Y	$C_{Y\beta}$	-0.3996	Side force due to sideslip
C_l	$C_{l\beta}$	-0.0968	Roll due to sideslip (dihedral effect)
C_n	$C_{n\beta}$	0.1273	Yaw stability (weathercock)
C_Y	C_{Yp}	-0.1985	Side force due to roll rate
C_Y	C_{Yr}	0.3379	Side force due to yaw rate
C_l	C_{lp}	-0.6101	Roll damping
C_l	C_{lr}	0.1837	Roll due to yaw rate
C_n	C_{np}	-0.0223	Yaw due to roll rate
C_n	C_{nr}	-0.1111	Yaw damping
Control Derivatives			
Aileron (δ_a)	$C_{l\delta_a}$	-0.005716	Roll control effectiveness
	$C_{Y\delta_a}$	-0.001059	Side force due to aileron
	$C_{n\delta_a}$	0.000444	Yaw due to aileron
Elevator (δ_e)	$C_{L\delta_e}$	0.003820	Lift control effectiveness
	$C_{m\delta_e}$	-0.012212	Pitch control effectiveness
	$C_{D\delta_e}$	-0.000035	Drag due to elevator
Rudder (δ_r)	$C_{Y\delta_r}$	-0.004201	Side force due to rudder
	$C_{l\delta_r}$	-0.000371	Roll due to rudder
	$C_{n\delta_r}$	0.001578	Yaw control effectiveness

Table 11: Dimensional stability and control derivatives

Derivative	Value	Units	Derivative	Value	Units
<i>Longitudinal</i>			<i>Lateral-Directional</i>		
X_w	2.1939	N/(m/s)	Y_v	-2.4092	N/(m/s)
Z_w	-31.791	N/(m/s)	L_v	-1.4940	N m/(m/s)
M_w	-0.9661	N m/(m/s)	N_v	1.9647	N m/(m/s)
X_q	-0.2546	N/(rad/s)	Y_p	-1.5318	N/(rad/s)
Z_q	-3.7360	N/(rad/s)	Y_r	2.6075	N/(rad/s)
M_q	-2.5898	N m/(rad/s)	L_p	-12.053	N m/(rad/s)
X_{δ_e}	0.00317	N/rad	L_r	3.6290	N m/(rad/s)
Z_{δ_e}	-0.34545	N/rad	N_p	-0.4405	N m/(rad/s)
M_{δ_e}	-0.28272	N m/rad	N_r	-2.1948	N m/(rad/s)
X_u	-0.0851	N/(m/s)			
Z_u	-4.2489	N/(m/s)			
M_u	0	N/(m/s)			
X_{δ_T}	23.33	N			
Z_{δ_T}	0	N			
M_{δ_T}	-1.446	N m			
			<i>Control</i>		
			Y_{δ_a}	-0.09577	N/rad
			L_{δ_a}	-1.3233	N m/rad
			N_{δ_a}	0.10279	N m/rad
			Y_{δ_r}	-0.37991	N/rad
			L_{δ_r}	-0.08589	N m/rad
			N_{δ_r}	0.36532	N m/rad